

Meat and Meat Products Technology

(Including Poultry Products Technology)

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to
the Young Members
of
the Profession

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Preface

The need for a book on meat technology in the Indian context was being felt since a long time. It became all the more necessary with the introduction of meat technology in the syllabus for veterinary graduates. More than half of the veterinary students hail from vegetarian families and they have very little exposure to meat as such. This book deals with the science of meat and meat products in a systematic, clear, concise form but yet in a comprehensive manner. The basic and emerging concepts in fresh meat technology, processed meat technology and quality control systems have been incorporated in right earnest. Due importance has been given to the poultry products technology in order to have an integrated approach in a single volume. The book will be useful to undergraduate students specialising in veterinary science, food science and technology, postgraduate students and research fellows in concerned specialisation at the university level. This volume will serve as a handbook for meat plant managers, quality control supervisors and processing technicians at the industry level.

This book could not have been written but for the inspiration provided by Dr OS Tomer, Director, IVRI, Izatnagar who constantly encourages his scientists to do their best in research, teaching and extension. I have been fortunate enough in having the benefit of the advice of leading meat professionals especially Dr AK Chatterjee, Dr PC Panda, Dr N Sharma and Dr Sushil Kumar. Besides, cooperation extended by Dr R Somvanshi, Dr JS Berwal, Dr VK Rao and Dr SP Singh has been a big help.

I shall be satisfied if this book serves the intended purpose. There is no limit for the betterment and in that spirit, suggestions extended by senior professionals and colleagues for the improvement of the book will be highly appreciated.

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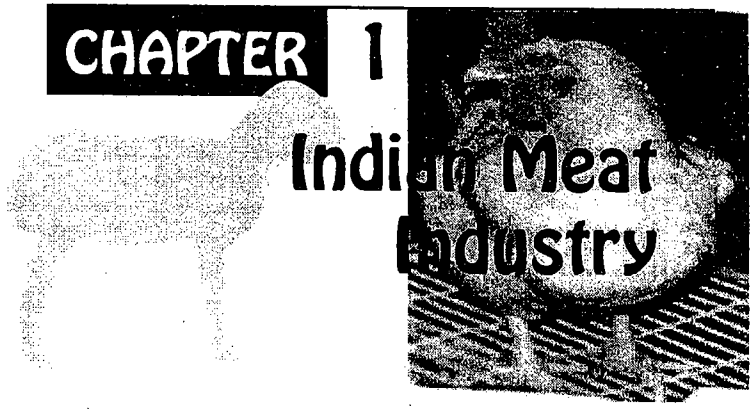
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CHAPTER 1

Indian Meat Industry



LIVESTOCK RESOURCE

India has the largest livestock population in the world. There are 192.9 million cattle, 78.8 million buffaloes, 44.8 million sheep, 118.3 million goats and 117.8 million pigs and 467 million chickens in the country (FAO, 1994). However, Indian Livestock Census (1992) shows slight variation in the livestock population (Tables 1.1 and 1.2)

Our country shares about 50% of the buffaloes and nearly 15% each of cattle and goat population of the world. India ranks first in the world in buffalo and goat population and sixth in the sheep population. The contribution of livestock to the national economy is estimated to be over 11%. Livestock output as a percentage of agricultural output comes to around 26%. National sample survey has reported that in India livestock activities are carried out by over 90% of small cultivators and low wage earners to supplement their income. This is in contrast to the concept of large sized livestock farms in the developed countries. It is also noteworthy that 75% of our livestock population does not conform to the specific breed characteristics and has significantly reduced their production potential.

For a long time, meat industry has remained confined to a very small section of people in our country. These people had little knowledge of clean meat production and effective utilisation of valued slaughterhouse by-products. The scene is now changing. However, industry is still largely based on spent animals except for pig and farm poultry. Most animals are utilised for meat production

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after loosing their economic viability in the primary field. Cow (not bullock) slaughter is banned in India except in West Bengal and Kerala. The concept of meat type animals is yet to take roots in our country, although an awakening in this regard is discernible. Of late, particularly due to export potential, buffalo is emerging as a prospective meat animal.

Table 1.1: Livestock population—1951-92 : All India species-wise

(in million numbers)

Species	*1951*	*1956*	*1961*	*1966*	*1972*	*1977*	*1982*	*1987*	*1992*
Cattle	155.30	158.70	175.60	176.20	178.30	180.00	192.45	199.69	204.53
Buffalo	43.40	44.90	51.20	53.00	57.40	62.00	69.78	75.97	83.50
Sheep	39.10	39.30	40.20	42.00	40.00	41.00	48.76	45.70	50.80
Goat	47.20	55.40	60.90	64.60	67.50	75.60	95.25	110.21	115.28
Pigs	4.40	4.90	5.20	5.00	6.90	7.60	10.07	10.62	12.79
Total Livestock	292.80	306.60	335.40	344.10	353.40	369.00	419.59	445.28	470.14
Poultry	73.50	94.80	114.20	115.40	138.50	159.20	207.74	275.32	307.07

Table 1.2: Livestock population growth rate—1951-92 : All India species-wise

Species	*Annual Growth Rate (%)*							
	1951-56	*1956-61*	*1961-66*	*1966-72*	*1972-77*	*1977-82*	*1982-87*	*1987-92*
Cattle	0.43	2.04	0.07	0.24	0.19	1.35	0.74	0.48
Buffalo	0.68	2.66	0.69	1.61	1.55	2.39	1.71	1.91
Sheep	0.10	0.45	0.88	-0.97	0.50	3.53	-1.29	2.14
Goat	3.26	1.91	1.19	0.88	2.29	4.73	2.96	0.90
Pigs	2.18	1.20	-0.78	6.65	1.95	5.79	1.07	3.79
Total Livestock	0.93	1.81	0.51	0.53	0.87	2.60	1.20	1.09
Poultry	5.22	3.79	0.21	3.72	2.82	5.47	5.79	2.21

Source : Livestock Census, Directorate of Economics and Statistics, Government of India

Meat Production

Meat is an important livestock product which in its widest sense includes all those parts of the animals that are used as a food by man. Though meat has a very high biological value, its production and processing has always been the subject of social prejudice. This factor has adversely affected the growth of meat industry. In many cases, social resistance and ignorance have resulted in inordinate delay and deferment of abattoir modernisation schemes. An important milestone in this area was the establishment of a

modern abattoir at Mumbai in 1973. Further, in the Fourth Five-Year Plan, eight bacon factories were established with the foreign assistance. A few meat corporations were also formed to take up the development of slaughterhouses.

Table 1.3 shows the population, slaughter rate and meat production figures of our traditional meat animals. At present, other than poultry, almost 91 million animals are slaughtered annually yielding 3.98 million tonnes of meat (Table 1.4). It may be noted that nearly 60.6% is contributed by the cattle and buffaloes and 15.6% by sheep and goats. Nearly, 99% pig population is slaughtered annually contributing 9.9% of the total meat production. Poultry with a population of 467 million contribute 0.44 million tonnes of meat (10.7% of total meat production). There has been an impressive rise in the share of poultry and pig meat over the years and the same trend is likely to continue in future also.

The traditional form of meat industry is characterised by unorganised sector in the hands of butcher-workers with very little knowlege of personnel hygiene. At present, there are 3600 licensed slaughter houses in the country. A large number of them are outdated and of substandard according to the present production and processing technology specifications. These slaughterhouses operate as service abattoirs where butchers slaughter the animals for a fee and both edible and non-edible parts of the carcasses are delivered to the butchers. Most of them need modernisation with facilities for lairage, slaughter hall, chilling room, rendering plant

Table 1.3: Food animal population, slaughter and meat production during 1994

Species	*Population (million)*	*Slaughtered (million)*	*Quantity (OOO MT)*	*% Share in total meat production*
Cattle	192.9	12.5 (6.5%)	1292	31.4
Buffaloes	78.8	8.7 (11.0%)	1204	29.2
Sheep	44.8	14.2 (31.8%)	171	4.2
Goat	118.3	47.0 (39.7%)	470	11.4
Pigs	11.8	11.6 (98.9%)	408	9.9
Chicken	467	–	440	10.7

Source : FAO Production Year Book (1994), Vol. 48

etc. While it is imperative to have all these facilities in big cities, a semi-modern approach with mechanical hoist facility is the workable proposition for medium and small sized towns.

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Table 1.4: Value of meat and meat products from live stock sector (1994-95)

Products	Value (Rs. in crores)
Beef and buffalo meat	1923
Mutton	6350
Pork	584
Poultry meat	6407
Meat products	526
Hides	628
Skin	609
Other by-products	233
Total	17260

During the last decade, ten modern abattoir complexes have come up in public sector. An equal number have become functional in private sector also. Eight new projects on modern mechanised abattoirs were initiated in 1990-91. In the Eighth Plan, five private sector export abattoirs are nearing completion. These developmental activities are necessary to improve the image of the Indian meat sector.

World Scene

Meat industry, although in a very developing stage in India, is the top food industry in the world. An analysis of world meat scenario reveals that Europe leads in production followed by Asia. Developed continents (N. America, Europe and Oceania) contribute about 60% to total meat production but they have a monopoly in meat exports as their share was as high as 84%. Nearly 55% of all world meat exports are being shared by European countries alone. The share of Asia in world meat export is very low (6.5%) but it is on the rise.

It is disheartening to note that India with a vast raw material base, contributes less than 1% to the world meat production. Our share in the export of meat is also of the same magnitude. The export of meat from India mainly comprises of fresh chilled meat, frozen meat and frozen meat products (Tables 1.5 and 1.6). A major chunk of meat exports amounting to Rs.400 crores is contributed by buffalo meat (1994-95). Malaysia and UAE are the principal importers of buffalo meat from India (Table 1.7). However, Indian exports of meat constitute barely 0.8% of global export of this commodity. A great potential exists for exporting buffalo meat, beef and poultry in view of increased demand in gulf countries and higher cost of meat from

developed countries. India has additional advantage of geographic proximity to gulf countries. There is an urgent need to tap the world meat export market by establishing modern and hygienic slaughterhouses with chilling facilities solely for export purposes.

Table 1.5: Growth of meat and meat product export

Year	Value (Rs in crores)	Year	Value (Rs in crores)
1985-86	74	1991-92	231
1986-87	76	1992-93	287
1987-88	88	1993-94	245
1988-89	94	1994-95	403
1989-90	114	1995-96	627
1990-91	140		

Source : Economic Survey of India

Table 1.6: Export of meat and meat products (quantity in 000 MT value in Rs in crores)

Item	1990-91		1991-92		1992-93	
	Qty	Value	Qty	Value	Qty	Value
Buffalo meat	63.50	107.00	81.40	189.0	81.60	214.4
Sheep/Goat meat	8.30	31.00	7.60	32.5	13.70	75.0
Processed meat	0.16	0.82	0.19	1.0	0.15	1.0

Table 1.7: Product destination

Item	Major markets
Buffalo meat	Malaysia, UAE, Jordan, Oman and UAR
Sheep/Goat meat	Saudi Arabia, UAE, Oman, Baharain and UK
Processed meat	Jordan, Russia, Oman, Scychelles, Baharain
Poultry Products	Bangladesh, UAE, Oman, Saudi Arabia, Maldives

If the quality of Indian meat is strictly controlled, the country may boost its meat exports by selling to developing Asian, African and Latin American countries, that import about 25% of the world meat exports. These measures will also help in fetching better prices for our produce which is nearly 30% lower than the average world meat export price.

Processed Meat Industry

Most of the meat produced in the country is sold by the retail butcher shops to the consumers as fresh hot meat (unchilled). This meat is

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then cooked in the households in many different ways depending on their taste and preferences. A very small proportion—less than 3% is sold as processed meat. The production of processed meat products in the organized sector got a fillip with the establishment of bacon factories in the Fourth Plan. These bacon factories stimulated the establishment of many processing units in those areas. Nearly 220 licensed manufacturers under MFPO (1973) produce 22,000 tonnes of processed meat which comprises 50% cured products, 20% sausages and 20% canned products. Our production figures are in sharp contrast to developed countries where 65 to 80% of the total meat produced is sold in the processed form. Lately, however, Indian dynamics is changing in favour of processed meat products especially in metropolis and big cities.

Several traditional meat products like meat kabab, chicken biryani, tandoori chicken, meat curry etc. are popular in the non-vegetarian population for a long time. Some other foods products adopted in meat like meat samosa, meat tikka, meat kofta, meat pickle etc. have been able to create an impact on the urban consumer. Various region-specific meat products like Nihari (Delhi), Goa sausage (Goa), Pork pickle (Himachal Pradesh), Yakini and Gustaba (Kashmir), Rapka (Arunachal Pradesh) etc. have good acceptability in their traditional consumers.

Western type meat products like cured ham, bacon, sausages, frankfurters, hot dog, meat patties/burgers, luncheon meat and loaves, liver paste etc. have good demand in cities. Eight bacon factories, five meat corporations and a fairly good number of MFPO licensees in private sector have taken up the production of a wide range of these products. They are catering to the requirements of defence, restaurants and household consumers. Canned meat products are relatively new entrants in the domestic market and are primarily being manufactured for defence supplies. The prices of canned meats are comparatively high rendering them beyond the reach of common consumers, although their presence can be noticed in the departmental stores in the metropolitan cities.

Processed meat products are poised for continuous growth in the country. In big cities, there is an ever increasing demand for 'heat and serve' and 'ready to eat' convenience or fast foods. These are delicious, nutritious and if required, easy to carry home. The growth of fast food parlours and restaurants is attributed to the rapid urbanisation, changing life styles and upwards in the number of

women entering the outdoor work force. It may be pointed out that increase in consumption of value added processed meat products is closely linked with increase in disposable income and growth of urbanisation. Thus, convenience type meat products are going to have spectacular growth in the coming years. Due to nutritional awareness and liberal food habits of the newer generation, the adoption of western type products with indigenous flavour profiles is bound to take place at a rapid rate.

We must strive to export processed meat products rather than live animals and fresh meat. There is a need to study the consumption pattern of meat products in importing countries, so that we can tailor our products according to their requirements. A shift from primary products to value added products besides fetching more profits will decrease the transportation cost and generate more employment. It will also encourage more efficient utilisation of meat by-products.

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CHAPTER 2

Structure, Composition and Nutritive Value of Meat Tissues

Meat is predominantly composed of muscle tissue alongwith various types of connective tissue. The skeletal muscle is the principal muscle tissue in meat, although very little of smooth tissue is also present. The main connective tissue types are adipose tissue (fat), bone and connective tissue proper.

STRUCTURE OF MUSCLE TISSUE

Animal musculature is mostly of mesodermal origin. There are more than 300 muscles in the animal body. These muscles constitute about 30-45% of the live weight or 35-60% of the carcass weight of meat animals. In addition to the skeletal muscle, which forms the bulk of meat, a little of smooth and cardiac muscles are also present in blood vessels and heart respectively. Smooth and cardiac muscles are involuntary in nature. Skeletal and cardiac muscles are sometimes referred as striated muscles due to their specific microscopic appearance.

Skeletal Muscle and Associated Connective Tissue

In general, skeletal muscles are directly attached to the bones, although some attach indirectly via ligament, cartilage, fascia and skin. Each muscle is surrounded by a sheath of connective tissue known as epimysium (Fig. 2.1). From the inner surface of epimysium, a septum of connective tissue penetrates into muscle and surrounds

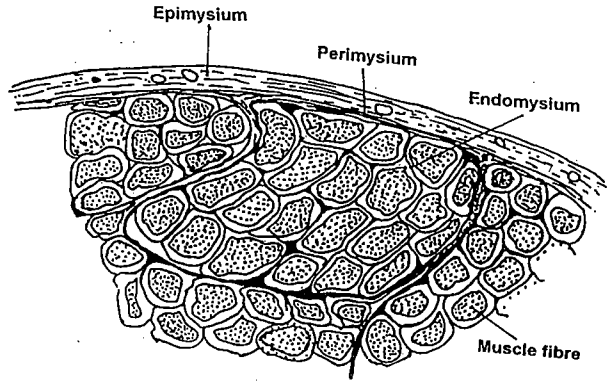


Fig. 2.1: Cross-section of a typical striated muscle depicting arrangement of connective tissue and muscle fibres

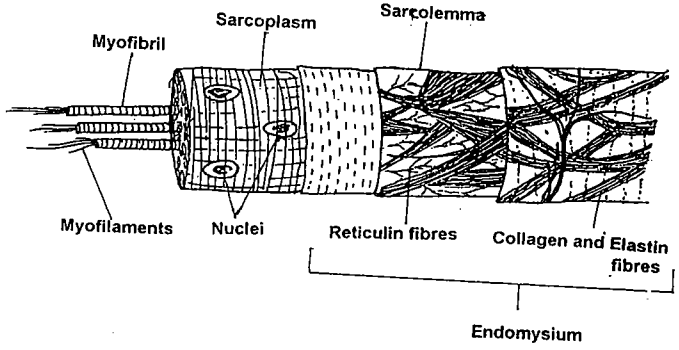


Fig. 2.2: Schematic diagram of a muscle fibre

the bundles of muscle fibres or fasciculi. This connective tissue is called perimysium. It contains major blood vessels and nerves. Muscle fibres or specialised muscle cells are the structural units of the skeletal muscle tissue. Each muscle fibre (Fig. 2.2) is surrounded by a connective tissue layer called endomysium, beneath which is delicate sarcolemma or muscle cell membrane. It transmits nervous signals along the surface of muscle fibre.

Skeletal muscle fibres are long, narrow, almost tubular multinucleated cells which may extend from one end to the other

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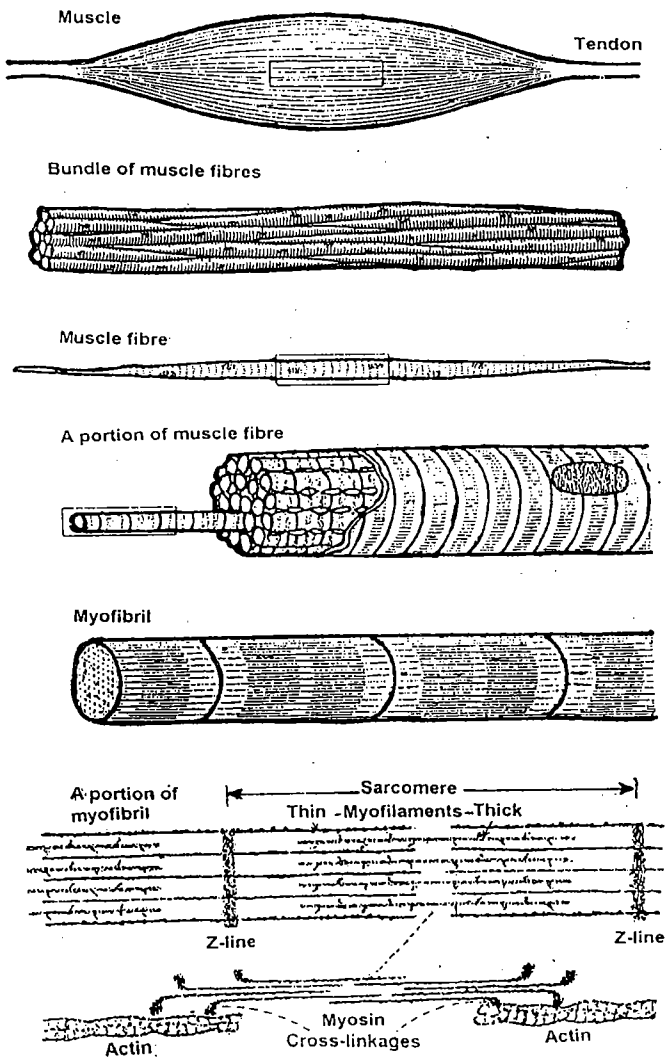


Fig. 2.3: Structure of a skeletal muscle (schematic)

end of the muscle (Fig. 2.3). The nuclei are distributed peripherally close to the sarcolemma. Muscle fibres are usually 10-100μ in diameter with conical or tapering ends and their length ranges from 1-40 mm. The individual fibre may also be classified as red, intermediate and white. Most animal muscles contain a mixture of

these three types. Red muscle fibres have smaller diameter, lower glycolytic metabolism and ATPase activity but higher oxidative metabolism as compared to white muscle fibres.

Myofibrils have a number of elongated unbranched contractile muscle fibre that occupy almost 80% of its volume. They are responsible for the cross-striated appearance of the muscle fibre. Each myofibril is about 1μg in thickness and may run the length of muscle fibre. The cross-striated myofibrils remain embedded in the cytoplasm of the muscle fibre called sarcomplasm. The myofibrils are surrounded by a complex system of membrane tubules. The longitudinal tubules called sarcoplasmic reticulum run parallel to myofibrils. Another series of tubules run transversely as invaginations of the sarcolemma. The sarcoplasmic reticulum and T-tubules are arranged in a sequence and play an important role in generating Ca^{++} fluxes in the excitation-contraction mechanism. Sarcoplasm also contains glycogen particles, lipid droplets etc.

At low magnification (2000×), myofibrils, the intracellular contractile elements, show characteristic banded or striated pattern (Fig. 2.4). This situation arises due to the orderly arrangement of dark or A-band and light or I-band. A clear area in the centre of dark band called H-zone is bisected by a dark M-line. The light or I-band is also bisected by a dark Z-line. The distance between two adjacent Z-lines is called sarcomere. In fact, the sarcomere is the functional unit of myofibril.

At 20,000× magnification, the myofibril itself is seen to be composed of a number of thick and thin filaments. Thick filaments traverse the entire width of A-band whereas thin filaments extend from Z-line to the edge of H-zone. Thus, only thick filaments are present in the H-zone. These thick and thin filaments consist of contractile proteins myosin (Fig. 2.5) and actin (Fig. 2.6) respectively.

Connective tissue serves as the major supportive element of the animal body. It envelops the muscle fibres (endomysium) and bundles (perimysium) and finally the entire muscle (epimysium) connective tissue fibres form the bulk of tendons and ligaments. The tendons attach muscle with bone whereas ligaments connect two bones or support organs. Connective tissue consists chiefly of a mucopolysaccharide matrix in which fibres of collagen, elastin and reticulin are embedded. Collagen fibres predominate over those of reticulin and elastin. Adipose tissue is a specialised type of connective tissue which is primarily made up of cells storing fat droplets. It is seen around kidneys, omentum and in and around various muscles and organs.

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Smooth Muscles

Smooth muscles are found in the gastro-intestinal tract, blood vessels, lymphatics and skin in close association with the connective tissue layers. These are involuntary in nature. Smooth muscle fibres are long, unevenly thickened in the centre and tapering on both the sides. The myofibrils are homogenous and do not show alternating dark and light bands like those of skeletal muscle. There are no Z or M-lines. The sarcoplasmic reticulum is also not much developed.

Cardiac Muscles

The cardiac muscles found in the heart are also involuntary. Their muscle fibres are rounded to irregular in shape and give off branches which get mixed up with those of nearby fibres. The nuclei are placed in the centre of the fibre. Myofibrils depict striations similar to skeletal muscle. The sarcoplasm shows numerous and much more mitochondria than the skeletal or smooth muscles. The intercalated discs are present at the position of Z-lines.

COMPOSITION OF MUSCLE TISSUE

Muscle tissue contains approximately 75% water and 25% solids, of which 19% are proteins. Lipids constitute about 2.5 to 5% of muscle. Chemical composition of a fresh animal muscle is presented in Table 2.1. For simplification, meat can be taken as the postmortem aspect of a muscle.

Water

This is the largest component comprising two third to three fourth of the muscle tissue. Due to polar behaviour, water molecules are attached with the electrically charged groups of muscle proteins. About 40.5% of the total water in muscle is so tightly bound that it is almost impossible to dislocate it. The attraction of molecules keeps on decreasing as the distance from the reactive groups increases. Thus, most of the water exists in immobilised and free forms. When pH of meat is more than isoelectric point, the enhanced negative charge increases the interfilamental space resulting in retention of excess water. It may be noted that almost 70% of water content in fresh meat is located within the myofibrils. Further, an increased water holding capacity is associated with juiciness and tenderness of cooked meat.

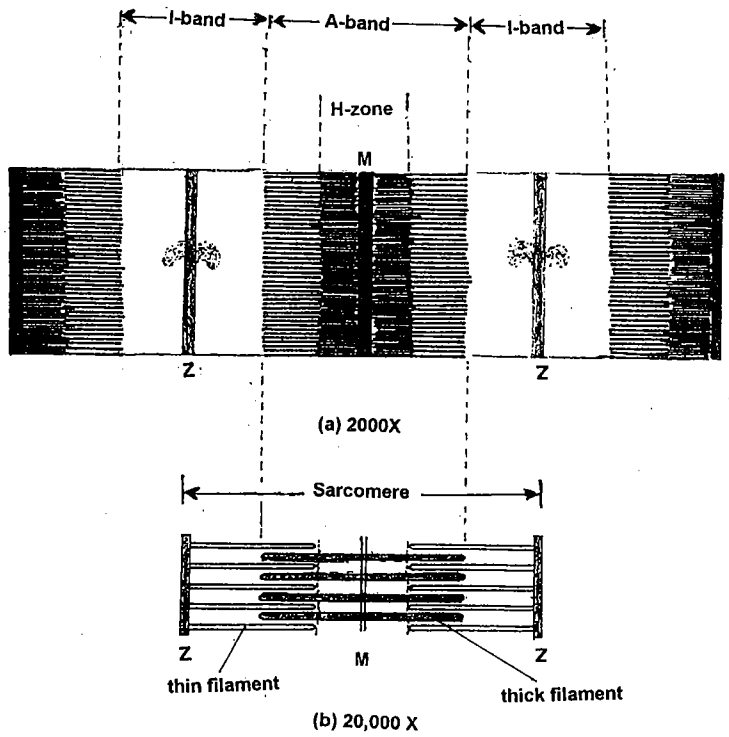


Fig. 2.4: Myofibril depicting
(a) Typical dark (A) band, light (I) band, H-zone, M-Line and Z-lines.
(b) Thick and thin filaments

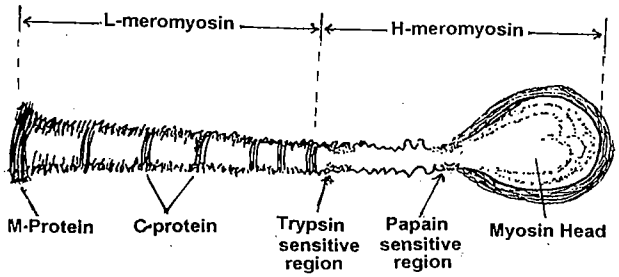


Fig. 2.5: Sketch showing one myosin molecule

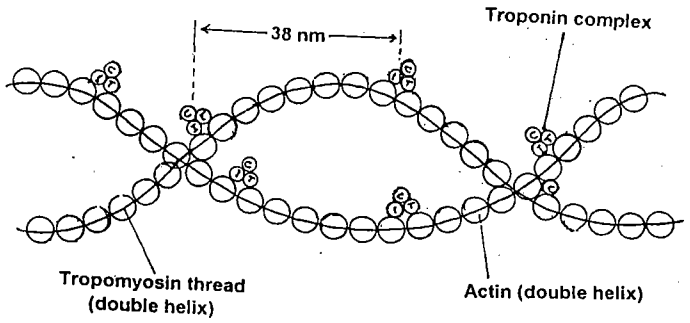


Fig. 2.6: Sketch showing arrangement of actin molecules

Protein

Muscle proteins have been broadly classified into three categories:

i. Myofibrillar proteins – soluble in dilute salt solution
ii. Sarcoplasmic proteins – soluble in water or very dilute salt solution
iii. Stroma or connective tissue proteins – almost insoluble.

Myofibrillar proteins These proteins constitute contractile part of the muscle and make up about 60% of the total protein in the skeletal muscle. Thick filaments constitute the A-band of the sarcomere and consist of the protein myosin. There are 00-400 molecule of myosin in each thick filament. Myosin is a long asymmetrical molecule containing a globular head and two identical polypeptide chains. It has a relatively high charge and shows a strong affinity for the divalent cations, calcium and magnesium. Tryptic digestion splits myosin into two large pieces—heavy and light mesomyosin. Heavy meromyosin head portion carries the ATPase activity and possesses actin binding ability. This ATPase activity of myosin is stimulated by Ca^{++} ions and inhibited by Mg^{++} ions.

The thin filament constitute I-band of the sarcomere and extent on either side of the Z-line beyond I-band also into the A-band between the thick myosin filaments. Actin is the main protein of the thin filament. Actin occurs in two different forms. Globular or G-actin is a monomere form, each molecule of which binds one molecule of ATP or ADP with high affinity. Further, each molecule of G-actin binds one Ca^{++} ion very tightly. At high ionic strength and

usually in the presence of ATP, G-actin is polymerised to a high molecular weight fibrous or F-actin. At low ionic strength, F-actin depolymerises to yield G-actin usually with bound ADP.

Table 2.1: Chemical composition of a typical animal muscle

Component	*Per cent (wet basis)*
1. Water	75
2. Protein	19
a) Myofibrillar proteins	11.5
Myosin	
Actin	
Tropomyosin	
Troiponins C, I and T	
Connectins	
Desmin	
b) Sarcomplasmic proteins	5.5
Myoglobin	
Glycolytic enzymes	
c) Storma or connective tissue proteins	2.0
Collagen	
Elastin	
Sarcolemma	
Sarcoplasmic reticulum	
3. Lipids	2.5
Neutral lipid	
Phospholipid	
Cerebrosides	
Cholesterol	
4. Carbohydrates	1.2
Glycogen	
Glucose-6-phosphate	
Glucose	
Lactic acid	
5. Miscellaneous soluble non-protein subs	2.3
a) Nitrogenous subs	
Creatine	
Ionosine monophosphate (IMP)	
Nucleotides	
Carnosine, Anserine	
b) Inorganic subs	
Total soluble phosphorus	
K, Na, Mg, Ca, Zn and Trace elements	
6. Vitamins	Minute quantities
Fat soluble vitamins	
Water soluble vitamins	

Source : Lawrie (1975)

Relatively small quantities of other proteins generally referred as regulatory proteins are associated with major myofibrillar proteins. Tropomyosin is a fibrous protein which occurs as a double helix. These helical strands are present in close association with in filaments, extending through the grooves of action helix. Troponin is another important regulatory protein which is present in association with thin filament cementing the long chain of tropomyosin thread into the grooves of actin at a regular interval. Troponin is composed of three sub-units:

Troponin T	–	binds to tropomyosin and links it to F-actin filaments.
Troponin C	–	binds to calcium ions
Troponin I	–	inhibits or prevents the interaction between actin and myosin in relaxed state. It allows their interaction only in the presence of calcium ions.

Actinin is a globular protein having similar amino acid composition as actin. It has two subunits. The alpha-actinin is a constituent of Z-line and has been shown to accelerate the polymerisation of G-actin to F-actin. The beta-actinin regulates the length of thin filament.

Myofibrillar proteins are of special interest to the technologists because they contribute to approximately 95% of the water holding capacity, 75% of the emulsifying capacity and to a large extent the tenderness of meat.

Sarcoplasmic proteins These proteins make up about 30-35% of the total proteins in the skeletal muscle. They contain hundred of enzymes for the normal functioning of muscle cell. Sarcoplasmic proteins have mostly glycolytic enzymes and associated proteins like creatine kinase, lactic dehydrogenase, myoglobin, aldolase etc. In general, these proteins are very susceptible to heat.

Myoglobin is a conjugated protein consisting of a prosthetic heme moiety and a protein moiety (globin). It provides red colour to the muscle and serves as a carrier of oxygen to the muscle fibre. It is the most important pigment of meat colour. Cytochrome enzyme, flavin etc. contribute very little to meat colour. The amount of myoglobin present generally shows considerable variation. In loin muscle of different species, the concentration of myoglobin (%) are:

Rabbit	–	0.02
Pig	–	0.06
Sheep	–	0.25
Cattle	–	0.50
Blue whale	–	0.91

At a given time, meat colour depends on the chemical state of myoglobin.

Stroma or connective tissue proteins The connective tissue is composed of an amorphous ground substance or matrix in which formed elements mostly fibres and a few cells are embedded. The ground substance is a viscous glycoprotein solution. The extra-cellular connective fibres are:

a. collagen
b. reticulin
c. elastin

Collagen is the main fibrous protein the muscles and significantly influence the meat toughness. It makes upto 40-60% of the total stroma protein and 20-25% of the total protein in the body. A fine network of collagen fibres is present in almost all tissues and organs including skeletal muscles. It is the most common constituent of tendons. White coloured collagen fibres are straight, inelastic and non-branching. These fibres shrink or shorten at a temperature of 60°C but higher temperatures or boiling causes transformation to water soluble gelatin. Acid or alkali treatment results in the marked swelling of these fibres. Collagen is the only protein possessing a fair amount of hydroxyproline (approximately 14%) and low concentration of hydroxylysine. The smallest unit of collagen molecular structure is tropocollagen which aggregate to form more massive structures—the fibril, primitive fibre and mature fibre. High tensile strength and insoluble nature of mature collagen fibres is due to increased inter-molecular linkages.

Reticulin is composed of small fibres which resemble that of collagen except for its intimate association with a lipid containing myristic acid. Reticular fibres form a network around blood vessels, neural structures, epithelium etc.

Elastin is common in ligaments and its yellow fibres can be easily stretched. Elastin fibres are branched and do not hydrolyse on boiling. Elastin contains two unique amino acids—desmosine and isodesmosine which contribute to its highly insoluble nature. The nutritive value of elastin is practically nil due to its resistance to digestive enzymes.

Lipids

Lipid is a major component of the carcass of a meat animal. It is highly variable and is inversely proportional to the moisture content.

Animal lipids are mainly neutral lipids and phospholipids. The neutral lipids are mostly glycerol esters of the straight-chain carboxylic acids or triglycerides, with small amounts of mono- and diglycerides. Though animal lipids generally contain fatty acids of even carbon atoms (simple triglycerides), mutton and beef have odd number and branched chain fatty acids (mixed triglycerides). The saturated and monounsaturated fatty acids predominate in meat lipids. The principal saturated fatty acids are palmitic and stearic acids with little amount of other fatty acids, whereas most commonly occurring unsaturated fatty acids are oleic, linoleic and linolenic acids. The composition and distribution of meat lipids depend on the diet of meat animal.

Phospholipids are found in muscle tissue in small percentages (0.5-1%). They are principal structural and functional constituents of cell membranes and have a key role in the flavour and shelf stability of meat and meat products. Phospholipids are generally found in meat as phosphoglycerides, the less common are phosphatidyl choline, phosphatidyl ethanolamine, phosphatidyl serine and sphingomyelins. Meat phospholipids are more susceptible to oxidation than triglycerides. They may cause discolouration and off flavour in meat products when exposed to air and heat.

Cholesterol is a minor but important component of animal tissues. Most of it is in unesterified or free form. It has been noticed at high levels in patients with heart disease. A reduction in animal fat intake is usually advised as precaution, since cholesterol is formed in the body even without any dietary intake.

Carbohydrates

Immediately after slaughter, muscle normally contains a very small amount (nearly 1%) of glycogen. It is a macromolecule of glucose residues which serves as a reserve polysaccharide of animal tissue. However, it gets worked up before the completion of rigor mortis and plays a key role in attaining the ultimate muscle pH. Both the rate and amount of glycolysis influence the colour, tenderness and water holding capacity of meat. Insulin deficiency results in decreased tissue glycogen. Glucagon administration causes rapid degradation of liver glycogen to glucose. Besides, several mucopolysaccharides are widely distributed in animal body. Their quantity is less but many of them like heparin, hyaluronic acid, chondroitin sulphate, keratosulphate and glycoproteins are biologically important substances.

Minerals

About 3.5% of the total body weight is inorganic matter. Most of the total body inorganic material is located in skeletal tissue primarily as salts of calcium and phosphorus and some other minerals especially magnesium. In living meat animal, essential minerals like calcium, phosphorus, sodium, potassium, sulphur, chlorine, magnesium, iron etc. and trace elements like manganese, copper, iodine, zinc, cobalt etc. serve a variety of important functions. These functions may be physical, chemical or biological depending on the chemical form and the location in body tissues and fluids.

In the conversion of muscle to meat, inorganic elements play an important role. Their main function relates to development of rigor mortis and alteration of fluid balance which cause a drop in pH and water holding capacity. Inorganic constituents also influence the meat colour and tenderisation. Several inorganic ions act as catalysts during oxidation of meat fat, enhancing the process of rancidity development.

Vitamins

The vitamin content of meat is variable, depending on the species and age of the animal, the degree of fatness and type of feed received by the animal. Water-soluble vitamins are localised in lean tissues whereas fat-soluble vitamins in fatty tissues. Variety meats have substantial amounts of B-complex vitamins. Pork contains 5-10 times more thiamine content as compared to mutton. The exudate from cut meat surfaces and drip loss during thawing of frozen meat contain an appreciable amount of B-complex vitamins and amino acids. Most of the vitamins in meat are relatively stable during processing or cooking. However, thiamine or to some extent vitamin B_6 are susceptible to heat treatment.

NUTRITIVE VALUE OF MEAT TISSUES

Meat is a very nutritious food. It is almost fully digestible. It is appealing to the eyes and pleasing to the sense of olfaction. The nutritive value of meat is attributed to its abundant high quality

proteins, essential fatty acids, some important minerals and B-complex group of vitamins (Table 2.2).

Meat proteins Meat is a concentrated source of proteins which are far superior to the plant proteins due to very high biological value. Most lean meat cuts contain 16.5 to 20% protein. This protein is rich in essential amino acids (Table 2.3). Essential, because there is no provision in the body for the synthesis of these amino acids and a deficient diet will lead to protein malnutrition. In fact, among meat proteins, myofibrillar and sarcoplasmic proteins are of very high quality because they contain enough of essential amino acids. Connective tissue proteins have lower levels of tryptophan and sulphur containing amino acids. Collagen is especially poor in lysine content.

Table 2.2: Proximate composition and caloric value of fresh meats

Meat	*Percentage*				*Calorie (per 100 g)*
	Moisture	*Protein*	*Fat*	*Ash*	
Lamb, composite cuts of trimmed, good grade (lean 79%, fat 21%)	62.5	16.8	19.4	1.3	247
Lamb leg - separable lean, good grade	73.8	19.9	4.7	1.6	127
Pork ham, trimmed thin (lean 77%, fat 23%)	59.2	16.7	23.2	0.8	281
Pork ham, thin separable lean	72.0	20.4	6.6	1.1	147
Beef carcass - total edible, good grade (lean 66%, fat 34%)	54.7	16.5	28.0	0.8	323

Source : USDA Handbook No.8

Table 2.3: Essential amino acids as percentage of crude protein in fresh meats

Amino acid	*Lamb*	*Pork*	*Beef*
Lysine	7.6	7.8	8.4
Methionine	2.3	2.5	2.3
Cystine	1.3	1.3	1.4
Tryptophan	1.3	1.4	1.1
Leucine	7.4	7.5	8.4
Isoleucine	4.8	4.9	5.1
Phenylalanine	3.9	4.1	4.0
Valine	5.0	5.0	5.7

Source : Schweigert and Payne (1956)

Meat fats Meat fats contain ample amount of essential fatty acids and the nutritional demand of the body is easily met by intramuscular

fat itself (Table 2.4). The caloric value of fat in meat is attributed to fatty acids in triglycerides. The number of calories from lean meat is frequently less than those derived from equal weights of many other foods. In fact, the caloric value of a particular meat depends on the amount of fat in the meat cuts. The most abundant fatty acid in meat fat is oleic acid (an unsaturated FA) followed by palmitic and stearic acids (saturated FA). The essential fatty acids in human diets are linoleic, linolenic and arachidonic acids. Pork and organ meats are relatively good sources of linoleic and linolenic acids. It may be noted that excess dietary linoleic acid is converted to arachidonic acid in human body to meet its demand.

The phospholipids are essential components of the cell wall as well as mitochondria and play a vital role in cellular metabolism. Meat fat always contain some quantity of cholesterol and blood cholesterol level increases after ingestion of cholesterol in food. However, it is now well-known that our body is capable of synthesising more cholesterol than is normally ingested. Organ meats have remarkably high cholesterol content as compared to skeletal meat.

Minerals In general, meat is a good source of all minerals except calcium (Table 2.5). The minerals are in close association with lean tissues in meat. Of these, quantitatively potassium is most abundant followed by phosphorus. Meat is a good source of iron which is required for the synthesis of haemoglobin, myoglobin and certain enzymes and thus plays a vital role in maintaining good health. Since human body has a very limited capacity to store iron, mainly in liver, it has to be a part of regular dietary intake. Meat provides this important mineral in a form that is easily absorbed in the system.

Table 2.4: Occurrence of fatty acids as percentage of total meat fat

Fatty acid	*Lamb*	*Pork*	*Beef*
Palmitic acid (C16)	25	28	29
Stearic acid (C18)	25	13	20
Palmitoleic acid (C16:1)	-	3	2
Oleic acid (C18:1)	39	46	42
Linoleic acid (C18:2)	4	10	2
Linolenic acid (C18:3)	0.5	0.7	0.5
Arachidonic acid (C20:4)	1.5	2	0.1

Vitamins Lean meat is an excellent source of B-complex group of vitamins. It has only traces of fat soluble vitamins which are restricted

Table 2.5: Mineral and vitamin (selected) contents of raw meat (mg/100 g meat)

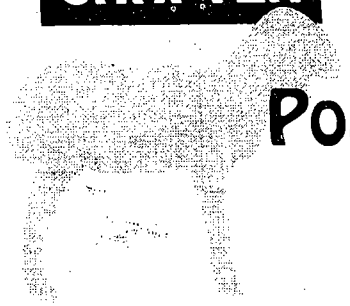
Mineral/vitamin	*Lamb*	*Pork*	*Beef*
Sodium	75	70	65
Potassium	295	285	355
Magnesium	15	18	18
Iron	1.2	2.3	2.8
Calcium	10	9	11
Phosphorus	147	175	171
Thiamine	0.15	0.76	0.06
Riboflavin	0.20	0.18	0.13
Niacin	4.7	4.1	3.6

Source: Watt and Merrill (1963)

to body fat. Vitamin C is almost absent in lean meat, although certain organs contain it in minor quantities. Among the B-complex group of vitamins thiamine, riboflavin and niacin concentrations are quite high. It may be noted that pork surpasses several meats as far as B-complex vitamins are concerned. In fact, lean pork has 5-10 times more thiamine than other meats. It has been noted that in monogastric animals like pig, intake of vitamins in feed is directly reflected in their tissues. The stability of vitamins to cooking or processing is discussed in chapter 22.

Several organ meats have slightly less protein and fat than skeletal meats. However, these are quite often more economical sources of protein and vitamins than retail cuts of skeletal meat. Liver is a very rich source of iron, riboflavin, niacin and vitamin. A nutritive value of skeletal and organ meats should be properly utilised to alleviate the malnutrition.

CHAPTER 3



Postmortem Changes

Slaughter of food animal is followed by a series of physical and chemical changes over a period of several hours or even days resulting in the conversion of muscle to meat. There is immediate loss of oxygen supply to the muscle due to exsanguination (bleeding). As the stored oxygen in myoglobin gets depleted, there is inhibition of aerobic pathway through citrate cycle as well as cytochrome system. The store of creatine phosphate (CP) used for rephosphorylation of ADP to ATP (creatine phosphate + ADP = ATP + creatine) gets soon exhausted. Energy metabolism is then shifted to anaerobic pathway resulting in the breakdown of glycogen to lactic acid. This process continues till all the glycogen stored in the muscle is exhausted. This resynthesis of ATP by anaerobic pathway is not enough to maintain the required ATP level and as it depletes, there is formation of actomyosin resulting in the onset of rigor mortis. The important changes that take place during postmortem period are as follows.

Loss of Homeostasis

Homeostatic mechanism, a system for the physiologically balanced internal environment which helps the body to cope up with the stresses of oxygen deficiency, extreme variation in temperature, energy supply, etc. is lost. The homeostasis is controlled by nervous system which ceases within 4-6 minutes after bleeding. In the absence of blood supply, there is loss of body heat and temperature starts declining.

Postmortem Glycolysis and pH Decline

In the absence of oxygen, anaerobic glycolysis leads to the formation of lactic acid from the glycogen reserves:

$$\text{Glycogen} \xrightarrow[\text{conditions}]{\text{Anaerobic}} \Rightarrow \text{lactic acid} + 2\ \text{ATP}$$

The accumulation of lactic acid lowers down the muscle pH which is an important postmortem change during the conversion of muscle to meat (Fig. 3.1). The rate and extent of pH decline are variable, being influenced by the species of food animal, various preslaughter factors, environmental temperature etc. In most species, a gradual decline continues from approximately pH 7 in the living muscle during first few hours (5-6 hours) and then there is a little drop in the next 15-20 hours, giving an ultimate pH in the range of 5.5 - 5.7. The rate of pH decline is enhanced at high environmental temperature. A low ultimate pH is desired to have a check on the proliferating microorganisms during storage.

A sharp decline in postmortem pH even before the dissipation of body heat through carcass chilling may cause denaturation of muscle proteins. So, the muscles depict pale, soft and exudative (PSE) condition. Contrary to this, muscles which maintain a consistently high pH during postmortem conversion to meat depicts a dark, firm and dry (DFD) condition. Both the conditions are undesirable.

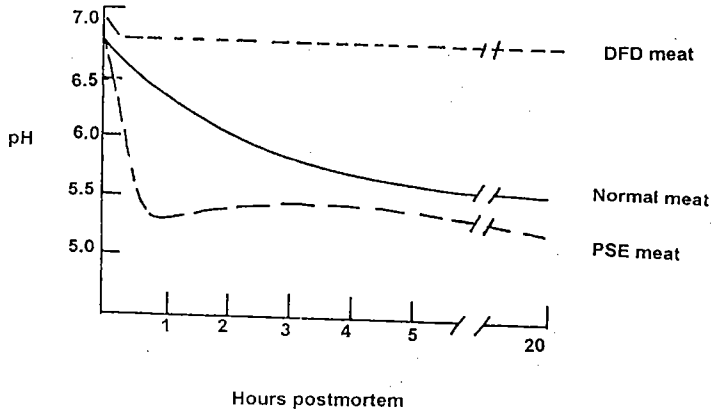


Fig. 3.1: Postmortem pH decline

पुस्तक खराब करु नये

Rigor Mortis

It refers to stiffening of muscles after death and is another important postmortem change in the process of conversion of muscle to meat. It is now very well-known that a particular level or concentration of ATP complexed with Mg^{++} is required for breaking the actomyosin bond and bringing the muscle to a relaxed state and as it drops, permanent actomyosin crossbridges begin to form and muscle gradually becomes less and less extensible under an externally applied force.

During the period immediately following exsanguination, the actomyosin formation proceeds very slowly at first and the muscle is relatively extensible and elastic. This period is called the delay phase of rigor mortis. Then actomyosin formation picks up and the muscle begins to loose extensibility. This phase is called the fast or onset phase of rigor mortis. When all the creatine phosphate (CP) is depleted, ADP can no longer be phosphorylated to ATP, muscle becomes quite inextensible and stiff. This stage marks the completion of rigor mortis. When postmortem pH decline is very slow or very fast, the onset and completion of rigor mortis is rapid. The onset of rigor mortis is enhanced at ambient temperature above 20°C.

The phenomenon of rigor mortis resembles that of muscle contraction in a living animal muscle except that rigor mortis is irreversible under normal conditions. The resolution of rigor mortis takes place due to microbial degradation of muscle structure in due course of time.

Pre-rigor meat is quite tender but its toughness keeps on increasing until rigor mortis is completed. It continues to be tough for some more time. However, with the resolution of rigor due to denaturation or degradation or ageing, meat again becomes tender. The onset of rigor mortis is also accompanied by a decrease in water holding capacity. This is true even when rigor mortis takes place at a high pH due to disappearance of ATP and consequent formation of actomyosin.

Loss of Protection from Invading Microorganisms

During postmortem period, body defence mechanism stops operating and membrane properties are altered. So, during conversion to meat, muscle is quite susceptible to invading

microorganisms. Except for low pH, most of the other postmortem changes favour bacterial growth. Hence, utmost handling precautions are necessary to prevent contamination of meat.

Degradation due to Proteolytic Enzymes

Several autolytic lysosomal enzymes called cathepsins which remain inactive in a living muscle tissue, are activated as the muscle pH declines. These enzymes initiate the degradation of muscle protein structure. In fact, catheptic enzymes are capable of breaking down even collagenous connective tissue of the muscle and cause tenderisation of meat during aging.

Loss of Structural Integrity

Postmortem alteration of membrane properties initiates the degradation of muscular proteins. There is a progressive disruption of myofibrillar structure. The resolution of rigor mortis is known to occur due to disintegration of Z-line structure. A rapid decline in muscle pH also causes denaturation of collagenous connective tissue.

CHAPTER 4



Some Meat Quality Parameters

Fresh meat can be referred as a product which has undergone imminent postmortem changes following slaughter but has not been subjected to any processing. However, fresh meat which has undergone freezing can be conveniently termed as raw meat. Some characteristics of fresh and raw meat need to be properly understood in order to achieve the best results in processing.

Meat Colour

This is the total visual perception of meat. The hue (primary colour), chroma (intensity) and the value (brightness) of meat colour are based on the quantity of principal muscle pigment—myoglobin and its chemical state. It is for this reason that meat colour varies with species, sex, age and even among different muscles of the same species. Myoglobin content of more active species and muscles is higher than the passive ones. Typical colour of meat from various species is:

Mutton and chevon	:	Light to dark red
Pork	:	Greyish pink
Poultry	:	Grey white to dull red
Buffalo meat and beef	:	Cherry red

Myoglobin constitutes about 80-90% of the total meat pigments. The role of haemoglobin in meat colour is negligible in a properly bled muscle. Catalase and

cytochrome enzymes are of little consequence as far as meat colour is concerned. Myoglobin molecule has a protein portion (globin) and a heme (iron containing) ring. It is one-fourth in size as compared to structurally similar haemoglobin molecule. In intact meat, iron in the heme ring of myoglobin exists in the reduced form. Upon cutting, grinding or exposure to air, myoglobin is oxygenated to form oxymyoglobin within 30-45 minutes. Oxymyoglobin has a bright red colour (bloom) which is very much desired by the consumers. However, this pigment is comparatively unstable. In conditions of less oxygen, partial vacuum or semipermeable package, myoglobin as well as oxymyoglobin is oxidised to brown coloured metmyoglobin. At the time of meat purchase, brown colour is usually associated, by the consumers with meat that has been stored for long although it is not always true. In order to prevent the formation of brown colour, fresh meat is often packed in films with very good gas (oxygen) transmission rate.

Water Holding Capacity

Water constitutes about 76% of fresh meat. It is a universal solvent and takes part in a large number of biological reactions. In muscles, water molecules carry positive and negative charges. The location of these molecules allows water to exist in three different forms—free water, immobilised water and bound water. The water molecules held by capillary forces on the surface make up free water which can be removed by application of even minor physical force. The middle layer of water molecules remain in contact with proteins and make up immobilised water, a large part of which can be removed by application of severe physical conditions. However, 4-5% of water molecules are so tightly bound to the charged hydrophilic groups on the muscle proteins that they do not allow this bound water to escape by application of any physical force.

The capacity of meat to retain its water during the application of physical forces is known as water holding capacity (WHC). This property of meat has a special significance because it contributes to the juiciness of cooked meat besides influencing the texture and colour. Fresh meat with a good water holding capacity is less prone to shrinkage during storage. WHC of meat is very important in processing where meat is subjected to physical forces such as cutting, grinding, filling, pressing, heating etc.

Marbling

It refers to the intramuscular fat which can be visibly detected when the muscle surface is cut. The solidification of this fat during chilling contributes to the firmness of meat. Marbling prominently figures in the USDA quality grades for meat because of its merchandising value. During handling of chilled meat, some special retail cuts like chops and steaks retain their uniform thickness and typical shape due to marbling. Besides, marbling also enables meat to bear the impact of comparatively high cooking temperature. During thermal processing, moderately marbled meat yields a juicy and flavourful product whereas too little marbling yields a dry and flavourless product. Excess marbling neither enhances the eating satisfaction nor desired in a fat conscious society.

Quantum of Connective Tissue

The amount of connective tissue in meat has a direct bearing on its textural characteristics. During animals life time, more active muscles tend to deposit more connective tissue to gain strength. This factor accounts for the coarse texture of biceps femoris and tenderness of loin eye muscle postmortem. The quantum of connective tissue per unit muscle does not increase with age and is not responsible for tough meat of older animals. In fact, it is the increase in muscle fibre diameter and consequent increase in muscle fibre bundles which account for the coarse texture of such meat.

Most meat cutting practices are based on separation of coarse textured meat from the tender meat, so as to facilitate the right kind of cooking procedure and derive maximum palatability pleasure.

Firmness

The firmness of meat is a good quality parameter which plays an important role in carcass setting, fabrication, aging, processing, slicing and product display. During carcass chilling, the firmness increases due to loss of extensibility associated with the completion of rigor mortis. Fresh meat having a high water holding capacity shows good firmness and tight structure. It can be objectively measured by shear force apparatus or penetrometer. Meat with a good degree of firmness yields a comparatively better quality processed meat products.

Meat Storage Conditions

Cold shortening When prerigor meat is fast chilled below 15°C, there is shortening of muscles. This shortening or contraction is more at 0°C and still more at -2°C. However, it is minimum at 14-21°C. At 40% shortening, which is quite common, meat becomes very tough and large quantity of meat juices are exuded. However, in case cold shortening exceeds more than 40%, Z-lines are disrupted and meat becomes soft and tender. Otherwise also, cold shortening is a reversible phenomena and is resolved when glycogen content of muscle is exhausted. In the meat plant, cold shortening can be avoided by keeping the meat above 14°C for sufficient time to pass the rigor stage.

Thaw rigor When prerigor meat is frozen, a severe type of rigor mortis ensues during thawing. The shortening so produced may be 60 to 80% of the original length of the unrestrained muscle. Although shortening is less in a muscle attached to skeleton, the condition results in a tough meat and heavy drip losses.

Antemortem Factors Affecting Meat Quality

Various stress factors such as extremes of environmental temperature, overcrowding, preslaughter transportation, struggle during immobilisation and bleeding etc. have ultimate bearing on the quality of meat. Exposure to low temperature may cause shivering which results in reduction in muscle glycogen level. During any environmental stress, susceptible pigs, show porcine stress syndrome which is characterised by muscle tremors, anxious behaviour and reddening of skin. In such animals antemortem temperature rise, lactic acid build up and ATP depletion are the general features and postmortem changes are rapid. So conversion from muscle to meat is also fast due to a sharp fall in pH and muscle denaturation. Ultimately, meat becomes pale in colour, soft in texture and exudative or moist during chilling itself. Stress resistant animals are able to withstand exercise, fasting, fatigue, fight etc. but at the expense of their glycogen reserves. Slow and limited glycolysis often result in high ultimate pH and excellent water binding capacity. So meat appears dark, firm and dry.

Preslaughter handling such as long distance transportation and overcrowding in trucks is also stressful to the animals. This treatment causes shrinkage of muscular tissue and comparatively low dressing percentage. So holding of such animals for resting and feeding can be helpful in restoring their depleted glycogen level. However, basic principle of feed withdrawal and adequate water supply for 24 hr. before slaughter has to be followed for ease of evisceration and to reduce microbial contamination of carcass from intestinal contents.

Like other livestock products, meat is also quite prone to the absorption of off-odours from the surrounding environment. Hence, meat should not be stored in the presence of other strong smelling substances.

CHAPTER 5

Meat Cutting and Packaging

Meat cutting refers to the skill of separation of carcass into wholesale primal cuts in order to facilitate requirements of meat trade, cater to the consumer preference and convenient handling by the butchers. It is a specialised work which requires expert butcher to have a fair understanding of animal conformation, a good knowledge of carcass components and breakup, consumer preference and on-the-job experience. Different cutting methods are followed in various countries. Hence, terms, like British cutting method, American cutting method, French cutting method etc. are common. Besides, regional variations within a country are also not uncommon. The primal cuts are divided into subprimals which are further made into retail cuts. Any technical bulletin on specialised meat cutting can give further details.

Whatever system is adopted, the underlying principles remain the same. In many developing countries a centralized meat cutting room is attached to the slaughterhouse where primal cuts are prepared as in line operation and packed. Basic requisites in meat cutting are:

i. The carcass has to be essentially chilled for proper meat cutting and trimming job.
ii. Meat cutting room should be maintained at a temperature of 15-20°C and relative humidity of 80%. This environment is wholesome for meat and convenient to workers.
iii. All meat cutting equipments and machinery such as meat cutting tables, various types of knives, manual as well as electrically operated saws should be made up of stainless steel and be sufficiently sharp.
iv. Meat cutting operation has to be done by adequately trained and experienced butchers. It is important to maintain uniformity in cuts and economy of merchandising.

v. Approved meat cutting method should be followed step by step as per standard specifications.
vi. Thumb rules for meat cutting techniques are:
   a. More valued primal cuts are separated from the less valued counterparts.
   b. The muscular portion is cut with a sharp knife.
   c. The bony structure is severed with a manual or mechanised saw.
   d A limited force is applied while disjointing whenever joints are involved.
vii. In line operations, in a meat cutting room should be fully synchronised. Different cuts, fat, trimmings etc. should be transferred to their natural destinations.

Wholesale Cuts of Lamb Carcass

USDA and many other international standards specify the division of lamb carcass into fore saddle and hind saddle by cutting between the last two ribs. The right and left sides are not separated (Fig. 5.1). However, BIS specify the division of carcass into right and left sides.

Fore saddle (53%)

1. Neck – cut at last cervical vertebrae where it blends with shoulder
2. Shoulder – cut between 5th and 6th ribs
3. Rack – portion from 6th to 12th rib
4. Breast – cut forward from midway of the last rib to 1/2" above elbow joint
5. Foreshank – cut containing foreshank bones

Hind saddle (47%)

1. Loin – cut hind quarter by sawing in front of hip bone in between last two lumbar vertebrae
2. Leg – remaining portion of hind quarter
3. Flank – thin meat without bone from the natural seam starting from breast.
4. Suet and kidney

In India, people generally go for six cuts only—neck, shoulder, rack, foreshank and breast, loin and leg (Fig. 5.2).

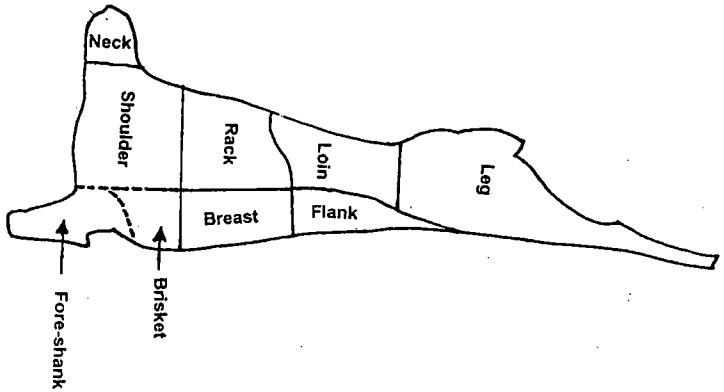


Fig. 5.1: Goat and Lamb Carcass–Primal cuts (USDA)

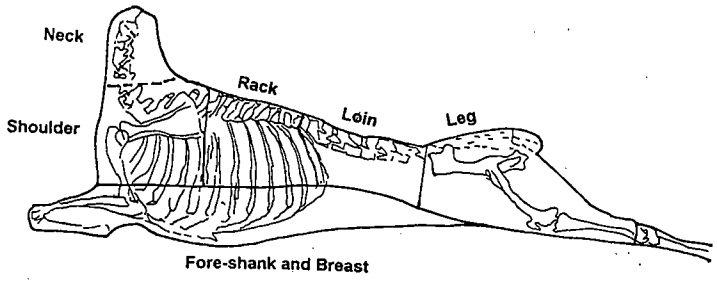


Fig. 5.2: Goat and Lamb Carcass–Primal cuts (Indian Method)

Wholesale Cuts of Pork Carcass

Pork carcass is divided into right and left sides. Then front feet are removed 1" above knee and hind feet at the lower edge of the hock and each side is subjected to six cuts (Fig. 5.3).

Anterior part called rough shoulder is separated from the posterior by cutting between 2nd and 3rd ribs. This is made into three wholesale cuts—jowl, butt and picnic.

1 Jowl – cut close to the neck line
2. Boston butt – upper 1/3 of the skinned shoulder
3. Picnic shoulder – lower 2/3 of the shoulder
4. Ham – cut between 2nd and 3rd sacral vertebrae at right angle to the line of leg

5. Loin – upper middle portion
6. Belly – lower middle portion between picnic shoulder and ham.

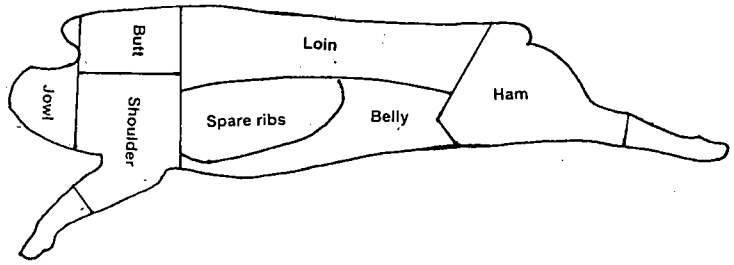


Fig. 5.3: Pork carcass–Primal cuts

Wholesale Cuts of Buffalo or Beef Carcass

A. Carcass sides are separated. The right side is called closed (kidney close) side whereas left side is called open (kidney free) side (Fig. 5.4).

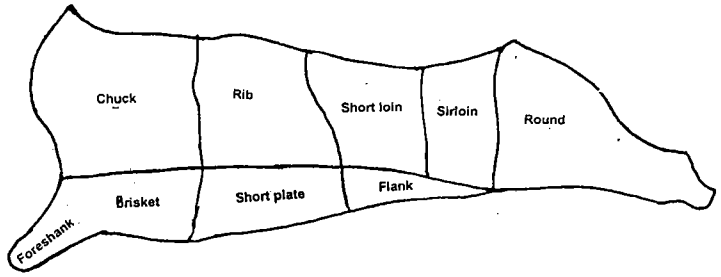


Fig. 5.4: Buffalo or beef carcass–Primal cuts

B. Each side is now subjected to quartering or ribbing. The fore quarter and hind quarter are separated by making a cut between 12th and 13th ribs.

C. Fore quarter is cut between 5th and 6th ribs to have shank, brisket and chuck in the anterior part and rib and plate in the posterior part.

1. Rib and plate – From the posterior part separate the upper rib from the lower plate by a straight and parallel cut to the backbone

2. Shank and Brisket – Place the anterior part on the table with rib side down. The shank is removed by cutting parallel to the underline and just dorsal to the lower extremity of the humerus. The brisket is removed by continuing the same cut through the breast bone and lower ends
3. Chuck – This is the remaining large square cut with ribs.
4. Flank – Cut below the aitchbone following the natural seam upto thick muscle of the flank and then extending straight upward cut upto 4 cm below the eye muscle.
5. Round – Start with a cut 4 cm anterior to the pelvic bone and go parallel to the rib and loin.
6. Loin – Left over cut which can be further—sectioned into anterior short loin and posterior sirloin by making a cut between lumbar and sacral vertebrae.

D. From the hind quarter first remove kidney knob i.e. kidneys alongwith the adjoining fat.

Grading

There is lot of variation in the carcass conformation, size and meat quality depending on the breed, age group, body conformation and health status of meat animals. Meat grading refers to the sorting or grouping of meat carcasses and cuts on the basis of their conformation, finish and overall quality. Although this concept is yet to develop in our country, it is beneficial to the animal raiser at the farm, processor at the packing plant, purchaser at the retail outlet and above all to the quality conscious consumers. Carcass conformation, trimming, meat to bone ratio, colour etc.play an important role in grading. Carcasses of buffalo, goat, sheep or calf may be graded for meat quality as prime, choice, good or poor. Pork carcasses are generally graded on the basis of yield in most of the developed countries.

Grading of meat carcasses and wholesale cuts is still confined in India at the export points, that too as per the agreement between the exporter and importer. Directorate of Marketing and Inspection,

Government of India has already evolved its grades but the same are still to be implemented.

PACKAGING

Packaging refers to the scientific method of containing a food for optimum protection till it reaches the ultimate consumer. Modern concept treats packaging as an important marketing tool also. Proper packaging helps in maintaining the quality of a product during storage, transport and provides convenience for easy handling by the consumers.

Packaging of fresh meat is required for the following reasons:

i. To prevent moisture loss during storage
ii. To offer meat in a most desirable colour to the consumers
iii. To prevent further microbial contamination
iv. To check the pick up of foreign odours by meat
v. To prevent lipid oxidation.

The following packaging materials and techniques are in vogue for primal and retail cuts of fresh meats.

Overwraps

Primal and subprimal cuts of fresh meat are overwrapped with thermoplastic films having excellent optical properties. Low density polyethylene (100 gauge) is the most widely used and cheapest film for this purpose. However, rubber hydrochloride, nylon - 6 or 11 film, highly plasticised PVC film (70 gauge) can also serve this purpose. Cellophane coated with nitrocellulose on one side has been in use for wrapping fresh meat for a considerable period. The uncoated side is kept in contact with meat. Another grade of cellophane coated with polyethylene can be used for irregular shaped meat.

Tray with Overwrap

The most common packages for retail fresh meat cuts in western countries are polystyrene foam or clear plastic trays overwrapped with a transparent thermoplastic film. These trays offer an aesthetically appealing background. Provision of absorbant cotton within the pack eliminates the chances of excessive meat juice accumulation. This meat has a shelf life of 10 days at 0°C. However, it can retain the desirable bright red colour for 5 days only.

Shrink Film Overwrap

These films are biaxially oriented to stay stretched at room temperature but shrink on exposure to hot air or hot water for a few seconds. Shrink films are good water vapour barriers and have high structural strength. These films are used for wrapping large and uneven cuts of fresh meat. These are also frequently recommended for storage of carcass quarters under frozen condition. Shrink films offer neat appearance, contour tight package and are easy to handle. Heat shrinkable polypropylene, irradiated polyethylene or polyvenyledene chloride (PVDC) can be used for this purpose.

Vacuum Packaging

This technique is recommended for long term storage of primal and subprimal cuts of buffalo meat and beef. It ensures a shelf life of 8-10 weeks at 0°C. Vacuum packaging of lamb and pork is avoided for different reasons. Lamb may have a shelf life of 3 weeks only because of comparatively high pH. Pork starts with a large load of bacteria and pork cuts are reported to have a shelf life of 2 weeks only at 1°C.

Vacuum shrink packaging in cryovac barrier bags may provide a means of storage and transport of frozen carcasses, sides or quarters to overseas destinations. Vacuum packaging is done either in laminates or co-extruded films. Typical laminates in use are:

Aluminium foil/Polyethylene
Polyster/Polyethylene
Polyamide/Polyethylene
PVDC/Polyester/Polyethylene

Modified Atmosphere Packaging

In this technique, atmosphere inside the package is modified to extend the shelf life of meat while retaining its colour, flavour and weight. The package air can be suitably replaced by gases usually nitrogen, oxygen or carbon dioxide alone or in combination. Different meats have varying modified atmosphere requirements. Buffalo meat and beef need a high oxygen content to maintain a bright red colour. Pork needs less oxygen due to high fat content. Nitrogen serves as an inert filler to balance a gas mixture. However, its use increases the cost of packaging.

CHAPTER 6

Principles of Various Preservation Techniques

Meat is a highly perishable commodity due to nearly neutral pH (low acid food), high moisture content and rich nutrients. Contamination with spoilage organisms is almost unavoidable which makes the preservation of meat more difficult than most other foods. Unless proper preservation methods are adopted, deteriorative microbial activity, enzymatic and chemical reactions alongwith physical changes are bound to occur. Efforts should be made to attain asepsis by avoiding contamination as much as possible. However, once meat is contaminated with microorganisms, their removal is difficult. Hence, preservation of meat is usually accomplished by the use of low temperature, high temperature, moisture control, direct microbial inhibition etc. Various methods employed to prolong the shelf-life of meat are :

1. Chilling/Refrigeration
2. Freezing
3. Curing
4. Smoking
5. Thermal processing
6. Canning
7. Dehydration
8. Irradiation

Chilling/Refrigeration

This is the most widely used method of preservation for short term storage of meat because chilling or refrigeration slows down the

microbial growth and enzymatic as well as chemical reactions. Storage of fresh meat is done at a refrigeration temperature of 2 to 5°C. Refrigeration of meat begins with the chilling of animal carcasses and continues through the entire channel of holding, cutting, transit, retail display and even in the consumer household before ultimate use. The relative humidity is generally kept 90% in order to check excessive shrinkage due to loss of moisture. Carcasses are first held in chill coolers (15°C) to remove their body heat and then passed on to holding coolers (5°C). It is important to maintain proper spacing between carcasses so as to allow thorough air circulation.

The refrigerated storage life of meat is influenced by species of origin, initial microbial load, packaging and temperature as well as humidity conditions during storage. Pork and poultry start with a comparatively high microbial load. Irrespective of species of origin, utmost care should be taken during handling of meat in order to check further microbial contamination. Since convenience of meat plant workers is also important, the temperature in cutting and packing halls generally exceeds 5°C. As such, operations should be accomplished by specialised hands within the minimum possible time. Refrigerated temperatures favour the growth of psychrophilic organisms causing spoilage of meat in due course of time.

Generally, fresh meat is maintained in good condition for a period of 5-7 days at a refrigerated temperature of 4±1°C. Processed meat products are also stored under refrigeration till these are finally consumed. These meat products are less perishable as compared to fresh meat. The refrigerated shelf life of these products depends on the processing steps followed in each case.

Freezing

Freezing is a method of choice for the long term preservation of meat. It stops the microbial growth and retards the action of enzymes. It has the advantage of retaining most of the nutritive value of meat during storage, although a very little loss of nutrients does occur in the drip during thawing process. Since drip is not possible in cooked meat products, proper freezing conditions result in retention of most of the nutritional and sensory properties.

It is utmost important to wrap fresh meat in suitable packaging film before freezing, otherwise meat undergoes freezer burn. This

abnormal condition occurs due to progressive surface dehydration resulting in the concentration of meat pigments on the surface. This discolouration in frozen meat due to sublimations of ice crystals, is irreversible condition. On cooking, freezer burn meat is quite tough and lack juiciness.

The quality of frozen meat is also influenced by freezing rate. In slow freezing, extracellular water freezes more quickly due to low solute concentration as compared to intracellular water. Thus, there is formation of large extracellular ice crystals which may cause mechanical damage to the muscular tissue, giving it a distorted appearance in the frozen state. Contrary to this, in fast freezing, numerous small ice crystals are formed uniformly throughout the meat tissue. Thus, problem of muscle fibre shrinkage and distorted appearance is not there. Besides, drip losses during thawing are considerably low as intracellular water freezes within the muscle fibre itself. Numerous small ice crystals on the surface of fast frozen meat also impart it a desired lighter colour as compared to slow frozen meat.

Various type of freezers are employed to freeze meat and meat products. In plate type freezers, meat is placed in trays which remain in direct contact with metal freezer plates. A temperature of –10°C or so is achieved. Blast type freezers are used in large meat plants. Such freezers render fast freezing of meat products due to rapid air movement. A temperature range of –10 to –30°C is generally achieved.

The quality of meat and meat products can be preserved for months together during frozen storage at –10°C. However, a storage temperature of –18°C is recommended because at this level almost all water in meat is frozen and minor temperature fluctuations can be taken care of. At –18°C, storage life of buffalo meat, beef, mutton and chevon is approximately 6 months, while that of pork and poultry is less (4 months) because of associated unsaturated fat, prone to rancidity development. Storage life of cured and salted meat products is still limited (2 months) as salt is a pro-oxidant. However, at –10°C, storage life of these meats is reduced to half or even less. Thawing of meat should be done within the package itself preferably in a refrigerator so as to minimise the drip losses. However, if thawing is to be accomplished at a short notice, warm air or lukewarm water may be used. Refreezing a thawed meat is not suggested in tropical countries and repeated freezing should not be practised.

At times, freezing and thawing of young chicken may pose the problem of bone darkening due to leaching of haemoglobin from the marrow of porous bones to the adjoining muscle tissue. This tissue appears grey or black after cooking, although other sensory attributes are not affected.

Curing

Preservation of meat by heavy salting is an age old practice. It was applied as a thumb rule because refrigeration facilities were not available. Later, curing by common salt and sodium nitrite resulted in comparatively improved products. These days mild curing of meat products is practised mainly for specific flavour and colour development and preservative effects of curing ingredients is an added advantage. Sodium chloride, sodium nitrite, sodium nitrate and sugar are the main curing ingredients.

Sodium chloride (common salt) exerts its preservative action as follows:

i. It acts by dehydration and alteration of osmatic pressure that inhibits the growth of spoilage bacteria.
ii. Chloride ions in the salt directly act on the microorganisms.
iii. It slows down the action of proteolytic enzymes in meat.

Besides, sodium chloride interacts with fatty acids to enhance the flavour of the cured products. It also contributes to the tenderness of the product.

Sodium nitrates and nitrite Serve to stabilise the attractive curec meat colour and impart characteristic cured meat flavour. Colour reactions of cured meats can be summarised as follows:

Nitrate reducing organisms
Nitrate ——————————> nitrite

Absence of light and air
Nitrite ——————————>Nitric oxide (NO)
NO + Mb ——————————>Nitric oxide metmyoglobir (NOMMb)

Favourable conditions
NOMMb ——————————> Nitric oxide myoglobin (NOMb) (unstable cured pigment)
NOMb + heat/smoke ——————————>Nitroso haemochromoger (stable pink pigment)

Nitrates and nitrites at permitted levels of 500 ppm and 200 ppm respectively act as preservatives by inhibiting the growth of a number of bacteria especially *Clostridium botulinum*. These chemicals also retard the development of rancidity. Cured flavour develops due to reaction between fatty acids and sodium nitrite resulting in the formation of benzonitrile and phenylacetonitrile. However, nitrite has been found to be involved in the formation of nitrosamine which is supposed to be carcinogenic. Sugar counteracts the harsh hardening effect of salt, adds to the flavour development and also serves as an energy source for nitrate reducing bacteria in the curing solution or pickle. Sucrose or dextrose is mainly used for this purpose. Traditionally curing of meat is limited to pork (esp. ham and belly) and beef (esp. brisket and leg muscles).

There are several methods of curing:

i. *Dry cure* Dry ingredients are rubbed to meat, e.g. curing of bacon

ii. *Pickle cure* Meat cuts are immersed in ingredient solution (pickle), e.g. curing of pork shoulder

iii. *Injection cure* Concentrated solution of the ingredients is pumped into the meat through artery or injected by needles in the muscular pork, e.g. curing of pork ham.

iv. *Direct addition method* Curing agents are added directly to finely ground meat, e.g. luncheon meat

The temperature of curing room is maintained at 3 ±1°C and curing process is allowed for 3 to 4 days depending on the strength of curing pickle.

Smoking

Meat smoking was known to man as an aid in preservation for a long time, although its chemical basis was a mystery. It is now well-known that smoke contains a large number of wood degradation products such as aldehydes, ketones, organic acids, phenols etc. which exert bacteriostatic effect besides imparting characteristic smoky flavour. Preservation of smoked meat is also due to surface dehydration, lowering of surface pH and antioxidant property of smoke constituents. Curing and smoking of meat are closely inter-related and these days, curing is usually followed by smoking. Besides, smoking and cooking operations are accomplished simultaneously.

Smoke is produced in the specially constructed 'smoke house' where saw dust or hardwood and sometimes both are subjected to combustion at a temperature of about 300°C. High temperature is desirable to minimise the production of carcinogenic compounds. Smoke generation is accompanied by formation of numerous organic compounds and their condensation products. Aldehydes and phenols condense to form resins, which constitute 50% of the smoke components and contribute most of the colour of smoked meat products. Phenols act mainly as bacteriostatic and formaldehyde as the chief bactericidal compound.

These days many liquid smoke preparations are commercially available in the developed countries. Liquid smoke is generally prepared from hardwood wherein polycyclic hydrocarbons are removed by filtration. Application of liquids smoke on the product surface before cooking imparts it a smoky flavour which is very much liked by the consumers.

Thermal Processing

Unlike refrigeration of meat that slows or stops microbial growth, thermal processing as a preservative method is employed to kill the spoilage microorganisms. Two temperature regimes, that of pasteurisation and sterilisations are generally used. Pasteurisation refers to moderate heating in the temperature range of 58°C to 75°C whereby most of the microorganisms present including trichinae occasionally found in pork are killed. Incidentally, this is also the cooking temperature range of most processed meats. This heat treatment significantly extends the shelf life of meat, although such products also need to be stored under refrigeration. Sterilisation refers to severe heating at temperatures above 100°C whereby all spoilage microorganisms in meat are killed or their microbial cells are damaged beyond repair. This heat treatment renders the meat products commercially sterile because some bacterial spores may still survive. Such meat products have a recommended shelf life of two years in cans and one year in retort pouches at ambient temperature in tropics. However, exposure of meat to high temperatures imparts sulphydril flavour in cans and modifies texture also.

Various meat products differ in water content, amount of fat and consistency. These factors have a definite bearing on the therma

processing schedule. For example, moist heat is much more effective in killing microorganisms and spores as compared to dry heat, so a meat product with higher moisture content will require comparatively less heat for sterilisation.

Canning

It is a process of preservation achieved by thermal sterilisation of a product held in hermetically sealed containers. Canning preserves the sensory attributes such as appearance, flavour and texture of the meat products to a large extent. Besides, canned meat products have a shelf life of atleast 2 years at ambient temperature. Conventional canning is done in the following steps:

i. ***Preparation of meat and gravy*** Carcass is deboned and 4 cm meat chunks are prepared. Meat gravy is prepared using condiments, tomatoes, dry spices and salt etc.

ii. ***Precooking*** Meat and gravy, both are precooked at 70°C for 15 minutes. It causes the inevitable shrinkage of meat chunks and reduces the initial microbial load.

iii. ***Filling*** Filling in cans may be done manually or mechanically leaving proper headspace as per BIS specifications. Half of the gravy is filled first followed by meat chunks and finally the rest of the gravy. Special care is taken to avoid trapping of air during this operation.

iv. ***Exhausting*** It refers to the removal of air from the container before it is closed. It is necessary to minimise the strain on the can seams due to expansion of air during heat processing. Mechanical exhausting may vacuum seal the cans.

v. ***Seaming*** This is usually done by a double seamer machine.

vi. ***Retorting or thermal processing*** The product is subjected to high temperature under pressure for sufficient duration to achieve commercial sterility.

vii. ***Cooling*** Retorting is followed by very fast cooling upto 30-40°C to give a shock to the thermophilic bacteria.

viii.. ***Storage*** Cans should be stored in a cool and dry place preferably at a temperature of about 20°C.

Dehydration

Removal of water from meat concentrates the water soluble nutrients making them unavailable to the microorganisms. The extent of availability of water to microbial cell is expressed as water activity (a_w). Dehydration lowers the water activity considerably to prevent the growth of spoilage organisms. Sundrying of meat chunks as a means of preservation was practised even in ancient days but rehydration of such meat chunks used to be limited. Mechanical drying process involves the passage of hot air with controlled humidity but here also there is difficulty in rehydration.

Freeze drying of meat is a satisfactory process of dehydration preservation due to better reconstitution properties, nutritive quality and acceptability. Freeze drying involves the removal of water from a food by sublimation from the frozen state to vapour state by keeping it under vacuum and giving a low heat treatment. Freeze drying of meat is carried out in three stages:

i. Prefreezing
ii. Primary drying
iii. Secondary drying

Meat is first frozen at –40°C. Then it is dried under vacuum for 9-12 hours at low temperature in plate heat exchangers at 1 to 1.5mm pressure of mercury. Ice crystals get sublimated to water vapour and there is no rise of temperature. In the first phase of drying, free and immobilised water of meat, which is freezable and constitutes about 90-95% of total moisture, is removed. Secondary drying is done at high temperature to remove remaining 4-8% bound water. Freeze dried products are packaged under vacuum and have very good storage stability. The process has been largely used for the preparation of dehydrated meat soup mixes.

Irradiation

Radiation is the emission and propagation of energy in the material medium. Electromagnetic radiations are in the form of continuous waves. These are capable of ionising molecules in their path. These radiations can destroy the microorganisms by fragmenting their DNA molecules and causing ionisation of inherent water within microorganisms. Since microbial destruction of foods takes place without significantly raising the temperature of food irradiation is many times referred as cold sterilisation. Among radiations, alpha

and beta-rays are charged particles and have limited use in food irradiation. However, lambda rays are electronic waves of short wave length and not the charged particles. These are easily obtained from isotopes like ^{60}Co and ^{137}Cs and have excellent penetration power. Gamma radiations produce desired effect only during food irradiation and have no effect after removal of source. These are widely used in food preservation. A dose of 50-100K rad (radurisation) can enhance the shelf-life of fresh meat cuts and poultry products by 19 days whereas a dose of 4-5 Mrad (rad appertisation) can sterilise pork, poultry and fish. Among the non-ionising radiations, ultraviolet radiations of 2650A° are most bactericidal in nature, but due to poor penetration power, these are used only for surface sterilisation of meats. It may be mentioned that certain chemicals like ascorbates have been found to increase the sensitivity of the microorganisms to radiation.

In addition to the above mentioned preservation techniques, there are many chemicals which prevent microbial growth in foods and act as preservatives. Several organic acids have been generally recognised as safe (GRAS) for use as chemical preservatives. Citric acid, propionic acid, benzoic acid, sorbic acid and their salts are effective mold inhibitors. Acetic acid and lactic acid prevent bacterial growth, whereas sorbate and acetate are capable of arresting the growth of yeast in foods. It may be noted that modern meat food processors do not rely on any single preservative factor or technique. They employ a combination of preservative factors (hurdles) in a balanced manner to derive maximum benefit. Technologists have exploited the hurdle concept in the development and keeping quality enhancement of intermediate moisture and shelf stable food products.

CHAPTER 7

Processing of Meat and Meat Products

Basic meat plant operations such as cutting, trimming, deboning and grinding do not constitute meat processing. In fact, processing refers to any treatment including salting which brings about a substantial chemical and physical change in the natural state of meat. Processing invariably imparts considerable shelf stability to meat. As a matter of fact, many processing techniques evolved in the pursuit of preservation.

Basic Processing Procedures

i. ***Comminution*** All processed meats can be classified as either non-comminuted or comminuted products. Non-comminuted products are generally processed from intact cuts. These products are usually cured, smoked and cooked, e.g. ham and bacon. Comminution refers to subdivision or reduction of raw meat into meat pieces or particles. The degree of comminution or particle size varies with the processing characteristics of products. Such meat particle size reduction helps in the uniform distribution of seasonings and eliminates the toughness associated with meat of old animals and lowers the fuel cost for cooking (Padda *et al*, 1987). Comminution is done with the help of meat mincer for coarse ground products whereas bowl chopper is also employed for making fine meat emulsion.

ii. ***Emulsification*** A mixture of two immiscible liquids where one liquid is dispersed as droplets in another liquid is called emulsion. An emulsion has two phases—a continuous phase and a dispersed or discontinuous phase. These phases remain immiscible due to the existence of an interfacial tension between

them. The emulsion remains unstable if interfacial tension is very high. The emulsion can be stabilised by reducing the interfacial tension with the help of emulsifying agents or emulsifiers. Homogenised milk is a good example of true emulsion in which fat droplets are dispersed in an aqueous continuous phase. The size or diameter of dispersed fat droplets in a true emulsion ranges from 1 to 5 micrometer (μm).

Meat emulsion comprises of a dispersed phase of solid or liquid fat droplets and a continuous phase of water containing salt and proteins (Fig. 7.1). Here, continuous phase can also be referred as a matrix in which fat droplets are dispersed. Due to the presence of

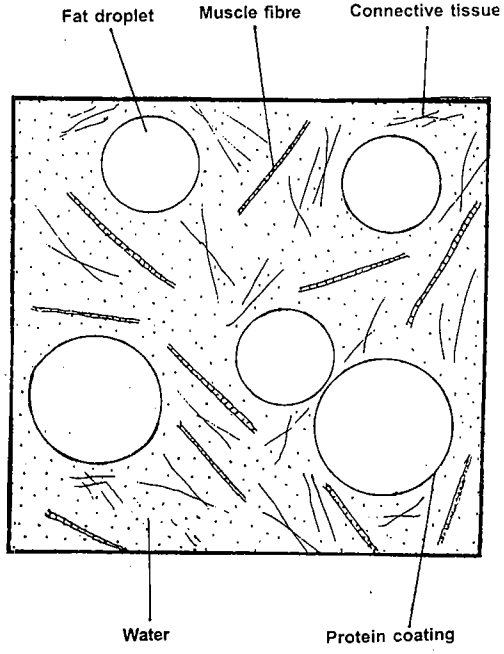


Fig.7.1: Meat emulsion (Ultra view)

matrix, many people call meat emulsion as a multiphase system. For practical purposes, meat emulsion is an oil-in-water emulsion where solubilised meat proteins act as emulsifiers. The fat droplets are usually larger than 50 μm in size and remain coated with a soluble protein—either myofibrillar or sacroplasmic. The amount of

fat that can be incorporated in a stable emulsion depends on fat particle size, meat pH, temperature during emulsification and the amount and type of soluble proteins. It is very important to maintain low temperature during emulsion formation in order to avoid melting of fat particles, denaturation of soluble proteins and lowering of viscosity. This is done by adding ice flakes instead of chilled water during chopping.

For the preparation of a good meat emulsion, lean meat is first chopped with salt to extract salt soluble proteins and then fat and other ingredients are added. Salt soluble proteins have a relatively high emulsifying capacity. Once a good meat emulsion is formed, it has to be protected during cooking or heat treatment. The emulsion breakdown can occur due to sudden exposure to high temperature because of coalescence of finely dispersed fat particles into larger ones (fat pockets). The encased or moulded emulsion is first exposed to heat at 55°C so as to coagulate the coating proteins and stabilise the emulsion.

iii. ***Meat extension*** A lot of non-meat food items can be incorporated in meat products. These are generally termed as extenders, although these may be specifically referred as fillers, binders, emulsifiers or stabilisers depending on the purpose of their incorporation in the basic meat formulation. In developing countries, soy products, potato starch and flours of wheat, rice, pea, corn etc. are used as fillers to reduce the cost of formulations. Several milk products such as skim milk powder, dried whey, sodium caseinate etc. are frequently used as binders. Some gums like sodium alginate, carrageenan, gum arabic etc. may be used to stabilise fragile meat emulsions. Due to high cost, extension of meat should be taken up on a large scale in order to ensure the availability of meat products to the masses.

iv. ***Preblending*** It refers to the mixing of a part or all the curing ingredients (salt, nitrite, nitrate etc) with ground meat in a specified proportion. This process allows better extraction of proteins which in turn helps in the formation of stable emulsion. It permits control of product composition by adjusting the desired fat content. Besides, processors get enough time for the analysis of meat samples.

v. ***Hot processing*** It refers to the processing of carcass as soon as possible after slaughter (certainly within 1-2 hours) without undergoing any chilling. The term pre-rigor processing is used when muscular meat is processed in a pre-rigor condition. Though hot processing of meat has been a common practice in India, it is rather a new development in western countries. This technique has many advantages. It accelerates the processing steps and entire processing time is reduced to a great extent. There is improvement in the cooking yield and sensory quality of the product. In addition, there are financial benefits due to reduced chiller space and labour requirement. Thus, lot of energy is saved if hot processing is adopted at a pilot scale.

vi. ***Cooking*** Meat and meat products are cooked by any one or a combination of three methods—dry heat, moist heat and microwave cooking. Dry heat cooking is an accepted method for relatively tender cuts of meat such as pork chops, leg and chops of lamb, ground and comminuted meats etc. The product yield is relatively high due to comparatively less shrinkage. Dry heat cooking involves either broiling, roasting or frying. In broiling, meat held on a wire grill, is exposed to heat from above as in electric and gas oven or below as in charcoal broiler. Meat is required to be turned for uniform and sufficient cooking of all sides. Roasting is also practised on tender cuts of meats such as pork shoulder and loin; shoulder, rack and loin of lamb and cured ham etc. The roast piece, at least 8 cm thick, is adjusted in open roasting pan with fat side up and placed in hot-air oven at 115-150°C. Cooking temperature and time varies according to the cut. Roasting generally gives good browning and improves the flavour of the product. Frying—deep fat or shallow pan is also classified under dry heat cooking. This method is especially suitable for thin cuts of meat such as sliced steaks, mutton chops, chicken meat pieces etc.

Moist heat cooking is recommended for relatively tough cuts of meat. In this method, hot water or steam is continuously kept in contact with meat for cooking, so that moisture loss does not take place beyond a particular stage. Pressure cooking, stewing, simmering etc. are popular moisture cooking procedures. Higher cooking temperatures can be achieved in pressure cooking facilitating the tenderisation of tough cuts of meat. In stewing, tough

meat pieces are first browned in small amount of fat and then covered with water alongwith curry stuff and allowed to cook at simmering temperature in covered container. The final product becomes tender alongwith a curry. Simmering involves cooking in hot water at a temperature of 70°C for considerable time. Braising utilises both dry heat as well as moist heat for proper processing of meat products. Several meat cuts like pork chops and steaks, mutton breast and shanks etc. are first fried in a frying pan and then put in a covered container alongwith water and seasoning for cooking at 80-90°C.

Microwave cooking is relatively a recent development. Microwaves are high frequency, non-ionising electromagnetic waves which are generated by magnetron vacuum tube within the oven. These waves are channelised into the oven cavity through a wave guide. A stirrer fan distributes the microwaves evenly. The microwaves penetrate the food from all directions simultaneously upto a depth of 2-4 cm causing water, fat and sugar molecules to vibrate at a very high speed. The vibrations cause tremendous friction which produces heat for cooking the food. The spread of heat throughout the three dimensional space in the food itself is called volume heating. Contrary to conventional heating, food is first to be heated in the microwave cooking which then transmits heat to container and oven environment. It saves a lot of time, taking only 25% time as compared to conventional thermal oven. Microwaves can pass through glass, pottery, wood and paper but reflected by metal. So, metallic utensils cannot be used in the microwave oven. There are some other disadvantages also. Food has to be frequently turned to ensure proper heating and browning of food does not take place in this cooking.

Processing of meat products is divided into the following groups for further discussion:

1. Cured and smoked meats
2. Sausages
3. Intermediate moisture and shelf stable meat products
4. Restructured meat products
5. Other popular meat products

Cured and Smoked Meats

All meat products belonging to this class are cured, whereas only some of them are smoked. The primal cuts of pork especially ham

and bacon have been subjected to curing and smoking for a long time. These days, it is a general practice to accomplish cooking also during smoking except for Country ham, which is smoked without cooking.

Hams

These are classified in several ways:

a. According to weight : light, medium and heavy
b. According to trimming : rough, regular, skinned and skinless
c. According to presence of bone : bone-in, semiboneless and boneless

Commercial Processing of Ham

Irrespective of classification, most hams are processed in three steps—curing, smoking and cooking. Most commercial hams are pickle cured. A typical curing solution consists of:

Ingredients	*Quantity*
Table salt	850 g
Phosphate	225 g
Sucrose	175 g
Sodium erythrobate	15 g
Sodium nitrite	8 g
Water	40 litres

Curing is usually done by artery pumping or stitch pumping to 10% of the green weight. However, best results with respect to colour and flavour are obtained by keeping the hams in a cover pickle at 4°C for 5 days. The hams are now shifted to smoke chamber which is maintained at 75-85°C temperature and 30-40% relative humidity for 5-6 hours. Smoke generated from hardwood is preferred for good results.

The processing of some variety hams is given here under:

Cooked ham These hams are deboned and cured in the pickle but smoking is not done. Instead, these hams are stuffed tightly into metal moulds and cooked in a water tank at 75-85°C for 2-3 hours depending on the weight of the ham. During cooking, the core temperature must reach 65-70°C. After cooking, the mould-in hams

are chilled in a tank maintained at 0°C for 12 hrs. These hams are then sliced and packed.

Country ham These uncooked hams are manufactured in USA by dry curing method. The curing mixture usually contains 8 kg salt, 1 kg sugar and 100g sodium nitrite. It is rubbed thoroughly at the rate of 30g per kg of ham on 1st, 5th and 10th day. The entire production schedule is divided into three phases: (a) Curing is allowed to take place under refrigeration at a relative humidity of 70-90% for 30-40 days during which hams are overhauled atleast three times, (b) Smoking is done at low temperature for 2-3 days till the hams become amber coloured and (c) Aging is done for 6-9 months at a temperature of 20-30°C and relative humidity of 50-60%. During this period, Country hams become progressively harder and develop a unique flavour. These hams have a final salt level of 4-5% and a moisture content of 50-60%. The shrinkage loss during processing amounts to 18-20%.

Proscicutto These hams are manufactured in Italy from certified trichinae free hams and traditionally consumed without being cooked. These are dry cured like Country hams. Curing continues for 45 days at 4°C, followed by smoking for 2 days at 55°C and finally ageing for 30 days at 20°C at a relative humidity of 65-75%. There is a shrinkage of 35% in weight during the entire processing schedule.

Bacon

Pork bellies are generally processed as cured and smoked bacon. There is no fixed criteria for the classification of bacon. However, many processors grade them on the basis of weights of green bellies.

Commercial Processing of Bacon

Green bellies are first cleared of rind and stitch pumped with a curing pickle. These are now transferred to smoke chamber maintained at a temperature of 60-65°C and a relative humidity of 30-40% for smoking as well as cooking. The cooking time depends on the size of bellies although an internal temperature of 55°C must be achieved. Cooking also helps to stabilise the cured colour. After smoking and cooking, bacon is chilled to 0°C to allow it to retain proper shape

and facilitate slicing. These bacon slabs are processed in a forming machine to give uniform width and thickness. Bacon blocks are now sliced to 5-7 mm thickness with the help of slicer. The slices may be packed in a modified atmosphere if long term storage is desired.

Some variations in the processing of bacon in different countries are inevitable. Canadian bacon is not manufactured from bellies but from larger muscles of pork loin and sirloin. In Europe and U.K., Wiltshire bacon is produced from pork sides where shoulder, loin, ham and belly are processed as a single large piece.

Sausages

Sausage term was derived in the ancient times from the latin word 'salsus' meaning salt. It was literally coined to refer to ground meat which was salted and stuffed in animal casings. Presently, sausage may be defined as a meat product which is prepared from minced and seasoned meat and formed into cylindrical shape by natural or synthetic casings. Though sausages originated in the western world, these products acquired universal popularity due to variety and convenience to the consumers. Sausages are economical also because these are generally prepared from cheaper cuts of meat and by-products of industry.

Classification

Sausages are such a large number of varying kinds of products that it is not possible to cover them in any classification system. Some overlapping is always there. Some of the popular classification systems are:

i. Based on degree of chopping :
   a. Coarse ground sausage
   b. Emulsion type sausage
ii. Based on moisture content :
   a. Fresh sausage
   b. Smoked uncooked sausage
   c. Cooked sausage
   d. Dry and semi-dry sausage
iii. Based on fermentation :
   a. Fermented sausage
   b. Non-fermented sausage

Processing Steps

i. *Grinding or mincing* Lean meat and fat are minced separately in a meat mincer. The choice of mincer plate or sieve depends on the type of meat.

ii. ***Mixing*** Meat and fat to be used for the preparation of coarse ground sausage are mixed uniformly in a mixer. Extender, condiments and spices should also be run in the mixer for even distribution.

iii. ***Chopping and emulsifying*** For emulsion preparation, lean meat is first chopped for few minutes in a bowl chopper with salt to extract myofibrillar proteins. This is followed by addition of fat and running for a few minutes again to get desired emulsion consistency. Now, all other ingredients are added and chopper is run for sometime for uniform distribution. The entire operation is conducted at low temperature by addition of ice flakes in place of chilled water.

iv. ***Stuffing*** Sausage emulsion or batter is taken to stuffer for extrusion into casings. The casings are first collected on the stuffing horn or nozzle and released to coincide with the extrusion.

v. ***Linking and tying*** In small sausages, the encased mass is twisted to produce links either manually or mechanically whereas in large sausages, the encased mass is tied with thread at regular intervals.

vi. ***Smoking and cooking*** Sausage links are hung on the smokehouse trolley and transferred to smokehouse. The temperature of smokehouse is usually maintained at 68-70°C which is enough for coagulation of sausage emulsion, cooking and requisite drying of sausages.

vii. ***Chilling*** The cooked product is showered with chilled water to an internal temperature of about 4°C.

viii. ***Peeling and packaging*** While artificial or synthetic casings are peeled off before the product is packed, small sized natural casings need not be removed. The product is generally unit packed for retail outlets.

Formulations of Some Sausages With Special Features

i Fresh pork sausages (coarse ground type)

Ingredients	Quantity
Lean pork trimmings	35.0 kg
Backfat	15.0 kg
Table salt	1.0 kg
White pepper	125.0 g
Sugar	75.0 g
Ginger powder	15.0 g

ii. Frankfurters (Cured, emulsion type meat sausages)

Ingredients	*Quantity*
Beef trimmings	30.0 kg
Pork trimmings (50/50)	20.0 kg
Ice flakes	15.0 kg
Table salt	1.2 kg
White pepper	110.0 g
Nutmeg	25.0 g
Cardamom	25.0 g
Ginger powder	25.0 g
Monosodium glutamate (MSG)	25.0 g
Sodium erythrobate	20.0 g
Sodium nitrite	7.5 g

iii. Salami (Fresh, emulsion type sausages)

Ingredients	*Quantity*
Pork trimmings (50/50)	20.0 kg
Pork cheek meat	10.0 kg
Pork hearts	10.0 kg
Pork stomachs	5.0 kg
Ice flakes	5.0 kg
Table salt	1.2 kg
Black pepper	115.0 g
Sodium erythrobate	25.0 g
Garlic powder	25.0 g
Sodium nitrite	5.0 g

Batter is stuffed in broader casings, usually dried cattle oesophagus and hanged for 2 days at 4°C. It is smoked and cooked simultaneously in India.

iv. Chicken sausages (Emulsion type)

Ingredients	*Quantity*
Chicken meat	30.0 kg
Vegetable oil	7.5 kg
Ice flakes	5.0 kg
Condiments	3.0 kg
Bread powder	3.0 kg
Table salt	1.2 kg
Whole egg liquid	1.0 kg
Spices mix	800.0 g
Sodium nitrite	5.0 g

v. Thuringer summer sausages (coarse ground, fermented, semi-dry type)

Ingredients	*Quantity*
Pork trimmings (75/25)	25.0 kg
Beef trimmings	20.0 kg
Table salt	1.2 kg
Dextrose	400.0 g
Black pepper	115.0 g
Starter culture	60.0 g
Coriander	30.0 g
Sodium nitrite	5.0 g

The pH of sausage mix comes down to about 5.0 due to bacterial fermentation. Dextrose serves as a substrate for fermentation. Sausage mix prepared at 0°C is stuffed into casings and held in green or ripening room maintained at 25°C and 85-90% relative humidity until fermentation is complete (2-3 days). These are smoked and cooked at 60-65°C. The moisture content of finished product is nearly 50%.

vi. Dry sausages (coarse ground, fermented sausages, e.g. pepperoni)

Ingredients	*Quantity*
Pork trimmings (50/50)	20.0 kg
Beef trimmings	15.0 kg
Pork hearts	5.0 kg
Pork cheeks	5.0 kg
Table salt	1.5 kg
Dextrose	500.0 g
Black pepper	175.0 g
Fennel seeds	10.0 g
Sodium nitrite	7.0 g

Sausage mix is stuffed into 40-44 mm animal casings and held for 9-11 days at 4°C. It is transferred to green room for 2 days to be maintained at 20°C and 70% relative humidity and then smoked for 3 days at 35°C and 80% relative humidity. The product is not cooked but held for 21 days in the drying room at 35°C and 70% relative humidity during which it will have a shrinkage. The final product has only 30-35% moisture content.

vii. *Bologna*: It is an emulsion type sausage prepared from the meat of old animals.

viii. *Hot dog:* It is a fairly spicy sausage in broader casings, usually weasand in India.

ix. *Mortadella:* It is a dry sausage prepared in cattle bladder or artificial broader casings.

Intermediate Moisture and Shelf Stable Meat Products

Sundrying of meat was one of the earliest preservative techniques used by man. Such meat had meagre rehydration capacity resulting in poor juiciness and texture. Later studies revealed that meat products with 20-50% moisture had moderate juiciness and texture on rehydration. Such products were resistant to bacteriological spoilage and could be held without refrigeration. These products were referred as Intermediate Moisture Meats (IMM). The basic reason for the stability of these products lay in the reduced availability of water to the microorganisms, since water activity generally remains in the range of 0.6 to 0.85. These semi-moist meats are of special significance to the developing countries where refrigeration facilities are not always available. Such products can be easily carried in defence expeditions and stress situations like floods, famines etc. for air drop.

Humectants

Various additives employed for lowering the water activity of foods are known as humectants. Some of the most commonly used humectants are:

Glycerol
Propylene glycol
Sodium chloride
Polyhydric alcohols (e.g. sorbitol)
Sugars (e.g. sucrose, dextrose, corn syrup etc).

The humectants are generally low molecular weight compounds which are easily soluble in water. These are chemically inert and do not modify the normal sensory qualities of the product. Besides, these compounds are edible in large quantities without any adverse effect.

In addition to humectants, use of antimycotic agents like potassium sorbate, sodium benzoate, propylene glycol etc. is a must in the semi-moist meats because 0.6 to 0.85 water activity range specifically permits the growth of moulds.

Basic Processing Techniques

A. *Moist infusion or desorption* It involves soaking and/or cooking of meat chunks or cubes to yield a final product having desired water activity level, e.g. sweet and sour pork, Hungarian goulash etc.

B. *Dry infusion or adsorption* It involves initial dehydration of meat chunks or cubes followed by soaking in an infusion solution containing desired osmotic agents, e.g. ready-to-eat cubes of roast pork, chicken a la king etc.

C. *Component blending* In this process dry and wet ingredients or components are blended, cooked and extruded or otherwise mixed to give a final product of desired water activity.

Whatever process is adopted, the thumb rules for the preparation of IMM are: (a) reduction of water activity by addition of humectants, (b) retardation of microbial growth by addition of antimicrobial especially antimycotic agents and (c) improvement of sensory properties such as flavour and texture through physical and chemical treatments.

Composition of infusion solution developed by Brockmann (1970) for the preparation of sweet and sour pork ($a_w = 0.85$) is given below to give an idea about the balancing of various additives:

Stability of Intermediate Moisture Meats

IMF products are fairly stable at ambient temperature for several weeks or even months. However, prolonged storage may result in some quality deterioration due to the following reasons:

a. Limited breakdown of both myofibrillar and sarcoplasmic proteins. Collagen being more susceptible to degeneration results in more hydroxyproline formation.

Ingredients	*Percentage*
Glycerol	25.00
Catsup	23.55
Water	15.00
Vinegar	13.50
Sucrose	11.84
Starch hydrolysate	14.50
Salt	2.59
Corn starch	2.30
Monosodium glutamate	1.15
Potassium sorbate	0.30
Mustard powder	0.24
Onion powder	0.02
Garlic powder	0.01

b. Degradation of haemoprotein (myoglobin and haemoglobin) causing loss of colour.

c. Development of rancidity.

d. Non-enzymatic browning resulting in loss of colour, consumer appeal, nutritive value and possibly off-flavour.

e. Formation of lipid-protein crosslinks causing decreased water binding capacity and net protein utilisation of meat products.

Hurdle Concept

Intermediate moisture meat products mostly depend on lower moisture content and consequent decrease in water activity for their shelf stability. Use of high concentration of humectants including salt and sugar for desorption usually produces a disagreeable taste. This is true for Indian palate also. Leistner and Rodel (1976) coined the term hurdles for the parameters like chilling, heating, pH reduction, low water activity, enhanced Eh, use of preservatives and competitive microflora. Use of these hurdles or combination preservation technique in a balanced and judicious manner was named as hurdle concept and later the hurdle technology. It does not allow a single parameter to affect the product characteristics drastically. Thus, hurdle technology is the use of two or more factors, none of which is independently capable of sufficiently inhibiting the spoilage or pathogenic microorganisms to extend the shelf life of food products. Hurdle technology based meat products provide a desirable taste, juiciness, texture and safety.

Restructured Meat Products

It has now become possible to utilise less desired or secondary carcass cuts into the production of highly preferred meat products such as steaks, roasts, chops, cutlets etc. The less desired carcass cuts are carefully trimmed to remove sinews, excess fat and other connective tissue. Any one of the following three basic procedures can be adopted depending on the appearance and texture targeted in the finished product:

i. Chunking and forming
ii. Flaking and forming
iii. Tearing and forming

i. ***Chunking and forming*** The trimmed meat is put through a dicing machine and reduced to small chunks. One per cent common salt and 0.25% phosphate are added at this stage and meat is put in a tumbler run at medium speed. During tumbling process, impact energy is utilised for the extraction of salt soluble proteins. The extracted proteins serve as cementing material when this meat is restructured by stuffing and pressing into suitable moulds. Restructured meat is frozen and then sliced to obtain uniform slices of desired thickness.

ii. ***Flaking and forming*** The trimmed meat is passed through a flaking machine to get flakes which are then mixed with 1% common salt and 0.25% phosphate. The material is run in a massager. During massaging process, frictional energy is utilised for the better extraction of salt soluble proteins. The meat mass is stuffed and pressed or formed into desired shapes–steaks, cutlets or chops. The restructured product is frozen and thawed just before cooking. The products have relatively good tenderness and uniform texture.

iii. ***Tearing and forming*** In this procedure meat fibres are torn apart. So, there is less damage to the membrane resulting in less chances of autoxidation. Further, the structural integrity of meat tissue is maintained, though it becomes tender. Efforts are on to mechanise the process.

Other Popular Meat Products

i. ***Luncheon meat*** It is a canned product usually prepared from pork alongwith some cereal component. The product contains

not less than 80% pork including pork fat which should not exceed 30% in the final product. Besides, added water is limited to 3% and the cereal ingredients should not exceed 7% of the total formulation. Lean pork and pork fat are ground through 5 mm and 3 mm plate of the meat mincer, respectively. These are initially chopped alongwith chilled water or ice flakes, common salt and nitrite followed by other ingredients such as refined wheat flour, condiments and dry spices. A simple formulation of luncheon meat is given below:

Ingredients	*Percentage*
Lean pork	67
Pork backfat	15
Ice flakes	3
Table salt	2.5
Dextrose	1.5
Refined wheat flour	5
Condiments	4.75
Dry spices	1.5
Sodium nitrite	150 ppm

The batter or meat mix, maintained at 4°C, is filled compactly in cans which are sealed under vacuum. Commercial sterilization is done in retorts at 121°C for 75 minutes. Cans are now cooled with cold water shower until the contents reach 38°C. Canned luncheon meat is stable at ambient temperature for a period of two years.

ii. ***Meat patties*** Meat patty is one of the most popular products among the ground meat items and is generally used as filling for burger roll or sandwich. Some people prefer to consume it separately with tomato sauce or chutney. This product has a very good demand in big towns and cities in India. Patties are partially or completely emulsion based product, contain less than 30% fat and are moulded manually or mechanically. An optimum formulation is presented below:

Ingredients	*Percentage*
Lean meat	65
Fat	15
Table salt	2
Texturised soy protein	10
Condiments	6.5
Dry spices	1.5

Lean meat is minced twice through 6 mm plate and fat through 4 mm plate of a meat grinder. These are mixed thoroughly with all other ingredients in an electrically operated mixer or prepared into an emulsion. The batter weighing 80-100 g is moulded into 70-80 mm diameter and 15-20 mm thick patties. Raw patties may be frozen for future use or broiled in a preheated oven at 190°C for 20 minutes. The internal temperature must reach 72°C. These are deep fat fried in many commercial establishments. The patties are cooled and consumer packed.

iii. ***Meat loaves*** This important ready-to-eat comminuted meat product is prepared from coarse ground meat or meat emulsion or a combination of both. The formulation of a family loaf is given below:

Ingredients	*Percentage*
Lean pork	65
Pork backfat	15
Ice flakes	5
Table salt	2
Refined wheat flour	7.5
Skim milk powder	4
Dry spices	1.5
Sodium nitrite	150 ppm

The meat mix or batter is tightly filled in aluminium or steel loaf pans which may be rectangular, cubical or cylindrical in shape depending on the requirement of slices for making the sandwiches. The pan-in mix is cooked in hot water maintained at 80°C or steam without pressure or broiled in hot air oven at 165°C for 2.5 to 3 hours. The internal temperature of 70°C must be achieved. It is then given a cold shower and chilled at 4°C. The chilled loaves are either packed as such or cut into slices of desired thickness and packed.

iv. ***Meat balls*** Indian consumers are familiar with this food item by the name of meat kofta. The product, stored raw or cooked offers a great convenience to restaurants, hotels and housewives who can just put few balls in the gravy and serve the food within 10 minutes. The product is prepared from ground meat which is mixed with fat, bread powder, salt, condiments and spices in an electrically operated mixer. The dough portions of 15-20 g are rolled into balls manually or mechanically. These

are either stored raw or deep fat fried in refined vegetable oil at 135°C for three minutes to get brown colour and fried flavour. Alternatively, these are cooked in hot water maintained at 80°C for 10 minutes to get an internal temperature of 65-70°C. Water cooked balls may be subjected to light frying to get golden brown colour. These are packed in polyethylene pouches and can be kept at 4°C for a week. Whenever required, the cooked balls can be simmered in gravy for a few minutes and enjoyed with rice or bread.

v. ***Meat nuggets*** It is a ready-to-eat convenient product which is obtained by cutting cooked and cooled rectangular or cubical shape meat loaves into approximately 4 cm x 1.5 cm x 1.5 cm pieces. The product is packed in unit pouches and can be stored at 4°C for a week. It is usually shallow fat-fried before serving for breakfast or refreshment.

CHAPTER 8

Microbial and Other Deteriorative Changes in Meat and their Identification

Meat is a highly perishable food item. Hence, utmost precautions must be taken to safeguard it right from bleeding of slaughter animals till final consumption. Though muscle tissue of living animal is free of microorganisms, it gets contaminated by the body surface and visceral contaminants during slaughter and dressing operations. These organisms are bound to proliferate and cause deteriorative changes in meat unless proper measures are employed to retard their growth. Besides, deteriorative changes can also be brought about by endogenous enzymes, lipid oxidation and improper storage conditions.

Microbial activity plays a major role in deterioration and spoilage of meat. We should, therefore, concentrate our efforts to check the initial invasion of microbes during slaughter and subsequent handling as well as processing.

Sources of Microbial Contamination of Meat

There are a number of potential sources of contamination of meat within the abattoir itself. These include:

i. Hides/skin and feet
ii. Gastrointestinal contents
iii. Instruments such as knives, cleavers saws, hooks etc.
iv. Water used for washing carcasses and instruments
v. Airborne contamination
vi. Hands and clothing of the personnel

Plate 1

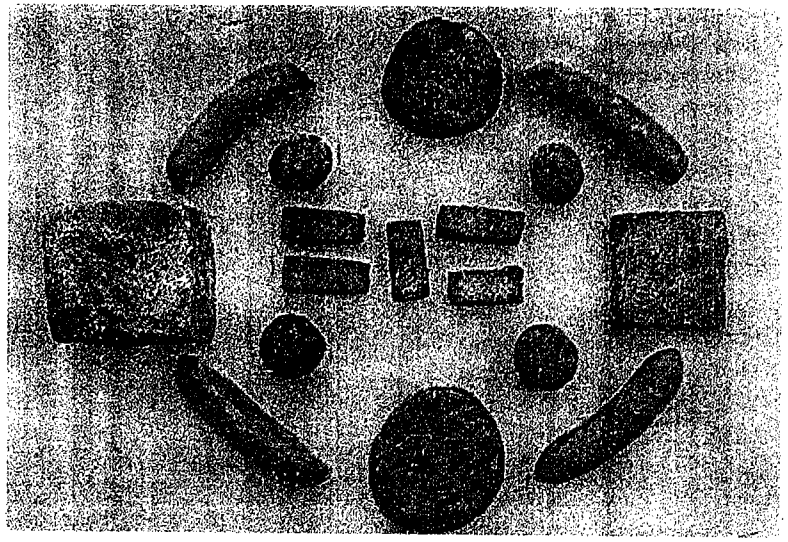


Fig. 7.2: Mutton products



Fig. 7.3: Meat sausages (*Courtesy* R Somvanshi)

Plate 2

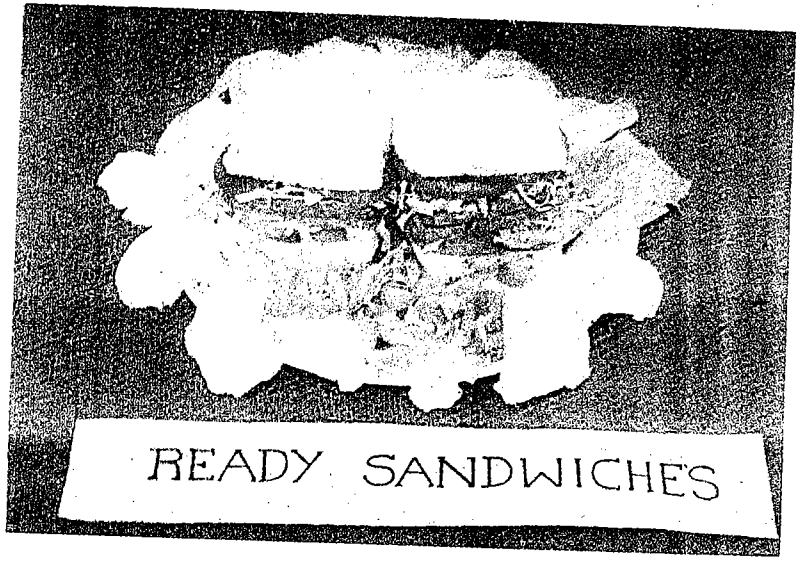


Fig. 7.4: Ready sandwiches

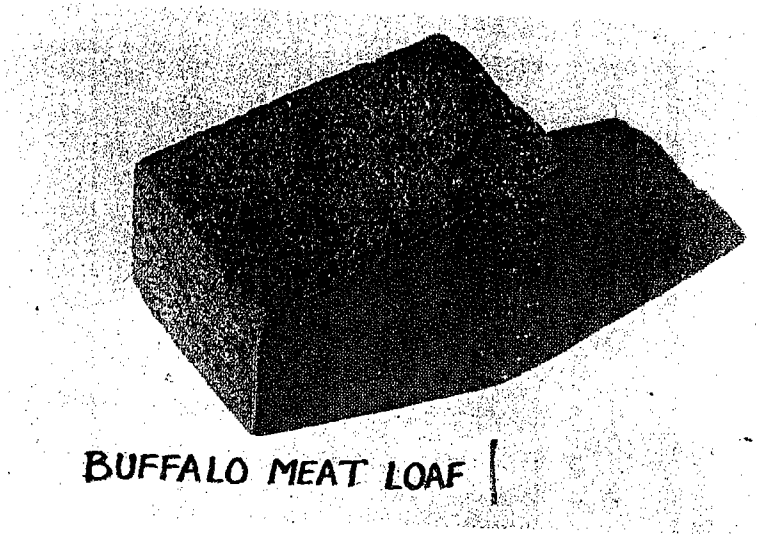


Fig. 7.5: Buffalo meat loaf

Plate 3

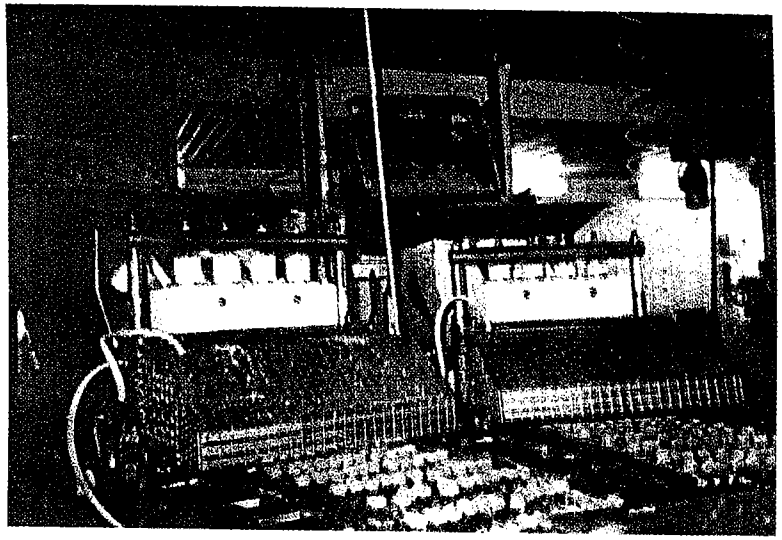


Fig. 7.6: Meat balls

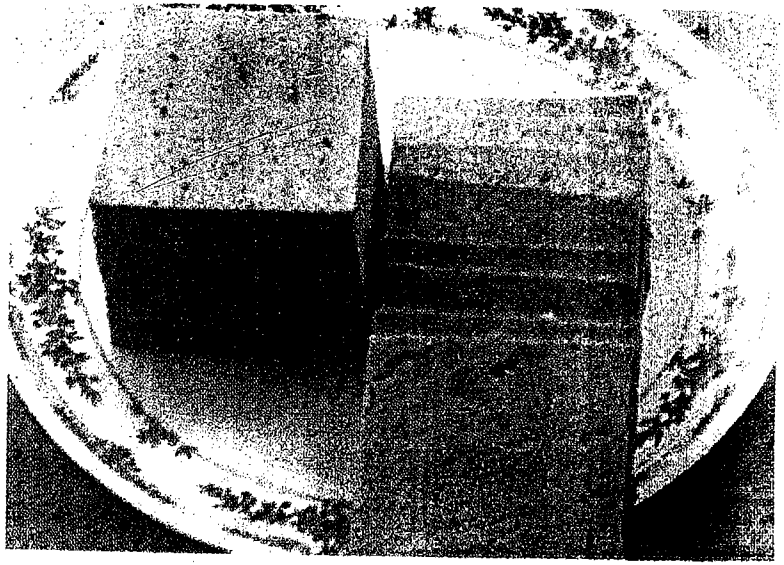


Fig. 7.7: Chicken loaf and slices

Plate 4

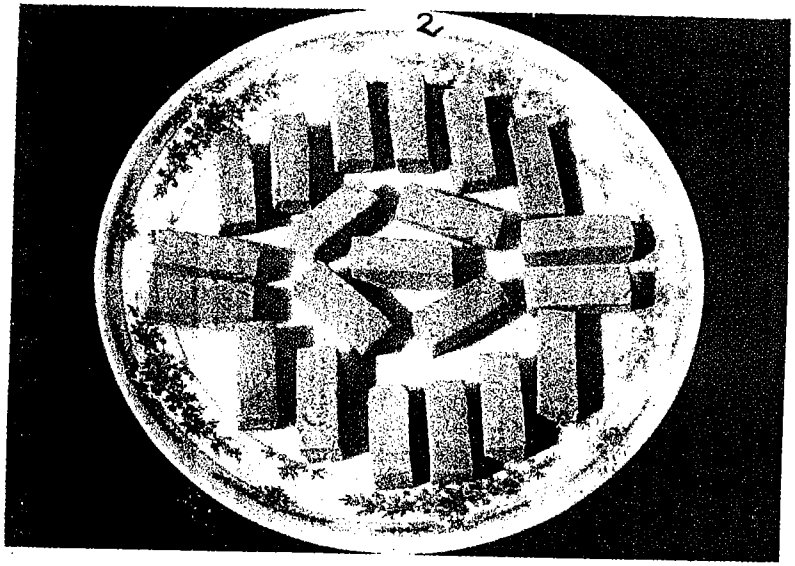


Fig. 7.8: Chicken nuggets

Contamination of meat may also take place during chilling, ageing, processing, packaging and distribution.

Growth of Microorganisms in Meat

The microorganisms that occur in meat may be bacteria or fungi. Fungi may be multi-cellular filaments (mold) or large single cells with buds (yeast). Molds are capable of producing minute spores under unfavourable conditions. Fungi gain upper hand over bacteria in meat when it is semi-dry. Bacteria are unicellular microorganisms which are spherical or ovoid or rod shaped and may occur in chains or clusters. Bacterial growth or multiplication takes place in phases:

Lag phase : Bacterial cells increase in size under favourable conditions

Log phase : Bacterial cells multiply and increase in number

Stationary phase : Growth rate becomes relatively constant due to environmental limitations

Decline or death phase : There is destruction of bacterial cells either due to nutritional depletion or application of some preservation technique.

An understanding of growth curve enables the meat technologists to apply suitable preservation technique to prolong the lag phase

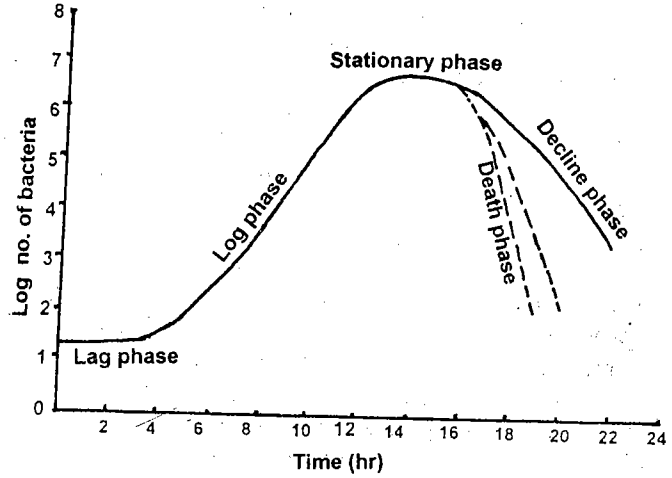


Fig. 8.1: A typical bacterial growth curve in meat

so that bacterial multiplication is retarded or if conditions have already favoured some growth, then to hasten the death phase.

Microbial growth activity in meat depends on various extrinsic and intrinsic factors. The extrinsic factors are temperature, oxygen and physical stage of meat. Different groups of bacteria have their own growth optima. Psychrophiles have their optimum growth temperature below 20°C, thermophiles above 45°C and mesophiles an optima ranging between these two. A refrigerated temperature of nearly 5°C greatly retards the growth of most psychrophilic organisms responsible for the spoilage of meat. These bacteria generally belong to genera *Pseudomonas, Achromobactor, Flabacteirum* (G^- rods), *Micrococcus, Streptococcus* (G^+cocci) in fresh meat and *Lactobacillus* (G^+ rods), in cured or vacuum packed meat products. The growth of bacteria on meat is usually characterised by slime formation. It should be noted that total bacterial population is above 10^7 when most signs of spoilage appear on meat. Effective freezing damages or kills most of the bacteria present on meat. It may be remembered that meat spoilage molds are also psychrophiles. On the contrary, food pathogens generally belong to the thermophilic group of bacteria.

Oxygen environment around the meat will determine the type of micro flora that will find favourable conditions for growth. Bacteria found in meat may be either aerobic or anaerobic or sometimes facultative. Bacteria that grow on the surface of fresh meat are generally aerobes, whereas it is a different flora in the interior of meat. All molds and yeast that grow in meat are aerobic in nature. Use of different barrier packaging films restricts the activity of aerobic microorganisms. Physical state of meat such as whole carcass or primal cuts or retail cuts or comminuted form also influence the rate of microbial growth. Microbial load increases with the increase in exposed surface area of meat.

Important intrinsic factors which affect the growth of microorganisms in meat are water activity, pH and redox potential. The amount of water available in a food system for the growth of microorganisms is generally expressed in terms of water activity. In fact, water activity (a_w) is defined as vapour pressure of the solution (p) in a food system divided by vapour pressure of pure solvent or water (p_o). Fresh meat generally has a water activity of 0.99 or more. Most meat spoilage bacteria can grow only upto a water

activity of 0.91 but most spoilage mold and yeast can grow upto a water activity of 0.86. This factor is commercially exploited in the production of intermediate moisture meat products. Meat pH is yet another intrinsic factor which influences the growth of microorganisms. Bacterial growth is best at neutral pH (i.e. pH 7.0). It keeps on diminishing as meat pH goes down. If ultimate pH is 6.0 or so, a large number of bacteria can still grow in meat. However, when normal ultimate pH of nearly 5.5 is achieved in meat, bacterial growth is reduced to a large extent and the growth of mold and yeast is favoured. Redox potential refers to the reducing or oxidising conditions prevailing in meat and this factor also influences the growth of microorganisms.

Deteriorative Changes in Meat

When meat depicts signs of decomposition and putrefaction, it is referred as spoiled and becomes unfit for human consumption. Besides microorganisms, intrinsic enzymes and insects also contribute to the spoilage of meat. Microbial spoilage of fresh chilled meat is generally on the surface whereas it is within meat at higher temperature. The causative agents and deteriorative changes are quite different in aerobic and anaerobic spoilage.

Under aerobic conditions, most significant symptom of meat spoilage by bacteria and yeast is the slime formation on the surface which results due to coalescence of a large number of individual colonies. There may be discolouration of meat due to oxidising agents produced by bacteria or growth of colonies of coloured organisms. The production of off-odours is also usually encountered. Bacterial action causes proteolysis of meat proteins and lipolysis of meat lipids. The end products of proteolysis are simple peptides and amino acids under aerobic conditions whereas sulphur dioxide, ammonia and other obnoxious compounds like amines and ketones under anaerobic conditions. Residual carbohydrates yield skatol and indole. Molds may grow on semi-dried meats causing surface stickiness and whiskers.

Under anaerobic conditions, meat decomposition is more offensive. There may be putrefaction in the deep tissues such as lymphnodes and bone joints, which is always accompanied by foul odours or taints. Souring may also develop due to accumulation of organic acids.

Canned meats may suffer chemical or biological spoilage or both. In the event of spoilage unopened cans may show distortion (swell) due to souring and putrefaction.

Identification of Meat Spoilage

The identification of meat spoilage is based on the deteriorative changes brought about by microorganisms, intrinsic enzymes and insects. During spoilage process, several utilisable substrates are consumed by microflora and new products are formed which can be measured or determined in meat. Thus, meat spoilage can be detected by any of the following physical and chemical methods:

i. Some physical observations such as discolouration, slime formation, stickiness, whiskers etc. give a clear indication of spoiled meats.
ii. At low temperature, meat spoilage is accompanied by the formation of many off-flavour compounds. Many of them owe their origin to free amino acids and related substances. The production of H_2S and mercaptans can be measured to ascertain meat spoilage. Chemical determinations for the presence of ammonia, indole, skatol, di- and trimethylamine etc. can be carried out to detect microbial spoilage in meats.
iii. The extract release volume (ERV) determination is particularly helpful in detecting the incipient spoilage in meats. ERV refers to the volume of aqueous extract released by a meat homogenate when it is passed through a filter for a given period of time. As meats undergo microbial spoilage, there is a complete hydrolysis of proteins which significantly decrease the ERV.
iv. Dye (usually resazurin) reduction test is many times used to detect spoilage in meats. Spoiled raw or cooked meat homogenate prepared from stomacher could bring about resazurin reduction within 2 hours. This method shows a very good correlation with bacterial numbers.
v. Incipient spoilage in meat shows a simultaneous rise in pH, bacterial counts and water holding capacity of meat proteins. At the time of incipient spoilage, pH value is more than 6.5 in ground meat but it may even increase to 8.5 in putrid meats.
vi. High thiobarbituric acid and peroxide values indicate chemical spoilage of meat and meat products.

CHAPTER 9

Standards and Quality Control Measures for Meat and Meat Products

Meat being a highly perishable commodity requires strict quality control right from slaughter operations till ultimate consumption. The basic objectives of quality control are :

i. Protection of public health
ii. Extension of product shelf life
iii. Provision of consumer satisfaction
iv. Compliance of regulatory legislation
v. Competitive edge in the trade

The general principles of meat product quality control involve:

i. Raw material control
ii. Control of processing operations
iii. Finished product inspection and control

It is very difficult to examine meat and meat products for every pathogenic, toxigenic and spoilage microorganisms. However, a product cannot be improved unless some objective assessment of its quality is available. But the methods adopted should be simple with quick results. Hence, the following indicator organisms are relied upon to determine the sanitary and safety status of these items:

Total Viable Counts

These are estimates of mesophiles and psychrophiles and serve as useful indicators of handling history and state of freshness or spoilage of meat. These counts provide meaningful guidance to

streamline the processing operations. However, if these counts are less, most probable number (MPN) are enumerated.

Coliforms

These consist of *E. coli* and *Aerobactor aerogenes*. Their presence indicates faecal contamination due to unhygienic handling during or after processing of meat products. Coliforms can be distinguished because of their property to produce gas from lactose at 44°C. However, out of two organisms in this category, *E. coli* is a better indicator of faecal contamination.

Enterococci

These are members of faecal streptococci (group D) which consist of:

S. faecalis (and its varieties)
S. faecium (var durans)
S. bovis
S. equinus

These organisms indicate poor hygienic quality of frozen meats and inadequate heat treatment of canned meats.

Other Indicators

Besides above indicators, specific organisms like *Staph. aureus, Salmonella*, yeast and mold counts are also important. Heat treated meats should also be screened for the presence or absence *of B. cereus* and Clostridia.

Canned meat products are generally subjected to sterility test. For this purpose, cans are incubated at 30°C and 55°C for 15 days. Swollen or disfigured cans show the product spoilage.

STANDARDS IN MEAT INDUSTRY

Quality control departments frequently utilise the reliable methods and techniques for establishing the standards. A standard can be referred as carefully drawn specification with respect to a food product. The specifications give comprehensive instructions to ensure correct and reliable process control. The compliance of specifications increases the confidence of top executives in the production and marketing of perishable food items.

Bureau of Indian Standards (BIS), established in 1947 as Indian Standard Institution (ISI) as a joint venture of Government of India and Industry took up the responsibility of preparing and promoting the general adoption of standards in the country. The erstwhile ISI constituted the Meat and Meat Products Sectional Committee, AFDC 18 under the Agricultural and Food Products Division Council in 1958 to prepare Indian Standards for the meat industry. This committee represents the scientists, technologists, manufacturers, government agencies and consumers. The standards are prepared keeping in mind the needs of industry protecting the interests of both producers and consumers and are reviewed periodically. A list of relevant standards is given below:

IS:4393-1979	Basic requirements for an abattoir (first revision)
IS:1982-1971	Code of practice for antemortem and post-mortem inspection of meat animals (first revision)
IS:1723-1973	Specifications for pork (first revision)
IS:2474-1979	Specifications for smoked bacon (first revision)
IS:2476-1963	Specifications for ham
IS:2536-1963	Specifications for mutton and goat flesh—fresh, chilled and frozen
IS:2537-1963	Specifications for beef and buffalo flesh—fresh, chilled and frozen
IS:3060-1965	Specifications for pork sausages, canned (under revision)
IS:3061-1965	Specifications for pork sausages, fresh (under revision)
IS:4352-1967	Specifications for pork luncheon meat, canned
IS:3044-1973	Specifications for mutton and goat meat, curried and canned (first revision)
IS:4674-1975	Specifications for dressed chicken (first revision)
IS:4951-1975	Specifications for ham, canned (first revision)
IS:5558-1970	Specifications for chicken essence
IS:6659-1972	Code of practice for antemortem and post-mortem inspection of poultry.
IS:8182-1976	Code of hygienic conditions for processed meat products
IS:5960(Parts)-1970	Methods of test for meat and meat products

These BIS specifications are voluntary but an adherence to these guidelines definitely improves the quality of processed products. However, Meat Food Products Order which exclusively deals with quality control of processed meat products is mandatory.

Microbiological standards help to improve plant sanitation, ensure safety of the products and prevent losses due to microbial spoilage. Establishment of microbiological specifications is a very expensive and cumbersome task. Lot of data base is required in practical conditions at different locations, with a very good degree of reproducibility. Microbiological specifications for meat processing plants as suggested by Marks and Spencer, UK, are presented in Table 9.1.

The concept of quality control at few points in the entire production chain is now giving way to integrated quality management systems. HACCP and ISO-9000 series are of particular interest to meat processors (Table 9.2).

Hazard Analysis Critical Control Point (HACCP) System

HACCP is a comprehensive food safety system right from the point of production to the point of consumption. This system analyses the hazards of raw material, identifies the points of potential contamination, monitors the processing operations and checks the risks arising from consumer abuse. It is a systematic approach to the production of microbiologically safe foods. It consists of the following steps:

i. Assessment of hazards and risks associated with raw materials, ingredients, processing, packaging, distribution and consumption of meat product on the basis of flow chart.
ii. Identification of critical control points (potential contaminants and their sources) to control and minimise a hazard.
iii. Establishment of critical limits or tolerance levels (standards) at each control point.
iv. Outlining the procedures to monitor the critical control points.
v. Defining the corrective action if a deviation is noticed during monitoring.
vi. Maintenance of proper records or documents of HACCP plan.
vii. Verification of methods, procedures and tests to oversee the compliance of the plan.

Table 9.1: Microbioloeical specifications

Meat type	*Test*	*Satisfactory*	*Acceptable but investigate*	*Reject, stop production*
A and B	Total viable counts	Less than 0.5 million/g	More than 0.5 million/g but less than 2 million/g	More than 2 million/g
	Coliforms	Less than 500/g	More than 500/g but less than 5000/g	More than 5000/g
	Salmonella, Shigella	Absent in 25 g	—	Present in 25/g
	Staph. (coagulase positive)	Less than 100/g	More than 100/g but less than 1000 g	More than 1000/g
C	Total viable counts	Less than 200/g	More than 200/g but less than 10,000 g	More than 10,000/g
	Coliforms	Less than 10/g	More than 10/g	50% of the repeat sample showing 10/g
	Streptococcus faecalis	Less than 10/g	More than 10/g	More than 10/g
	Salmonella, Shigella	Absent in 10 g	—	Present in 50/g
	Staph. (coagulase positive)	Less than 10/g	More than 10/g	Any repeat sample more than 10/g
	Clostridium perfringens	Less than 10/g	More than 10/g	-do-

A = Fresh meats; B = Raw meat products, both types to be thoroughly cooked before consumption; C = Cooked meat products to be consumed cold or reheated.

Table 9.2: Recommended microbiological standards for meat and meat products (counts per g)

Type of meat	*Total viable counts*	*Coliforms*	*Enterococci*	*Staph. aureus*	*Salmonella*	*Reference*
Raw meat	10^5-10^7	Absent in 0.01g	—	Absent in 50g	—	Jay (1986)
Precooked meat	Not more than 10^4	Absent in 0.1g	—	—	—	-do-
Raw poultry	Not more than 10^5	—	—	—	—	Frazier (1978)
Precooked sausages and canned ham	Not more than 10^4	Absent in 0.1g	Absent in 1g	Absent	Absent	-do-
Cured ham	Less than 10^3	(Become suspicious at 10^4 and unsatisfactory at more than 10^4)				Japson (1957)
Cooked baked meat	Not more than 10^5	Absent in 0.01g	—	—	—	-do-
Frozen seafood	Not more than 10^5	Not more than 10^2	—	Not more than 10^2	Absent	Thatcher (1963)

HACCP system has proved very effective in identifying and preventing contamination. It is designed to check each step critically along the processing line to ensure the safety of a food instead of testing at the end.

ISO-9000 Standards

International Organization of Standardization (ISO), Geneva has issued ISO-9000 series of quality standards to facilitate world trade. In this series,

ISO-9000 provides guidance on the choice of specific model to be adopted for quality assurance in an organization.

ISO-9001 is meant for manufacturers having their own product research and development (R and D).

ISO-9002 is for contract manufacturers without any product research and development.

ISO-9003 is meant for commodity suppliers, having only final product inspection and testing.

Thus, ISO-9000 is a very detailed quality management system. It has twenty elements for compliance by the manufacturers. Some of the important ones are management responsibility, design control, purchasing, product identification, process control, inspection and testing, corrective action, packaging, proper documentation and internal audit. ISO-9003 requirements are comparatively less strict. Many food processing and packaging companies in the developed world have sought ISO-9003 certification. This certification can open up global market for food product suppliers. It has the potential for obtaining competitive advantage in the world food trade

CHAPTER 10

Meat Food Products Order (MFPO)

Meat is one of the oldest food items of human being. Initially, it was being taken raw but later it began to be processed to satisfy the palate and to provide preservative action. But meat is a highly perishable commodity and sanitary conditions and utmost hygienic measures are necessary to safeguard the quality of processed meat food products. With the advent of centralised meat processing units, the responsibility to maintain the quality of meat food products became a central concern.

In 1973, Government of India promulgated an Order to enforce strict quality control on the production and processing of meat food products under Essential Commodities Act 1955. The responsibility to enforce this order was entrusted to Directorate of Marketing and Inspection, Ministry of Agriculture and Rural Reconstruction. The Agricultural Marketing Advisor to the Government of India was made the ex-officio Chairman of Meat Food Products Advisory Committee with 10 members who aid and advise Ministry in any matter pertaining to this Order.

The Order aims at maintenance of sanitary conditions in the slaughterhouses, ensuring proper antemortem examination, postmortem inspection of carcasses, in-process inspection and final product checking. No person could carry on business as a manufacturer except under and in accordance with the terms and conditions of licence granted to him under this Order. Every application for grant of licence shall be made as per proforma set out in the First Schedule of this Order. All meat processing units which produce meat food products for sale within the country come

under its preview. Restaurants and hotels that prepare meat food products for consumption within their premises are exempted. Any food item which is made from flesh or any other edible part of carcass through the process of curing, smoking, cooking, drying or any other processing technique is referred as meat food product. The Order does not apply on the raw (chilled or frozen) meat. It may be noted that no dealer, agent, broker or vendor can sell or expose for sale or despatch or deliver any meat food products unless the same are manufactured by a MFPO licencee.

Powers of MFPO Officers

The veterinarian inspecting officers of the Directorate are issued necessary authority cards to seek compliance of MFPO. These officers are authorized:

i. to enter and inspect the premises of meat food products manufacturers with a view to satisfy themselves that the requirements of this Order are being complied with.
ii. to seize or detain on giving proper receipt raw materials, documents, account books or evidence connected with the manufacture of meat food products in respect of which they have reason to believe that contravention of the Order has taken place.
iii. to dispose of all meat food products and raw materials so seized or detained as they deem fit.

Categories

Meat Food Products Order, 1973 initially categorises the meat food manufacturers into the following three broad heads on the basis of source of raw meat:

Category A	:	Includes those manufacturers or licencees of meat food products who possess their own slaughterhouse.
Category B	:	Includes those manufacturers of meat food products who purchase meat from approved slaughterhouse.
Category C	:	Includes those manufacturers of meat food products, who purchase raw meat from any other source.

The licence fee for each category differs and is collected every year at the time of renewal of licence.

Schedules

Meat Food Products Order, 1973 contains four schedules:

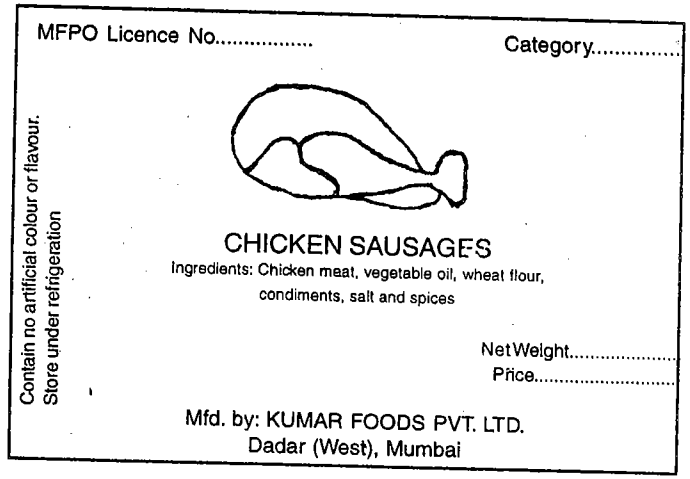
The first schedule: deals with application for licence or renewal of licence under MFPO. The information related to applicant, address of factory, source of raw material, description of meat food products which the applicants proposes to manufacture, installed capacity, a plan of factory and a list of equipments has to be provided. Application for renewal of licence should invariably contain the statements pertaining to the quantity and value of meat food products manufactured in the previous year.

The second schedule: deals with the minimum sanitary requirements to be complied with by a licencee. It contains detailed instructions regarding factory premises, construction, doors, windows and ceiling, plumbing and drainage system, equipment and manufacturing area, cold storage facilities, precautions against flies, rats and mice, water supply, personnel hygiene and vaccination of factory workers, provision of proper aprons and head gears etc.

The third schedule: deals with hygienic requirements to be complied with by a licensee who also slaughters animals in his factory. It contains detailed instructions regarding separation between clean and dirty sections within the slaughterhouse, provision of lairage, slaughter hall and refrigeration facilities, antemortem examination, humane slaughter, postmortem inspection and disposal of condemned carcasses or organs etc.

The fourth schedule: deals with the requirements to be complied with as regards to packaging, marking and labelling the containers of meat food products. It contains detailed instructions with respect to proper packing and sealing of flexible containers, use of internal lacquers and hermetic sealing in tin plate cans, use of bottles and jars. As per MFPO standards, canned meat food products should not contain poisonous elements viz. lead, copper, arsenic, tin, zinc in excess of 2.5, 20, 2, 250 and 50 ppm respectively by weight.

In process inspection of meat food product factories and premises is conducted by MFPO officers regularly. They conduct frequent surprise visits to licensed units so as to enforce the implementation of MFPO regulations. Samples of meat food products are collected



A Label conforming to MFPO specifications

and sent to regional and central Agmark laboratories for specified testing. At present there are more than 220 licensed meat food products units under MFPO, 1973 throughout India. These units manufacture as many as 185 different types of meat food products. Thus, MFPO is playing a major role in safeguarding the interest of meat food products consumers.

CHAPTER 11

Eating Quality and Sensory Evaluation of Meat Food Products

Consumers are the ultimate users of a food item. So, their perception and satisfaction is of paramount importance for the success of any food product. If a product is very palatable, it makes an excellent eating. Some of the important eating quality attributes are flavour, texture and tenderness, appearance or colour, juiciness etc. A knowledge of these attributes is of prime importance for their subjective and objective assessment.

Flavour

Flavour is a complex sensation comprising mainly of taste and odour. It is sensed jointly by the oral and olfactory senses. There are only four basic tastes—sweet, salty, sour and bitter. The receptor taste buds or papillae for each of these tastes are located on the tongue and the upper palate. Odour is sensed at the regio olfactoria inside the nose. Meat sample should be smelled first followed by tasting for a rational and sound flavour perception of several volatile components present in meat. These components are significantly marked when meat is cooked. Flavour has been shown to have a profound effect on the overall acceptability of meat product.

Flavour of meat depends on species, sex, age and method of cooking. In general, flavour of fresh raw meat is weak, salty and serum-like. Fresh meat fat also has almost indistinct taste and odour. It is during cooking that flavour get pronounced and becomes meaty.

Fresh cooked beef is metallic and astringent. Veal flavour is sweet and flat. Pork flavour is regarded sweet and bland. Sex odour is more pronounced in male. Pork from boar has a definite piggy odour or boar taint. During long term storage, most meats develop rancid odour due to fat oxidation. It may be muttony for mutton, tallowy for beef and stale, cheesy or fishy for pork. In case, meat is spoiled during storage, it emanates putrid odour due to protein decomposition. When cooked meat is stored for a long time, myoglobin catalysed fat oxidation takes place yielding a distinct warmed-over flavour. Canning imparts canned-meat flavour to meat products due to severe heat treatment.

Texture and Tenderness

Texture is one of the most important eating quality attributes in the acceptance of meat. The overall impression of texture is perceived by the senses of touch, sight and hearing. The texture of animal foods have different components. The mechanical component relates to the effect of stress, e.g. hard, brittle, gummy, chewy, elastic, cohesive etc. The geographical component relates to size, shape and arrangement of fibres in meat, e.g. coarse, grainy, gritty, fibrous etc. Some other components relate to the moisture and fat perception of meat, e.g. greasy, oily, watery, dry, moist etc. In general, three factors are considered to get an overall impression of meat tenderness. These are the ease with which teeth sink into meat, the ease with which the meat breaks into fragments and the left over residue after chewing.

The texture of meat is affected by many preslaughter factors like sex, age, heredity, diet, carcass grade etc. Several postslaughter factors like postmortem glycolysis, fat deposition, connective tissue, conditioning, freezing, cooking etc. have a profound influence on the texture of meat.

Several tenderising agents like salt, weak acids (vinegar and citric acid) and some enzyme preparations are used to make the meat tender. Plant enzymes like papain, bromelin and ficin have a proven tenderising effect on meat. During cooking, connective tissue (including collagen) becomes more tender.

Appearance and Colour

Appearance is a wider term than colour because it is influenced by other factors also. Appearance and colour have a definite appeal

when consumer goes out to buy meat or meat product. The colour of fresh meat is species specific. It is determined by the concentration of principal meat pigment—myoglobin and its derivatives (oxymyoglobin and metmyoglobin). Consumers relate the appearance and colour of meat with safety and healthfulness.

The colour of cooked meat is related by consumers to doneness. The final colour of cooked meat is dependent upon the pigment changes brought about by temperature, time and method of cooking. When meat is cooked, there is gradual change of colour from dark red or pink to a lighter shade and finally at higher temperatures to gray or brown colour. Pressure cooked or boiled meat will discern a gray colour whereas roasted, broiled or canned meat turns brown. The brown colour of thoroughly cooked meat is due to denaturation of heme pigments and polymerisation of some proteins and fats. The colour of fresh pork, mutton and buffalo fat is white and undergoes very little change during cooking. The colour of cured meats is due to nitric oxide myoglobin which is heat stable.

Juiciness

The juiciness of cooked meat comprises of two components—the first impression of juiciness comes from rapid release of meat fluids during initial chews whereas sustained juiciness is due to slow release of serum and stimulatory effect of fat on salivation. A good quality meat is more juicy than that of poor quality because of higher content of intramuscular fat. Fresh frozen meat with high ultimate pH is quite juicy.

Cooking temperature has a profound effect on the juiciness of meat. The degree of shrinkage on cooking is inversely proportional to the juiciness of meat. Juiciness and tenderness are closely related to meat attributes. In a tender meat, the juices are quickly released on chewing and left over residue is less.

Besides these, certain other eating quality attributes also influence the overall palatability of a meat product. Some meat products have a residue coating the mouth after swallowing. Mouth coating has been observed in chicken frankfurters, buffalo sausages and patties etc. Overall acceptability of a meat product is not the sum average of all the eating quality attributes. This is so because some attributes influence the overall acceptability of the product more as compared to others. Lawrie (1979) rated texture and

tenderness as the most important eating quality attributes whereas Bartholonew and Osulao (1986) reported that compared to texture and appearance, flavour had more effect on the overall acceptability of processed mutton products.

Sensory Evaluation

Sensory evaluation of foods including meat refers to their scientific assessment through the use of human senses. This evaluation is necessary for:

a. New product development
b. Analysis of competitive products
c. Product shelf life studies
d. Market level consumer tests
e. Aroma research etc.

Modern instruments available these days can analyse usually one and at times two eating quality parameters at a time but almost all attributes and the total impression of a food product can be properly assessed only by human sensory perception.

Specialised Panel

The sensory evaluation of meat and meat products may be product oriented or consumer oriented. Product oriented evaluation is done by specialised panel. When new products or formulations are being developed or earlier ones are being altered, specialised panel testing usually precedes consumer panel testing. In specialised panel testing, small number of trained panelists use their senses as biological detectors. Such panelists can easily identify even minor differences in the eating quality attributes of similar meat products.

Sensory panel members are selected on the basis of general health, appetite, sensitivity, willingness, availability etc. They are invited to attend an orientation session where testing facilities are shown and importance of this work is explained to them. Their sensitivity is tested by conducting recognition test and threshold test for four basic tastes. Selected individuals are then trained by making them familiar with the important meat traits and conducting serial dilution tests till they attain good performance on hedonic scale (Table 11.1).

The sensory panel members are presented test meat samples alongwith a control in a random order after assigning suitable codes.

Table 11.1: Proforma for sensory evaluation of meat products based on hedonic scale

Name ---------- Expt. No. ---------- Product : ---------- Date: ----------

Scoring guide

Attributes	Scale of descriptive attributes of product								
	8	7	6	5	4	3	2	1	
Appearance/Colour	Excellent	Very Good	Good	Fair	Slightly poor	Moderately poor	Very poor	Extremely poer	
Flavour	Extremely desirable	Very desirable	Moderately desirable	Slightly undesirable	Slightly undesirable	Moderately undesirable	Very undesirable	Extremely undesirable	
Juiciness	Extremely juicy	Very juicy	Moderately juicy	Slightly juicy	Slightly dry	Moderately dry	Very dry	Extremely dry	
Texture	Extremely desirable	Very desirable	Moderately desirable	Slightly undesirable	Slightly undesirable	Moderately undesirable	Very undesirable	Extremely undesirable	
Saltiness	Extremely desirable	Very desirable	Moderately desirable	Slightly undesirable	Slightly undesirable	Moderately undesirable	Very undesirable	Extremely undesirable	
Mouth coating	None	Practically none	Traces	Slight	Moderate	Slightly abundant	Moderately abundant	Abundant	
Overall acceptability	Extremely acceptable	Very acceptable	Moderately acceptable	Slightly acceptable	Slightly unacceptable	Moderately unacceptable	Very unacceptable	Extremely unacceptable	

Sample	Appearance	Flavour	Juiciness	Texture	Saltiness	Mouth coating	Overall acceptability
1							
2							
3							
4							

Remarks:

Signature

The ideal time for conducting sensory evaluation of meat and meat products is late morning or late afternoon. It is necessary to rinse the mouth between successive samples. Difference tests are usually applied for product testing. In paired difference test, there is direct comparison of a control sample and an analytical sample for a single attribute. In a triangle test, three meat samples are offered—two similar and one different. The odd sample has to be identified. In hedonic scoring or rating test, the degree of acceptance is inferred on the basis of numerical rating of meat samples. In a ranking test, panel members are assigned to rank the samples according to their preference.

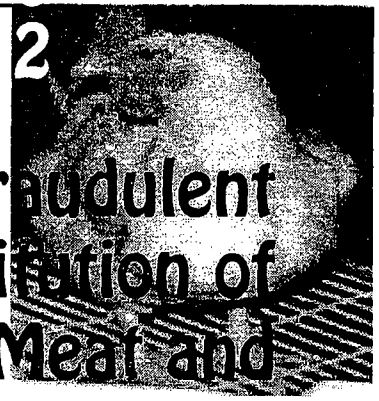
Consumer Panel

Consumer panel generally consists of 100 or more persons drawn from the geographical area proposed to be the market of a particular meat product. The panel members need not have any formal education or training. They may be factory workers, laboratory attendants or any body from the consumer population. This panel is of practical utility in predicting the future market performance.

The consumer panel members may be selected from the target population on random basis. Their likes and dislikes, eating habits, income group, psychological state or environment etc. are allowed to play its natural or justifiable role, since they are the potential product users. Consumer panel can be conducted at any central location—super market, retail outlet or even consumers homes. With little instructions and guidance, panelists can easily be involved in ranking test or hedonic score test.

The data obtained from specialised as well as consumer panel is statistically analysed to enable proper interpretation and application of results.

CHAPTER 12



Fraudulent Substitution of Meat and Its Recognition

Adulteration of meat involves substitution of costly or superior quality with cheaper, undesirable or inferior quality meat. It is a fraudulent practice that is objectionable on the grounds of health, religion and economics. It is punishable under Prevention of Food Adulteration Act, 1955.

The substitutions generally practised are mutton for goat meat (Chevon), beef for buffalo meat, rabbit meat for chicken etc. The instances of dog or cat meat or even veal as goat meat have also come to light. In United Kingdom, substitution of beef with horse flesh is the most likely one to be encountered whereas in Australia the possibility of substitution of beef with Kangaroo meat cannot be ruled out.

Recognition of Fraudulent Substitution or Adulteration

It is necessary to assure the wholesomeness of meat to the public, which besides other measures, may necessitate the authentic identification of species of meats. The following methods can be used for meat differentiation:

1. Physical Methods
   - i. General Appearance: Colour, consistency, odour
   - ii. General characteristics of the body fat: Colour, consistency, quality etc.

1. Physical Methods

S. No. Meat	General appearance			Characteristics of body fat			
	Colour	*Consistency*	*Odour*	*Colour*	*Consistency*	*Quality*
1. Chevon	Light red	Very firm	Goaty	Pure white	—	Practically no intermuscular fat
2. Mutton	Light to dark red	Firm and dense	Ammoniacal	White	Hard and firm	Abundant intermuscular fat
3. Buffalo meat	Dark red	Firm	—	Pure white	—	—
4. Beef	Red	Fairly firm	—	Yellowish white	Firm	Intramuscular fat
5. Bullock	Light to dark red	Firm	—	White to yellow	Firm to loose as per age	—
6. Pork	Light red	Very soft	Urine like	White	Soft and greasy	Subcutaneous fat deposition intramuscular also
7. Veal	Pale to white	Firm	—	White	Firm	No intramuscular fat
8. Horse flesh	Dark red	Soft	—	Yellow	Soft and greasy	No intramuscular fat
9. Dog meat	Dark red	—	—	White	—	Slight intramuscular fat
10. Poultry meat	White	Firm	—	Yellow	Loose	Mostly subcutaneous
11. Camel meat	Red	Fairly firm	—	—	—	—

2. Anatomical Methods
   i. Dentition
   ii. Bone percentage of carcasses
   iii. Rib numbers and their degree of curvature
   iv. Characteristics of long bones

Anatomical Methods

i. Dentition

S.No.	Species	Permanent dentition
1.	Cattle and buffalo	$2\frac{(0033)}{(4033)} = 32$
2.	Sheep and Goat	$2\frac{(0033)}{(4033)} = 32$
3.	Pig	$2\frac{(3143)}{(3143)} = 44$
4.	Horse	$2\frac{(3133)}{(3133)} = 40$

ii. Bone percentage of carcass

Proportion of bones in dressed carcasses can give indication of animal species

S.No.	Species	Percentage of bone
1.	Mutton	25
2.	Bobby calves	50
3.	Veal calves	25
4.	Pork	12-20
5.	Bull	15

iii. Ribs on the thorax

Paired ribs vary in number in different species of animals

S. No.	Species	Ribs in pairs	Sternal ribs
1.	Ox	13	8
2.	pig	14-15	7
3.	Sheep and goat	13	8
4.	Horse	18	8
5.	Dog	13	9

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iv. Characteristics of long bones

Bone	Species	Characteristics
Scapula	Sheep	Short and broad, superior spine thickness and bent back
	Goat	Possesses distinct neck spine, straight and narrow
Radius	Sheep	1.25 times length of metacarpus
	Goat	Twice the length of metacarpus
Ulna	Horse	Extends only 1/2th the length of radius
	Ox	Extends and articulates with carpus
Femur	Horse	There is no third trocanter. Fibula is only a small point projection.
	Ox	Possesses third trocanter. Fibula extends 2/3rd the length of tibia

3. Histological Methods
   i. Muscle fibre length
   ii. Muscle fibre diameter
4. Chemical Methods
   i. Composition of meat
   ii. Myoglobin content
   iii. Glycogen content
   iv. Composition of body fat
5. Immunological/Serological Methods
   i. Precipitation test
   ii. Double immuno-diffusion test
   iii. Single radial immuno diffusion test
6. Electrophoretic Methods
   i. Polyacrylamide disc electrophoresis
   ii. Polyacrylamide gel electrophoresis
   iii. Sodium dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE)
7. Iso-electric Focusing
8. Enzyme-Linked Immuno Sorbent Assay (ELISA)

Histological Methods

The diameter and number of muscle fibers, determined by a fibreoptic microscope, can also lead to species identification. Diameter of muscle fibres of buffalo is more than ox, whereas muscle fibres of buffalo are smaller in size and polygonal in cross section as compared to large and irregular muscle fibres of ox. As far as other species are concerned, the size of muscle fibres decreases in the following order : pig, buffalo, sheep, goat, poultry.

Chemical Methods

i. ***Composition of meat*** Muscular fat content varies according to the meat species. The intramuscular fat in mutton is higher (13.3 per cent) as compared to beef (2.6 per cent), buffalo meat (0.9 per cent), chevon (3.6 per cent) and pork (4.4 per cent). Vitamin A is present in beef and mutton but absent in buffalo meat, chevon or pork.

ii. ***Myoglobin content*** Horse flesh contains maximum myoglobin content (0.71 per cent), much higher than the musculature of other animals.

iii. ***Glycogen content*** Horse flesh is richer in glycogen as compared to most food animals.

Composition of Body Fats

a. *Refractive index*: Horse fat has higher refractive index (53.5) than ox (less than 40) and pig (less than 50) fat.
b. *Iodine number*: Iodine number of horse fat (70-85) is higher than even lard (50-70), whereas iodine number of ox and sheep fat is 35-46.
c. *Carotene content*: Buffalo fat is white due to absence of carotene whereas cow fat is cream to yellow in colour due to carotene content.
d. *Fatty acid analysis*: Horse fat contains 1-2 per cent linoleic acid whereas other fats contain less than 0.1 per cent linoleic acid.

Immunological/Serological Methods

These tests are based on the principle that a reaction between soluble antigen and its corresponding specific antibody in appropriate proportion yields a visible precipitate at the point of their interaction.

i. *Precipitation test or ring test:* In this test antigen is overlayered on to antiserum contained in a test tube. This test has now become obsolete because precipitate often diffuses in a short period of time.
ii. *Double immuno-diffusion (DID) test:* This test initially developed by Ouchterlony (1948) has been further improved and extensively used for the detection of meat species. The test is performed in wells punched in agarose. The buffalo anti cattle monospecific (BACM) serum and rabbit anti cattle monospecific (RACM) serum have been found to be most suitable from the point of view of simplicity and reliability. Antiserum should be checked for sensitivity before use. It should have a sensitivity of 10 per cent.
The precipitation lines produced by the interaction of homologous antigen and antiserum in this test remain distinct for quite sometime and can be preserved for future use. It is easy and cheap test for identification of raw or heated meat at 80°C for less than 10 minutes. Besides, it can detect adulteration upto 5 per cent. However, it is time consuming and ineffective for the detection of thoroughly cooked meats.
iii. *Single radial immunodiffusion (SRID) test:* In this test serum albumen content in meat extract is estimated on the ground that only this protein formed an immune precipitate in antiserum. Immunological species specificity of albumen is regarded as far superior than the globulin fraction.

Electrophoretic Methods

Electrophoretic methods have been found to achieve the separation of proteins by their differential migration through a supporting medium under the influence of an electric field. The protein bands thus resolved are visualised for characteristic pattern by direct observation or densitometeric scanning.

i. *Polyacrylamide disc electrophoresis* In this method, mitochondrial preparation of goat, sheep, cattle and buffalo meat has been used for the identification of the particular species of fresh meat. The species are identified according to the band pattern.
ii. *Polyacrylamide gel electrophoresis* Initially introduced as starch gel electrophoresis, it was improved with Polyacrylamide gel. Here buffer consisting of 0.5M NaCl and 0.034M EDTA is used at pH 5.4. After electrophoresis and staining, band pattern is observed for identification. It is applicable to meat cooked at less than 80°C for 10 minutes. However, every time we have to run the standard alongwith the sample.
iii *Sodium dodecyl sulphate polyacrylamide gel electrophoresis (SDS-PAGE):* When electrophoresis of different meat samples

is performed in a polyacrylamide gel alongwith sodium dodecyl sulphate, proteins run according to their molecular weights. The resultant band patterns can be observed for species specificity. SDS-PAGE electrophoresis yields not only excellent results for globular proteins in native state but also for the highly helical rod shaped molecules like myosin. This method is useful for cooked meat and meat products. However, complexity of bands in high molecular weight region hinder the identification especially in closely related species.

Isoelectric Focusing

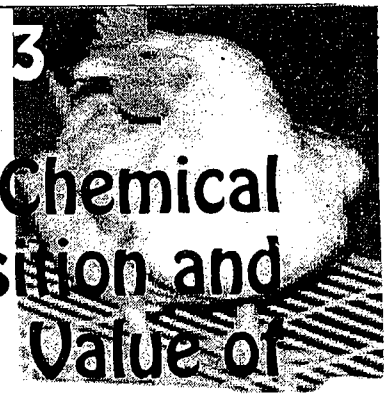
This method utilises differences in the isoelectric point of fresh meat proteins for meat differentiation. Tissue sections are placed directly on the surface of agarose gels and the proteins are eluted electrophoretically. This is a speedy method with high resolving power.

Enzyme-Linked Immuno Sorbent Assay (ELISA)

ELISA is an important qualitative immunological tool which is not monitored by precipitation. In this test, antigen-antibody interaction occurs in a monomolecular layer immobilised on an inert surface and is followed by means of an enzyme chemically bonded to one of the immuno-reagents. This is a rapid test and the results are obtained in 2-3 hours. The test is very sensitive also because even 2 per cent adulteration can be recognised by this test.

It should be noted that no single test is good enough to differentiate all types of meats. Physical, chemical and anatomical methods are more suitable for raw meat whereas comminuted meat products require sophisticated techniques such as Ouchterlony method, SDS-PAGE, Isoelectric focusing, ELISA etc. However, Ouchterlony method can not distinguish between closely related species such as sheep and goat, cattle and buffalo etc. The effectiveness of ELISA and SDS-PAGE is hampered by the cumbersome process of isolating species-specific serum. A new method called "Random Amplified Polymorphism DNA (RAPD) Fingerprint Technique" generates specific DNA fingerprint pattern for differentiation of red meats in a short time. Besides, some developed countries have patented field identification test kits. Such a kit is very much required in our field conditions also.

CHAPTER 13



Chemical Composition and Nutritive Value of Poultry Meat

Poultry meat is a good source of protein. It has a balanced lipid content and low calorific value. It is palatable, tender and easily digestible. It is easy to prepare and can be served in a variety of ways.

CHEMICAL COMPOSITION

A quantitative proximate composition of poultry meat is presented in Table 13.1. The detailed composition of raw chicken meat is presented in Table 13.2, although it varies according to the age of the bird.

Moisture

This is the largest component of muscle tissue. It dissolves nutrients and serves as a medium for their transport. Raw chicken meat contains 70-74% moisture. In general, younger birds have a higher moisture to skeletal muscle ratio as compared to older ones. The fat content of meat is inversely proportional to the moisture content. So, as the fat increases, the moisture content decreases.

Protein

Poultry meat has a higher protein content than most of the red meats. This protein is of very superior quality with respect to biological

value and essential amino acid contents. Male birds generally have a higher protein content as compared to their female counterparts. Protein content of the body is less prone to change in comparison to moisture and fat contents.

Fat

Most fat in poultry remains confined underneath the skin in contrast to red meats where it is generally distributed throughout the tissues. The content varies widely depending on the species, age, sex and diet of poultry. The carcass fat invariably increases with a raise in dietary fat or high energy diet. The proportion of desired unsaturated fatty acids—oleic and linoleic acid is more than 60 per cent of the total meat fat. Poultry meat contains less cholesterol, a fatty alcohol associated with atherosclerosis, as compared to most other animal based foods.

Table 13.1: General composition of raw poultry meat

Species	*Moisture (%)*	*Protein (%)*	*Fat (%)*	*Ash (%)*	*Food energy (Cal/100g)*
Chicken					
i. Broiler (8 weeks)	74.0	18.5	6.0	0.80	125
ii. Spent hen	72.0	19.0	6.5	1.25	120
Quail (8 weeks)	70.5	20.5	5.5	1.20	125
Duck (8 weeks)	58.0	20.0	19.8	0.50	300
Turkey (medium fat)	60.0	19.5	18.0	1.00	270

Carbohydrate

Poultry meat has very little carbohydrate content, hardly 1-2 per cent of total edible tissue. Inositol, glucose and fructose are the major and mannose and ribose are the minor constituents of carbohydrate.

Vitamins

Poultry meat is a good source of many vitamins. Niacin is present in good quantity whereas thiamine (vitamin B_1), riboflavin (vitamin B_2) and ascorbic acid (vitamin C) are also present in fair quantity. Poultry liver is a rich source of vitamin A, vitamin B complex and vitamin C.

Minerals

Poultry meat contains nearly one per cent desired minerals. Some of the important ones are sodium, potassium, calcium, magnesium, iron, phosphorus, sulphur, chlorine etc.

Table 13.2: Detailed composition of raw chicken meat

1.	Amino acid content (% of protein)	
	Lysine	7.5
	Methionine	1.8
	Arginine	6.7
	Cysteine	1.8
	Tryptophan	0.8
	Tyrosine	2.5
	Leucine	6.6
	Isoleucine	4.1
	Histidine	2.0
	Valine	6.7
2.	Mineral content (mg/100 g)	
	Calcium	5.8
	Phosphorus	407
	Iron	0.7
	Sodium	46
	Potassium	248
	Magnesium	29
	Sulphurs	268
3.	Vitamin content (per 100 g)	
	Vitamin A	730 iU
	Thiamine (B_1)	0.07 mg
	Riboflavin (B_2)	0.38 mg
	Niacin	5.6 mg
4.	Lipid content (% of total lipids)	
	Phospholipids	48
	Neutral lipids	52
5.	Cholesterol content (mg per 100 g)	60

Source :
1. Scott, M.L. (1958). J. American Dietetic Assoc., 34(2)
2. McCance and Widdewson (1960). World's Poult. Sci. J., 36:25
3 and 5. Watt and Merril (1963). USDA Handbook No. 8
4. Katz *et al.* (1966). J. Food Sci. 31:717.

NUTRITIVE VALUE

Poultry meat is a food of high nutritional value. It is higher in protein content as compared to red meats (Table 13.3). Poultry meat proteins are classified under first class category because it contains all the

essential amino acids in balanced proportion. Such high protein diet ensures overall development of the body and plays an important role in tissue repairs.

Chicken meat with low fat content offers good quality food to the consumers. It provides the much desired essential fatty acids which form necessary constituents of the cell wall, mitochondria and other cell constituents. Thus, it helps in maintaining the health of the consumers. Due to its low energy value, chicken meat is a good food for weight control diets. Chicken meat contains more phospholipids and low cholesterol than other meats, which minimises risks due to diabetes and heart diseases.

Chicken meat is a good source of vitamins and minerals in human diet. It is rich in niacin and moderately rich in thiamine, riboflavin and ascorbic acid. Chicken meat is also a good source of iron and phosphorus. Due to high biological value and easy digestibility, it is a choice food for aged persons as well as children. Chicken meat carries a high class image because of its product variety and healthful nature.

Table 13.3: Nutritive value of roasted chicken meat (100g)

Protein	24.5 g
Fat	6.2 g
Ash	1.04 g
Food energy	154 Calories

Source: Scott, M.L. (1956). J. American Dietetic Assoc., 32(10)

In general, poultry meat contains all the essential amino acids, fatty acids and minerals in an appropriate quantity. It has the ability to alleviate the nutritional stress conditions in the human beings. It has a good aesthetic appeal. Poultry meat has no religious inhibition and its many products satisfy the variety quest of the consumers.

CHAPTER 14



Pre-slaughter Handling, Transport and Dressing of Poultry

Several poultry dressing plants have come up in the country and many more are in the offing where large number of birds are to be handled and processed every day. These birds should be handled properly before slaughter. It reduces the changes of bruises, cuts and tears on the dressed birds. In fact, pre-slaughter care contributes a good deal to the wholesomeness of dressed chickens.

Pre-slaughter Care and Handling

In the intensive housing system, a great care has to be exercised in catching and crating the birds. All feeders, waterers and other accessory equipment should be moved to one corner of the house before catching and assembling is undertaken. The broilers are generally caught at night under very dim light. Culled and spent hens are caught in the cooler hours of the day, preferably in the afternoon. The birds are caught manually by the shank in a humane way.

Transport

Crates, coops or cages are used to transport birds in vans from the farm to poultry dressing plant. Special attention is paid to prevent overcrowding and suffocation. The loading of birds is carried out in

dim light either early morning or late evening to avoid excitement and transported in the cool period without much exposure to sun to prevent excessive shrinkage. Bulk weighing of birds in crates is the general practice at the large sized dressing plants. A shrinkage of 3-4 per cent takes place during pre-slaughter handling and transport. Birds should be kept off feed for 12 hrs before slaughter but enough shrinkage water should be made available. This practice not only helps in early evisceration but risk of contamination of meat by the intestinal contents is also minimised.

Dressing of Poultry

Slaughtering

Slaughtering involves stunning and bleeding:

i. *Stunning:* Stunning prevents struggling and relaxes the muscles holding the feathers. However, it is generally not practiced in case of chicken. A low voltage electric stunning of 50 volts AC for 1 min has been found to be satisfactory.

ii. *Bleeding:* This process is carried out in an inverted cone shaped equipment to rest the body of the bird and keep the head out and down. There are several techniques of slaughtering poultry in order to seek proper bleeding. The technique most commonly used these days is "modified Kosher Method" in which jugular vein is severed just below the jowl taking care not to cut trachea and oesophagus. Another technique for slaughtering the birds is decapitation which is not so common. Still another method which involves piercing knife through the brain has become obsolete. In general, a bleeding time of 1.5 to 2.0 minutes is allowed. Incomplete bleeding retards the keeping quality of dressed chicken.

Scalding

Scalding refers to immersion of birds in hot water for loosening the feathers. It should be done when all reflexes have ceased. The birds are transferred into the scalding tank. Broiler and young birds are scalded at 55°C for 1.5 minutes whereas culled birds and spent hens are scalded at 60°C for 2 minutes.

Defeathering

The process is carried out in a feather plucker consisting of two drums with rubber fingers which revolve in opposite directions pulling of feathers from the carcass. Any remaining feathers are picked up manually.

Singeing

The carcasses are now singed over a blue flame for 5 to 10 seconds to remove hair like appendages called filoplumes.

Washing

The singed carcasses are washed with spray water to remove dirt and reduce the microbial load.

Removal of Feet and Oil Gland

The next step involves cutting of feet from tarsometatarsal joint with a sharp knife and removal of oil gland.

Evisceration

The carcasses are hung by hocks to the shackles for evisceration. By a slit opening from the tip of breast bone, abdominal cavity is opened by means of a transverse cut. A circular cut is made around the vent. The viscera is drawn outside but allowed to remain attached to the carcass for postmortem inspection. Meanwhile, a slit is made in the skin of the neck for easy removal of crop and neck. After postmortem inspection, inedible offals, including trachea, lungs, oesophagus, crop, intestines, gall bladder and kidneys are removed whereas giblet consisting of heart, liver and gizzard should be collected, cleaned and packed in a wrapper.

Chilling and Draining

After washing, the dressed birds are chilled in a chilling tank containing slush ice or crushed ice for 30-45 minutes in order to cool the carcasses to an internal temperature of about 4°C. The chilled birds are kept on the draining rack for 10 minutes to remove the excess water.

Table 14.1: Indian standards for dressed chicken

	Grade 1	Grade 2
• Conformation	Free of deformities that detract from its appearance or that affect the normal distribution of flesh. Slight deformities such as slightly curved or dented breast bones and slightly curved backs may be present.	Slight abnormalities such as dented, curved or crooked back, or mis-shapen legs or wings which do not materially affect the distribution of flesh or the appearance of the carcass or part.
• Fleshing	The breast is moderately long and deep, and has sufficient flesh to give it a rounded appearance with the flesh carrying well upto the crest of the breast bone along with its entire length.	The breast has a substantial covering of flesh with the flesh carrying upto the crest of the breast bone sufficiently to prevent a thin appearance.
• Fat covering	The fat is well distributed so that there is a noticeable amount of fat in the skin in the areas between the heavy feather tracts.	The fat under the skin is sufficient to prevent a distinct appearance of the flesh through the skin, especially on the breast and legs.
• Defeathering	Free of pin feathers, diminutive feathers and hair which are visible to the inspector or grader.	Not more than an occasional protruding pin feather or diminutive feathers shall be in evidence under a careful examination.
• Cuts and tears	Free of cuts and tears on the breast and legs.	The carcass may have very few cuts and tears.
• Discolouration	Free from discolouration due to bruising, free of clots; flesh bruises and discolouration of the skin such as "blue back" are not permitted on the breast or legs.	Discolouration due to bruising; free of clots; moderate areas of discolouration due to bruises in the skin or flesh.
• Freezer burn	May have an occasional pock marks due to drying of the inner layer of skin (derma), provided that none exceeded the area of a circle 0.5 cm in diameter on chickens.	May have a few pock marks due to drying of the inner layer of skin (derma), provided that no single area exceeds that of a circle 1.5 cm in diameter.

Washing

Dressed birds are thoroughly washed again with clean spray water preferably maintained at 15 ± 5°C. Special care should be taken to wash the interior and sides.

Grading

Dressed chickens are graded on the basis of conformation, degree of fleshing, bruises, cuts and other quality attributes. Indian standards for dressed chicken are given in Table 14.1

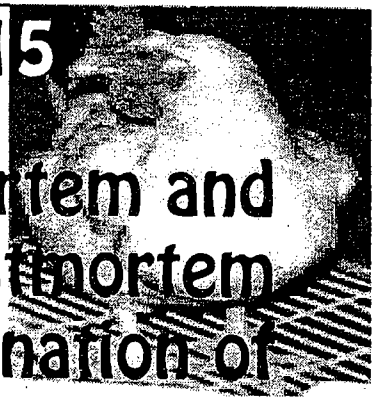
Packaging

Before packaging, dressed chickens having gizzard without mucosal layer, heart without pericardium and liver without gall bladder are placed in the abdominal cavity of the carcass and packed in polyethylene bags (200 gauge). Shrink packaging may be adopted if dressed chickens are to be stored in a frozen condition.

Storage

Dressed chicken can be stored in a refrigerator at 2°C for 7 days and deep freezer at –18 to –20°C for a period of 4-6 months.

CHAPTER 15



Antemortem and Postmortem Examination of Poultry

It is essential to conduct proper antemortem inspection of live poultry in order to ensure that they are not affected with any disease or condition which may render their meat unwholesome. Postmortem inspection becomes essential to detect dressed poultry which might have been diseased, thereby rendering them unfit for human consumption.

Antemortem Inspection of Poultry

Live poultry should be subjected to antemortem inspection in the holding pens by a qualified veterinarian on the day of slaughter. Enough space and water should be provided in the holding pens. Adequate light is an essential requirement during inspection. The birds are carefully examined and those in good health and alert condition are declared fit for slaughter. In general, birds with abnormal conditions are categorised as follows:

i. ***Unfit for slaughter*** Birds with morbid condition due to clinical evidence of a contagious disease, heat stroke or traumatic injury which cannot be treated are declared unfit for slaughter.

ii. ***Suspects*** Birds affected with disease conditions not advanced enough to declare unfit are passed for slaughter as suspect. Such birds are slaughtered separately and both ante and

postmortem findings are considered while taking a final decision.

Various poultry diseases and their antemortem significance is presented in Table 15.1.

Table 15.1: Antemortem significance of poultry diseases

Disease	*Unfit for slaughter*	*Passed for slaughter as suspect*
1. Ornithosis	All affected birds	—
2. Ranikhet disease	Birds in acute stage	All others
3. Chronic respiratory disease	-do-	Birds with slight respiratory distress
4. Infectious bronchitis	Birds in advanced stage	Recovered birds
5. Infectious laryngotracheitis	Birds in advanced stage with cyanosis	Birds with initial signs of disease or evidence of recovery
6. Infectious coryza	Acute stage of disease with debilitation	Highly affected birds
7. Neural lymphomatosis	All affected birds	—
8. Coccidiosis	Anaemic birds	Highly affected birds
9. Infectious enterohepatitis (Black head disease of turkey)	Birds showing symptoms	Slight involvement and recovered birds
10. Fowl typhoid	All affected birds	—
11. Fowl pox	Birds in debilitated and febrile condition	All others
12. Fowl cholera	Birds with septicaemic form	All others
13. Botulism (limber neck)	All affected birds	—

Postmortem Inspection

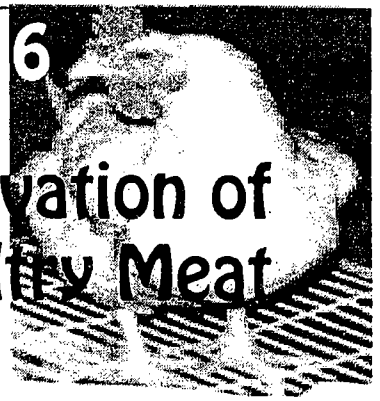
The body cavity of every dressed bird is opened through the transverse incision and visceral organs are drawn out. Now, the carcass is inspected externally for signs of disease, bone abnormalities, wounds, muscular atrophy, tumours etc. followed by body cavity. Liver is examined for consistency, texture, lesions and colour changes. Spleen is also palpated for texture and abnormalities. A cut is given on the hock to see synovial fluid for sinusitis. Common diseases encountered during postmortem examination along with their importance are presented in Table 15.2. Pathological lesions may be referred in *Diseases of Poultry* by Hofstad *et al* (1972).

Table 15.2: Postmortem significance of poultry diseases

Disease	Unfit for food/condemned	Partially condemned/passed for food
1. Avian leukosis complex	All affected carcasses	—
2. Erysipelas	-do-	—
3. Tuberculosis	-do-	—
4 Ranikhet disease	Carcasses with systemic involvement	If lesions are localised, only affected parts are condemned. Rest are passed for food
5. Infectious laryngotracheitis	-do-	-do-
6. Infectious coryza	-do-	-do-
7. Chronic respiratory disease	-do-	-do-
8. Fowl typhoid	-do-	-do-
9. Pullorum disease	-do-	-do-
10. Listeriosis	Carcasses with acute septicaemia	-do-
11. Salmonellosis	Carcasses with active septicaemic lesions	In chronic cases affected parts are condemned. Rest are passed for food
12. Fowl pox	Carcasses with progressive lesions and systemic changes	Recovered birds may be passed for food after removal of scabs
13. Fungal diseases	—	Only affected parts are condemned
14. Fowl cholera	All carcasses are condemned	—

All the condemned carcasses and parts thereof should be destroyed by chemical denaturing with crude carbolic acid or any phenolic disinfectant or completely destroyed by incineration.

CHAPTER 16



Preservation of Poultry Meat

The basic purpose of poultry meat preservation is to retard or prevent microbial spoilage and other physico-chemical changes which cause deterioration in quality. Thus, proper preservation safeguards the sensory quality and nutritive value of poultry meat. Various methods employed for preservation of poultry meat are as follows.

Chilling

Chilling extends the shelf-life of dressed birds by retarding the microbial growth. The efficiency of chilling depends on temperature, air circulation and moisture control. It is advisable to prechill the carcasses at 15°C to remove body heat. Dressed birds are usually chilled by immersion in ice water or chill packed in crushed ice for delivery to stores. Poultry meat can be safely stored at a temperature of 1 to 4°C and relative humidity of 80-85 per cent for a period of 5 to 7 days. The effectiveness of refrigerated storage can be enhanced to several weeks by applying vacuum packaging.

Freezing

Chilled poultry carcasses can be packaged and stored frozen for quite sometime. Freezing of poultry meat can be accomplished either using refrigerated plates or in air. Slow still air freezing generally accomplished in home freezer, takes 4 to 10 hours to freeze depending on the size of the product. Quick air blast freezing is widely used for long term storage in commercial enterprises. Here the prepackaged carcasses are frozen at an air velocity of 1400 rpm to –30 to –40°C in 1-2 hrs. Quick freezing has distinct advantage over slow freezing because intracellular ice crystals formed in this

case do not affect the appearance and other sensory attributes. In slow freezing there is formation of extracellular ice crystals which distort the musculature and increase drip loss upon thawing. The drip contains some water soluble proteins and vitamins. Quick frozen broilers could be conveniently stored for an year at –20°C, whereas cut up chicken and cooked chicken products could be stored for about four months at the same temperature.

Curing

Curing refers to the application of salt, sodium nitrite and cane-sugar to meat with or without ascorbates, phosphates, glutamates etc. Since chicken meat has a mild flavour, the amount of salt and other flavouring ingredients is kept comparatively low. Salt acts as a preservative besides influencing flavour and texture of meat. Curing salt has a high osmotic pressure which inhibits the availability of water to the microorganisms. Initially, there is outward flow of water and soluble proteins during curing. However, when salt diffuses inwards, it forms a complex with protein and achieves a higher osmotic pressure than the curing solution itself, causing some reverse flow of water. The final salt concentration attained is nearly 5 per cent. Sodium nitrite provides the much desired cured pink colour. Cane-sugar contributes to the flavour and counteracts the toughening caused by salt alone.

Chicken can be subjected to dry or wet curing. In dry curing, the carcass is thoroughly rubbed with curing mixture and then aged. In wet or pickle curing either the curing brine is injected in small quantities at several places in the carcass (stitch pumping) or the carcasses are immersed in a curing brine containing 15 per cent common salt, 7.5 per cent cane sugar, 150 ppm sodium nitrite plus nitrate and 0.05 per cent monosodium glutamate for 48 to 72 hours at 4°C. The chickens are overhauled every 24 hours to ensure uniform curing. The temperature of curing room is maintained at 4°C and lighting should be minimum to retard the chances of rancidity development. After curing excess salt on the surface is removed by proper washing. Cured chicken has a shelf life of 14 days at 4°C.

Smoking

Smoking is generally practised along with curing. Smoke obtained by the slow combustion of hardwood saw dust contains lower

alcohols, aldehydes, organic acids, carbonyl compounds, phenols etc. preserve meat by its bacteriostatic, bactericidal and antioxidant properties besides providing a protective film on the surface. Smoke also imparts characteristic flavour and stabilises the cured colour. The temperature of smoke chamber is maintained at 50°C to produce ready-to-cook chicken whereas it is kept at 80°C for 4 hours at 30-35 per cent relative humidity to produce ready-to-eat chicken. In order to eliminate the carcinogenic components obtained due to combustion of lignin especially enzpyrene, liquid smoke is produced these days through condensation. Liquid smoke can be directly sprayed over the cured chicken or added to meat emulsion to impart distinct flavour. Cured and smoked chicken has a shelf life of 1 month under refrigeration (4°C) and 2 to 4 months in a freeze (–18°C).

Dehydration

Cooked chicken meat is sometimes dehydrated for specific supplies. Chicken chunks may be dried in a rotary air drier at controlled temperature to a moisture content of 4 per cent. Finely ground cooked meat may be spray dried to yield chicken soup mix. However, the best results are obtained in freeze dehydration. In this process, chicken meat chunks are quick-frozen and vacuum dried at 1.55 mm Hg at low temperature for 12-24 hours. The final product containing hardly 2% moisture is packed in tins under nitrogen and has a shelf life of one year. The product retains its natural flavour and nutrients and can be reconstituted within minutes.

Canning

This process refers to extreme thermal processing of chicken meat in hermetically sealed cans. It involves precooking of chunks and gravy, filling in lacquered cans, exhausting, sealing under vacuum and cooking in retorts at 15 psi pressure for 35 minutes followed by rapid cooling. The canned chicken product has a shelf life of two years at ambient temperature.

Radiation Preservation

Poultry meat can also be preserved by using radiant energy. Radiation brings about lethal changes in the nuclear material of microorganisms and inactivate the enzyme system without raising

the temperature. Cobalt-60 gamma radiation is generally utilised because of its adequate penetrating power. This method, usually called as cold sterilisation, is used as a supplement to other preservation methods such as refrigeration and freezing. For poultry meat, radiation sterilisation dose of 4.5 Mrad alone and pasteurisation dose of 0.5 Mrad in combination with other preservation methods have been successfully used.

High radiation doses may initiate several undesirable changes in meat such as discolouration, off flavour, loss of water holding capacity, rancidity development and loss of nutrients like thiamine, vitamin B_{12} and vitamin C. Off flavour development may occur due to the production of ammonia, H_2S and mercaptans from free amino acids. These undesirable effects can be prevented by undertaking irradiation when the meat is in frozen condition and packaged under vacuum or inert gas atmosphere. Radiation preservation has the advantage of speedy operation. However, high cost restricts its use as a commercial practice.

CHAPTER 17

Processing of Some Convenience Poultry Products

Chicken is the major species of poultry in India. It is consumed far and wide in many forms of traditional and processed products. Convenience products do not require any preparation prior to consumption. The common traditional products are tandoori chicken, chicken sheek kabab, chicken shami kabab, chicken curry, chicken kofta, chicken tikka, chicken samosa etc. Other poultry products such as barbecue, chicken patties, chicken sausages etc. also have a good market in urban areas. The methods of preparation of some convenience poultry products have been described in this chapter.

Tandoori Chicken

Broilers at 6 weeks of age are preferred for tandoori chicken because of their tender meat and ability to sustain roasting. Dressed chickens with intact skin are rubbed with 4 per cent salt along with spices and seasoning and kept for 15 minutes. After draining, the carcasses are thoroughly marinated with sauce on the surface and in the interior. A marination time of 1-2 hours is allowed. The formulation of sauce depends on the consumers preference for taste and other sensory attributes. In general, dry and ground spices along with condiments are blended with vinegar (10%) and curd (10%).

The marinated chickens are roasted in a tandoori oven under smokeless, moderate and uniform heat for 20-30 minutes depending on the temperature of oven and size of the broilers. Care must be taken to keep the chickens away from the direct fire and avoid burning or blistering of the skin or extremities. During roasting, chickens are

occasionally removed from the oven and pasted with sauce or fat with the help of a brush. The doneness of tandoori chicken is tested by twisting one of the drumsticks when it dissociates easily from the joint. By this time, it also acquires slightly smoked flavour.

Chicken Barbecue

Broilers with about 750 g dressed weight are preferred for barbecuing. The dressed chickens are longitudinally halved for this purpose after removing the neck portion. The chicken halves are marinated with sauce containing spices, salt and seasonings according to the consumers taste and preference and allowed to stay for an hour. The sides are then placed on the oven for barbecuing during which these are periodically turned and basted with sauce with the help of a brush to avoid drying. The cooking should proceed slowly at moderate temperature so that tender, golden brown and slightly smoked flavoured barbecue is obtained.

Chicken Seekh and Shami Kababs

Culled or spent chicken meat can be utilised for preparing sheek kababs. Lean meat is minced through 8 mm plate of a meat grinder. Wheat flour (3 per cent) and whole egg liquid (5 per cent) should be incorporated as binders to provide sufficient strength to the mince. Fat, salt, dry spices and seasonings are added as per consumers preference. The mince is pasted around specially made iron bars (seekh) and cooked over moderate and uniform heat, turning the bars and basting with vegetable oil from time to time till doneness with brown colour is achieved.

In the preparation of shami kababs, meat chunks and water soaked black gram dal are simmered in water for nearly 15 minutes before grinding. It is seasoned with salt, dry spices and condiment paste. Some people also add liquid egg to the mince. It is made into round cakes which are shallow fried with edible oil on a girdle till both the sides are brown.

Chicken Kofta

Meat from spent or culled chicken can be utilised for preparing kofta (meat balls). Lean meat is coarse ground through 8 mm plate of a meat grinder. Ten to fifteen per cent vegetable oil is added to it.

Wheat flour (3 per cent) in combination with whole egg liquid (5 per cent) are incorporated to provide sufficient binding strength. Seasonings, salt and spices can be mixed as per consumer preference. The dough is rolled into 15 g balls with hands. The balls are deep fat fried for 5 minutes. Cooked balls, packed in polyethylene pouches have a keeping quality of 8 to 10 days at 4°C.

Poultry Pickle

Dressed chicken is trimmed off excess fat and deboned. Now meat is cut into 2.5 cm cubes, applied with 2 per cent salt and pressure cooked for 8 to 10 minutes. Cooked meat is taken out and fried at medium heat in mustard oil to get brown colour. Oil is decanted from the fried cubes and green curry stuff is fried in the same oil to get golden brown colour. This is followed by addition of dry spices, remaining 2 per cent salt and fried meat continuing frying for another 3-4 minutes. After some cooling, it is thoroughly mixed with 10 per cent vinegar. The product has a shelf life of 100 days at ambient temperature without any appreciable loss of quality attributes.

Chicken Samosa

Lean chicken is minced through 5 mm plate of a meat grinder. Condiments are fried in vegetable oil to get a golden brown colour and dry spices along with salt are added towards the end. Minced lean and cooked mashed potatoes are mixed with the fried spices and heating is continued for another 4 to 6 minutes. The fried stuff is ready for filling. Dough portions of about 30g are rolled out and divided into two halves. Each half is moulded into a triangular pouch and the fried stuff (20-25 g) is filled in. The pouch is closed and samosas are deep fried in vegetable oil at medium heat to obtain a crispy product.

Chicken Sausage

The tough meat from spent hens can be utilised for the preparation of chicken sausages. Deboned chicken meat is minced once through 9 mm and then through 4 mm plate of a meat grinder. Fat is minced separately through 4 mm plate. Lean meat, ice flakes (10 per cent), salt and sodium nitrite are run along with fat in a bowl chopper to prepare a fine emulsion. Other ingredients like spices, condiments

etc. are added to the emulsion in the final run for a minute. Meat emulsion is filled into casings with the help of a sausage filler and suitable links are made. The sausages may be cooked in water at 80°C for 15 to 20 minutes or steam cooked. Smoking along with cooking stabilises the colour and imparts a characteristic flavour to the sausages.

Chicken Patties

Raw deboned chicken meat and fat are minced twice through a meat grinder. Other ingredients like wheat flour or texturised soy protein (binding agent), salt, condiments, spices etc. are mixed to the ground meat in an electrically operated meat mixer. The blended mass is divided into 100g portions and moulded into patties. These are broiled in a hot air oven set at 200°C for 15 to 20 minutes to get a core temperature of about 72°C. Hot patties may be used to prepare burgers or chilled in a refrigerator for latter use.

Chicken Tikka

Deboned chicken is minced in a meat grinder. Forty per cent of the mince is pressure cooked for 2 minutes. Besides, peeled and shredded potatoes are partially cooked in boiling water separately. Now, mince meat (60 raw : 40 cooked), shredded potatoes, rice powder, bread crumbs, salt, spices and condiments are thoroughly mixed in an electrically operated meat mixer. The blended mass is divided into 70 g portions and moulded into tikkas. These are shallow fat fried in a girdle to achieve an internal temperature of 70°C. The product has a unique texture and is consumed as a hot snack.

The list of convenience chicken products is on the increase. These ready-to-eat products have a bright scope in India. These have already become quite common at the fast food corners and restaurants in the urban areas. With the vast availability of spent hens, comminuted chicken products are likely to surpass other meat products in near future.

CHAPTER 18



Utilisation of Poultry Industry By-products

Indian poultry industry has made rapid advances in the last two decades. A large number of poultry farms have come up as clusters resulting in the so-called poultry pockets in various parts of the country. There is a growing realisation to collect and beneficially utilise the by-products of poultry industry which have so far been dumped as waste, except for poultry manure. Establishment of a poultry by-product processing plant, within each cluster of farms having hatchery dressing units etc. will generate substantial revenue, minimise environmental pollution and help reduce the livestock feed cost. However, places where collection of poultry by-products is not enough to warrant a separate plant, services of carcass utilisation plant or renderer operating in the area can be requisitioned once or twice a week. By-products of poultry industry are:

A. Poultry dressing plant waste (on live weight)
   - i. Feathers : 6%
   - ii. Blood : 3.5%
   - iii. Offal
     - a. Heads : 3%
     - b. Feet : 4%
     - c. Inedible viscera : 9%
       (intestines, lung, pancreas, spleen etc.)

B. Hatchery waste : Infertile eggs, dead in germs, dead embryos, egg shells, unhatched chicks, unusable chicks etc.

C. Egg processing unit waste : Unsound eggs and egg shells.

D. Poultry manure : Used-up deep litter and wet droppings from cage houses.

E. Dead birds

Feathers

i. ***As livestock feed*** Poultry feathers are hydrolysed to yield feather meal in by-product processing plant or renderer at a steam pressure of 30 psi for 2 hours with continuous stirring. Feathers are sometimes crushed to break thick shanks and speed up the process. Since feather meal has a poor amino acid profile, feathers are generally processed with blood.

ii. ***As bedding*** Soft and fluffy down feathers are used to make comfort pillows and mattresses. These bedding articles have good resilience and warmth besides being easy to handle and store.

iii. ***As fertilizer*** Feathers are crushed and cooked under pressure to yield a good quality fertilizer. In places where cooking is not cost effective but irrigation is not a problem, feathers can be ploughed in the soil to slowly decompose and release nitrogen.

iv. ***Ornamental and sport use*** Feathers are used for making artificial flowers, toys and various decorative articles. Stiff feathers are used for making shuttle cocks.

Blood

i. ***As livestock feed*** Blood collected in the poultry dressing plant can be processed in a simpler cooker to yield dry blood meal with only 9% moisture. Since quantity of blood is generally inadequate and blood meal alone is not so palatable to the livestock, blood and feathers are usually cooked together in a by-product processing plant or renderer. Caking of blood on the renderer wall is prevented by putting an iron can in the renderer. Blood and feather meal has a rich amino acid profile and very good digestibility.

ii. ***As fertilizer*** Unwholesome or decomposed blood can be simply cooked for sometime to reduce moisture and used as a fertilizer in the fields.

iii. ***As fish bait*** Blood can also be used as a bait during fishing operations.

Offal

Poultry offal comprises of heads, feet and inedible viscera. Inedible viscera includes intestines, lungs, pancreas, spleen and the

reproductive organs. Irrespective of end use, offals, should be sufficiently cooked to destroy microorganisms. Generally, offals are processed for the following purposes:

i. ***As livestock feed*** Offals are cooked in the by-product processing plant at 15 psi pressure for half an hour and allowed to dry. After cooling, fat is extracted and material is ground to yield offal meal. Some renderers first hydrolyse feathers and then add offals to yield offal and feather meal without any fat extract.

ii. ***As pet food*** Poultry offal is utilised as a good quality pet food especially for dogs. For this purpose, offal is washed, cooked, ground and then mixed with other products to form pellets. Some processors prepare canned pet food from poultry offals.

iii. ***As a feed for fur bearing animals*** In cold countries, poultry offals have been conveniently used as a feed for fur bearing animals as a replacement for costly meals. Mince has been specially found to relish poultry offal and record fast growth.

iv. ***As fish food*** In fish hatcheries, poultry offal is cooked, ground and utilised as delicious fish food.

Table 18.1: Per cent yield and composition of poultry by-products meal

Constituents	*Feather Meal*	*Blood Meal*	*Offal Meal*	*Mixed by-product Meal*
Yield	33	18	55	33
Moisture	7	9	10	8
Crude protein	85	86	52	66
Crude fat	3	1	24	18
Ash	4	3	14	1.8

Mixed Poultry By-products Meal

Poultry offal, blood and feathers are mixed in their natural proportion and dry rendered at a pressure of 15 psi for 3 hours to ultimately yield a meal with only 8% moisture. Dead birds can also be included in the charge. MPBM serves as a fairly balanced livestock and poultry feed (Table 18.1). It can be used upto 5-7% in poultry ration.

Hatchery Waste

Entire hatchery waste, other than egg shells is utilised in the preparation of hatchery by-product meal which is comparable to

fish meal with respect to protein content. The meal can be used in poultry ration at 3% level.

The egg contents from infertile eggs, dead in germs etc. are collected and cooked at a pressure of 10 psi for 15 minutes. After cooking, the material is allowed to dry and then ground. The yield of hatchery by-product meal is about 25%. It contains nearly 35% crude protein, 40% crude fat and 4.5% total ash. Calcium and phosphorus contents are 0.05% and 1% respectively.

Egg shells are sterilised and powdered to pass through very fine sieve. Egg shell powder is used in mineral mixture as a calcium supplement. Unsterilised egg shells are sun-dried and ground for use as a fertilizer.

Poultry Manure

i. ***As fertilizer*** Poultry manure has been traditionally used as a fertilizer. Its nutrient value depends on litter to manure ratio. In general, dried poultry manure contains 15-18% moisture, 25-30% total protein, 6-8% uric acid (nonprotein nitrogen), 15-25% ash, 3-6% calcium and 1.5-2% phosphorus besides many other essential trace elements. It has a good fertilizing value for crops, lawns and gardens. It also improves the soil structure because of high organic content.

ii. ***As livestock feed*** Caged poultry manure can be used as feed for pigs and poultry whereas deep litter manure has been used a feed for ruminants. For this purpose, the manure is spread in the sun for drying to moisture content of 10% or less. If facilities are available, it can be autoclaved at 15 psi pressure for 30 minutes and taken to hot air drier for drying. Dehydrated poultry manure has been successfully incorporated upto a level of 15% in layer poultry ration.

Thus, poultry industry products in the form of poultry dressing plant waste, hatchery waste, egg processing unit waste, poultry droppings and dead birds should invariably be processed and utilised for a number of reasons. It will solve the waste disposal problem and reduce pollution in and around the farms and processing units. It will also economise the poultry production as such by yielding various by-product meals for feeding poultry.

CHAPTER 19



Structure, Composition and Nutritive Value of Eggs

An acquaintance with the formation and structure of eggs is necessary to effectively preserve its quality during storage and marketing. Egg is basically nature's device to produce a chick. So, it has the necessary infrastructure for the production and nutritional requirements of developing embryos and newly hatched chick.

There are four main components of hen's egg:

a. Shell
b. Shell membranes
c. Albumen or white
d. Yolk

The yolk develops in the functional left ovary of the hen as an ovum largely during the final 10 days before release. After ovulation or release, fully developed ovum or yolk is engulfed in the oviduct where a gel of albumin or egg white is secreted to surround the yolk for a few hours. Finally, the shell membranes and the calcareous shell are deposited in the oviduct for nearly 16 hours before the egg is laid.

Structure (Fig. 19.1)

i. ***Shell*** The outer protective covering of an egg is shell which comprises around 11% of its total weight. It is mainly composed of calcium carbonate. The shell contains numerous minute

pores on the entire surface, which are partially sealed by keratin. These pores allow loss of carbon dioxide and moisture from the eggs. However, a few of them (hardly 12-20) may permit bacterial penetration within the egg under specific circumstances. Thus shell structure consists of three basic units:

a. Outer cuticle made up of keratin
b. Middle spongy or calcareous layer
c. Inner mammary layer.

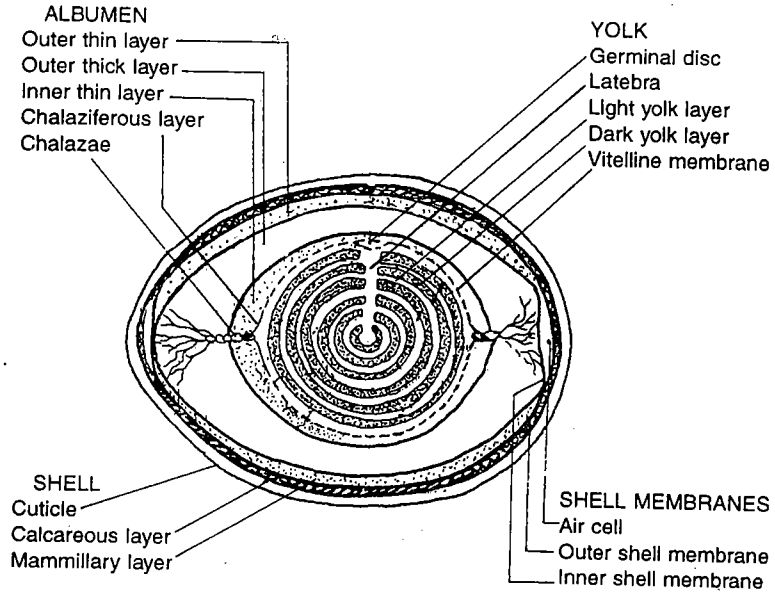


Fig. 19.1: Structure of an egg

ii. ***Shell membranes*** The shell is attached to the shell membranes. The outer thick and inner thin membranes are usually inseparable except at the broad end of the egg forming an air cell. The shell membranes are a part of in-built defence mechanism in the egg because of their role as an effective barrier against bacterial invasion. The air cell continues to increase in size during storage due to loss of moisture and shrinkage of egg contents.

iii. ***Albumen*** The white or albumen portion of egg constitutes about 58% of the total weight of an egg. It occurs in four layers as follows:

a. Outer thin layer
b. Outer thick layer
c. Inner thin layer
d. Inner thick white or chalaziferous layer.

The proportion of thin and thick white varies according to the breed and age of the hen. Thick content is comparatively more. The inner thick white layer surrounds the vitelline membrane of the yolk and its firm mucin like fibres continue as chalazae which has the specific function of keeping the yolk in the centre.

iv. ***Yolk*** The yolk constitutes nearly 31% of the total egg weight. It consists of the following four structures from outside:

a. Vitelline membrane
b. Concentric rings of light and dark yolk material
c. Latebra (cone like portion extending to the centre of yolk)
d. Germinal disc (located in the cone of latebra).

Chemical Composition

As mentioned earlier, an egg consists of 11% shell, 58% albumen and 31% yolk. The cuticle of egg shell is composed of a foaming layer of proteinaceous matter especially keratin. It covers the calcified portion of the shell which is made up of calcium carbonate (94%) with minor quantities of calcium phosphate (1%), magnesium carbonate (1%) and proteinaceous material especially keratin. The true cell membrane consist of protein fibres. The inner membrane is comparatively thick.

Table 19.1: Chemical composition of egg

Component	*Total (%)*	*Water (%)*	*Protein (%)*	*Fat (%)*	*Ash (%)*
Whole egg	100	65.5	11.8	11.0	11.7
Albumen	58	88.0	11.0	0.2	0.8
Yolk	31	48.0	17.5	32.5	2.0
		Calcium carbonate (%)	Calcium phosphate (%)	Magnesium phosphate (%)	Organic matter (%)
Shell	11	94.0	1.0	1.0	4.0

Source: USDA

Egg albumen or white contains approximately 88% water. Most of the solid content is protein. Lipid content is virtually absent. However, very minute quantity of carbohydrate (0.5%) may be present. Albumen may be regarded as a protein system consisting of microscopic fibres in a solution of numerous globular proteins. Some important albumen proteins and their characteristics are presented in Table 19.2.

Table 19.2: Important proteins in egg albumen

Protein	*Relative amount in albumen (%)*	*Characteristics*
Ovalbumin	54	Phosphoglycoprotein
Conalbumin	13	Binds metals especially iron
Ovomucoid	11	Inhibits trypsin
Lysozyme (Globulin G_1, G_2, G_3)	10	Lyses some bacteria
Ovomucin	1.5	Sialoprotein
Flavoprotein	0.8	Binds riboflavin
Ovoinhibitor	0.1	Inhibits several proteases
Avidin	0.05	Binds biotin

Source: Baker (1968)

Egg yolk contains more than 50% solids, which are mainly lipids (32%) and proteins (16%). Yolk lipid is composed of mostly triglyceride (65%), good amount of phospholipid (28%) and controversial cholesterol (5%). The ash content of yolk is about 1%. A little of carbohydrate, usually less than 0.5%, may also be present.

Nutritive Value

Table 19.3 shows the nutritive value of a chicken egg. An egg contains about six grams of protein. Egg protein is of such a high quality that its biological value has been taken as 100 and it acts as a standard for evaluating the biological value of other food proteins. All the essential amino acids required in human diet are present in egg proteins. An egg also provides five to six grams of easily digestible fat, wherein the proportion of much desired unsaturated fatty acids (especially oleic acid) is more as compared to most other livestock products. Egg is an important source of fat soluble vitamins (A, D, E and K) and water soluble vitamins of B-complex group. However, it does not contain vitamin C.

Table 19.3: Nutritive value of edible portion of a chicken egg

Component	*Fresh raw egg*		
	Whole	*Albumen*	*Yolk*
Weight (g)	50	33	17
Water (%)	73.7	87.6	51.1
Food energy (Cal)	81.5	16.83	59.16
Protein (g)	6.45	3.60	2.72
Fat (Total lipids, g)	5.75	trace	1.65
Total saturated FA (g)	1.65	–	1.65
Total unsaturated FA (g)	3.30	–	3.30
Oleic (g)	2.2	–	2.2
Linoleic (g)	0.5	–	0.5
Cholestrol (mg)	230	–	230
Carbohydrate (g)	0.36	0.264	0.1
Fibre (g)	0	0	0
Ash (g)	0.5	0.231	0.2890
Calcium (mg)	27.0	2.97	23.97
Iron (mg)	1.15	0.033	1.117
Magnesium (mg)	5.5	2.97	2.72
Phosphorus (mg)	102.5	4.95	96.73
Potassium (mg)	64.5	45.87	16.66
Sodium (mg)	61.0	48.18	8.84
Vitamin A (IU)	590	0	590
Choline (mg)	253.0	0.4	253.0
Inositol (mg)	16.5	–	–
Niacin (mg)	0.05	0.033	0.017
Riboflavin (mg)	0.15	0.089	0.076
Thiamine (mg)	0.055	–	0.037
Ascorbic acid (mg)	0	0	0

Source: American Egg Board (1974)

With very little carbohydrates, egg has a remarkably low caloric value which justifies its inclusion in the food for people on restricted diet. Egg is very good source of important minerals such as iron, phosphorus, potassium and trace elements which are necessary for the formation of blood, bone and soft-tissues. Though cholesterol content of egg yolk is comparatively high, it is not likely to significantly influence the blood cholesterol level unless taken indiscriminately because cholesterol is found in blood, nerve tissues and other parts of the human body as a normal constituent of the cell.

CHAPTER 20



Microbial Spoilage of Eggs

It was widely believed in nineteenth century that contents of fresh eggs were always sterile. Studies conducted afterwards revealed that microorganisms can gain entry into the egg congenitally. However, most of the contaminants of eggs are of extragenital origin and come in contact with egg shell at oviposition from the dust, soil and faecal matter adhered to the nesting material. Since the cuticle and pores of the egg shell are moist at this stage, the possibility of invasion of the shell by some contaminants through a few pores cannot be ruled out. The microorganisms on the shell surface usually belong to a mixed group, but those causing spoilage of egg (generally called rot) are gram-negative in nature which have very simple nutritional requirements.

The microorganisms have to pass through a series of in-built physico-chemical barriers in the egg—the shell, the shell membranes, the albumen before reaching the yolk where they could easily multiply causing rot. The mechanism of microbial spoilage can, thus, be divided into three serial steps:

1. Penetration of microorganisms through the egg shell and shell membranes.
2. Colonisation of microorganisms on the shell membrane.
3. Overpowering of the antibacterial factors present in the albumen.

Penetration of Microorganisms through the Egg Shell and Shell Membranes

Egg shell acquires a diverse microflora at the time of oviposition. Under normal conditions of handling and storage, shell gets dried soon and most of these microorganisms fail to survive. An egg shell contains more than 17000 pores. However, only ten to twelve pores

allow the microorganisms to pass through. The microorganisms either succeed in when the egg contents contract on cooling or gain entry due to capillary action through pore canals when the shell surface is moist. The role of microorganisms remain passive in both situations. It is due to capillary action that incidence of rotting are comparatively high in washed eggs which have been subjected to dry abrasion. The cuticular plugs on the pore canals are opened during the process of abrasion of eggs.

After gaining entry through the shell pores, microorganisms come across shell membranes. These membranes act as bacterial filters and offer maximum resistance to the offending organisms which have succeeded in penetrating the shell. Some researchers believe that membrane lysozyme also has a limited role.

Mold may also cause rot in eggs under humid storage conditions. In such case shell is generally covered with mycelium (whisker) and hyphae penetrate the pores to reach shell membranes.

Colonisation of Microorganisms on the Shell Membrane

Once the microorganisms have an access to shell membrane, they are able to multiply and form colonies. However, the colonisation is not instant. In the early stages, there is preferential selection of gram-negative organisms having low iron requirement from the initial population dominated by gram positive organisms which have high iron requirement. Thus initially there is a decline in the microbial numbers. In the later stages, multiplication of organisms takes place at a faster rate because by this time albumen becomes heavily infected. The pH of egg contents move towards neutrality and yolk comes in contact with inner shell membrane.

Overpowering the Antibacterial Factors Present in the Albumen

Egg white or albumen provides an unfavourable medium for microbial growth because of the defensive role played by many of its component proteins which have been listed under composition of albumen. The role played by lysozyme and conalbumen is particularly important. Lysozyme of albumen cause lysis of mucopeptide rich cell wall of gram positive organisms. This enzyme does not affect the complex cell wall of gram negative bacteria having coating of lipoprotein and lipopolysaccharide over mucopeptide.

Conalbumen which is uniformly distributed and constitutes more than 10% of albumen chelates iron and make it unavailable to the bacteria. Conalbumen is the principal antimicrobial factor present in the egg and its inhibitory action is more on gram positive as compared to gram negative organisms. This inhibition definitely delays the spoilage of eggs to some extent. However, as yolk contents migrate into albumen or get mixed, multiplication of organism is very fast which results in the rotting of eggs. Some general type of rots may be summarised as follows:

Type of rot	*Changes in egg*	*Organisms*
Green rot	Albumen becomes green	*Pseudomonas fluorescens*
Black rot (Type 1)	Blackening of yolk with "faecal odour"	*Proteus sp.*
Black rot (Type 2)	Green coloured albumen but yolk is black with "cabbage odour"	*Pseudomonas sp.*
Red rot	Albumen stained red throughout, yolk surrounded by custard like material	*Serratia sp.*
Fungal rot	Pink spots on egg contents	*Sporotrichium*
	Black spots on contents	*Cladosporium*
	Yellow or green spots on contents	*Penicillium*

Besides rots, eggs may develop various types of off odour due to bacteria without any apparent signs of spoilage. These off odours may be musty or earthy (*Achromobacter sp.*), hay like (*Enterobacter sp.*), fishy (*E. coli*) or that of cabbage water (*Pseudomonas sp.*).

CHAPTER 21

Preservation and Maintenance of Eggs

A freshly laid egg can be assumed to have a highest quality. Since egg is full of essential nutrients, deteriorative changes soon start taking place which may pose a danger to the excellent sensory attributes of this nourishing and satisfying food item. Cleanliness and soundness of shell is the first step to assure the quality of egg to the consumers. The shell quality deficiencies mostly relate to the production practices adopted at the farm. Proper handling of eggs can delay the decline in the quality. Following precautions should be taken during handling of eggs:

i. Eggs should be collected 3 to 4 times per day. This will result in less dirty eggs and fewer breakages.
ii. After collection, eggs should be shifted to holding room maintained at a temperature of about 15°C and 70 to 80% RH atleast for 12 hours.
iii. Eggs should be properly packed in filler flats with broad end up. Bulk packing should be done in fibre board cartons.
iv. Eggs should be rapidly moved through the marketing channel so as to reduce the period between production and consumption.

All preservation methods for shell eggs have been designed to retard one or more of the following physico-chemical alterations which lower the quality of egg as it ages:

i. As the surface of egg dries, the keratin cuticle shrinks and size of shell pores increases rendering it easier for gases and microorganisms to pass in and out of the shell.
ii. As the warm egg cools down, the egg contents also contract, resulting in the formation of air cell.
iii. The breakdown of carbonic acid causing loss of carbon dioxide from the albumen is rapid during the first few hours after an

egg is laid. The alkaline pH acts on the mucin fibres to disturb the thick gel of albumen making it thin or watery.

iv. As the egg ages, water migrates from the albumen to the yolk which may overstretch, weaken or even rupture the vitelline membrane.

Following preservation methods are employed to maintain the quality of shell eggs:

Egg Cleaning

Earlier, it was a general practice to dry-clean dirty egg shells by abrasive mounting on a mechanical wheel. This practice has now become obsolete because it weakens the shell. These days washing in warm water containing a detergent sanitizer is an effective way of cleaning the eggs with dirty shells. A temperature difference of 10-15°C between eggs and wash water is ideal, otherwise there may be problem of crack shells. Besides, eggs should not be immersed in warm water for more than 3-4 minutes. After washing, the eggs should be dried promptly. Wash water should be changed after washing every five to six baskets of eggs. It should be emphasised that only dirty eggs are subjected to washing. It not only reduces the microbial load on the egg shell surface but also improves the appearance and consumer appeal.

Oil Treatment

Oil coating spray of eggs has become very popular for short term storage of this commodity. Coating oil forms a thin film on the surface of the shell sealing the pores. It should be done as early as possible, preferably within first few hours after laying of eggs because loss of CO_2 is more during this period and evaporation of moisture is also more during the first few days. Egg coating oil should be colourless, odourless and conform to food grade. Coating is done by dipping the eggs in the groundnut oil whereas for oil spray, the eggs are arranged in the filler flats with their broad end up. If the eggs need washing, oil coating should be done after washing. It is important to drain out excess oil before packaging. The temperature of oil should be in range of 15 to 30°C for ideal results. Oil treatment safeguards the quality of albumen for atleast 7 days because it effectively seals the shell pores.

Cold Storage

This method of preservation is suitable for long term storage of clean eggs in the main laying season and abundant availability. The temperature of cold store is maintained at 0°C (32°F) and relative humidity between 80 to 85 per cent. An anteroom with intermediate temperature is generally provided to check condensation of water vapour on the eggs during removal. Use of new egg packing trays are advised for cold storage. Like all other animal products, eggs also pick up strong odour, so the same cold store cannot be used for storing onion, garlic or any other commodity with strong odour. The quality of shell eggs can be maintained for about 6 months in a cold storage. Oil coating of eggs prior to cold storage can further enhance their keeping quality. Such eggs could keep well at 14°C and 90% RH for a period of 8 months.

Thermostabilisation

This preservation method involves stabilisation of albumen quality by holding the eggs in an oil bath maintained at 55°C for 15 minutes or 58°C for 10 minutes. This process brings about coagulation of thin albumen just below the shell membranes, thereby blocking the passage of air and moisture. In addition, oil coating of shell pores also takes place. Thus keeping quality of eggs is maintained for sometimes and thinning of egg white is retarded. Alternatively, eggs are immersed in hot water at 71°C for 2 to 3 seconds. In this flash heat treatment, bacteria present on the surface of the shell are destroyed and a thin film of albumen just below the shell membrane is coagulated sealing the egg shell from inside.

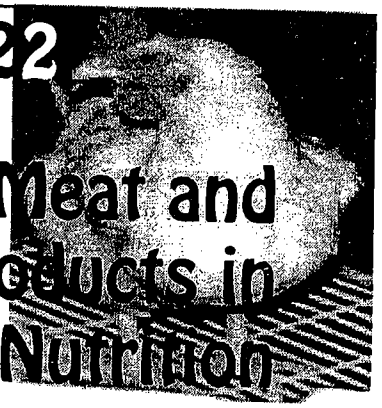
Immersion in Liquids

Under rural conditions, lime-water or water-glass immersion are most useful. In lime-water treatment, a litre of boiling water is added to 1 kg of quick lime and allowed to cool. Now 5 litres of water and 250 g of table salt are added to it. The solution is strained through a fine cloth when the mixture settles down. Eggs are dipped in the clear fluid overnight and then dried at room temperature. In this process, an additional thin film of calcium carbonate is deposited on the egg shell and seals the pores. Such eggs can be stored for a month at ambient temperature. In water-glass treatment, one part

of sodium silicate is mixed in 10 parts of water and eggs are dipped overnight. In this process, a thin precipitate of silica is deposited on the egg shell and partially seals the pores.

It is clear from the above discussion that eggs should be collected frequently, held initially at low temperature and then a suitable preservation method be employed to maintain its keeping quality for anticipated consumer acceptance.

CHAPTER 22



Role of Meat and Poultry Products in Human Nutrition

We have already discussed the nutritive value of fresh meat and chicken in chapter 2 and 13 respectively. Meat is a very well recognised nutritious food due to abundant high quality protein, B-complex vitamins and important minerals especially iron. However, all the nutrients contained in fresh meat do not reach the consumer. Several of them could be partially lost in the processing. The extent of nutrient loss will depend on the processing steps undertaken during the manufacture of a particular product. Hence, there is a need to have a fresh look at the nutritive value of meat and poultry products. Although variety range of processed meat products is very high, relevant information is available only generally prepared products (Tables 22.1 and 22.2).

Most processing procedures involve cooking which brings about a number of changes in meat. Cooking coagulates and denatures the meat proteins altering their solubility. It inactivates or destroys the indigenous proteolytic enzymes. Cooking invariably decreases the water content of meat lowering the water activity level. It intensifies the flavour and modifies the texture. In addition, considerable number of microorganisms are killed enhancing the storage life of meat.

Smoking and cooking take place simultaneously in most cured meat products. During smoking, carbonyl groups present in smoke react with amino groups of protein whereas phenols and polyphenols in smoke could react with sulphydril group of protein. Both the reactions cause some loss of available amino acids thereby decreasing the nutritive value of protein. Water soluble vitamins may

also be affected to some extent. In fact, some destruction of thiamine (Vitamin B_1) is inevitable, although effect on riboflavin (Vitamin B_2) and niacin may be very little. Smoking process can be nutritionally advantageous because it helps to stabilise the fat soluble vitamins due to antioxidant properties. Canning process is particularly detrimental to the water soluble vitamins present in meat. In canning, about 20-40% of thiamine, 10% each of riboflavin and niacin, 20% of biotin and 20-30% pantothenic acid are destroyed.

Processing changes the nutritional characteristics of fresh meat to some extent. The percentage of protein is slightly decreased whereas that of fat and minerals is increased. The percentage of minerals is generally increased due to added salt and seasonings. Besides, processed meats have more caloric values as compared to fresh meat due to the addition of fillers, binders and other extenders in the form of cereal flours or skimmed milk powder and frequently some fat.

Table 22.1: Proximate composition and caloric value of some processed meat and poultry products (per 100 g edible portion)

Product details	*Calories*	*Water*	*Protein*	*Fat*	*Carbohydrate*	*Ash*
Chicken, boneless, canned	198	65.2	21.7	11.7	–	1.4
Chicken with vegetables, canned	100	79.6	7.4	4.6	7.2	1.2
Turkey with vegetables, canned	86	81.3	6.7	3.2	7.6	1.2
Egg yolks with ham or bacon, canned	208	70.3	10.0	18.1	–	1.3
Lamb, strained, canned	107	79.3	14.6	4.9	–	1.2
Lamb, junior, canned	121	76.0	17.5	5.1	–	1.4
Liver, strained, canned	97	79.7	14.1	3.4	1.5	1.3
Pork, strained, canned	118	77.7	15.4	5.8	–	1.1
Bacon, cured, cooked and drained	611	8.1	30.4	52.0	3.2	6.3
Ham, cured, medium fat, cooked	289	53.6	20.9	22.1	–	3.4
Pork with gravy (90% pork, 10% gravy) canned	256	56.9	16.4	17.8	6.3	2.6
Beef, corned, boneless, medium fat, canned	216	59.3	25.3	12.0	–	3.4

Source : USDA Handbook No.8

Inspite of some processing losses, meat and poultry products are rich sources of vital nutrients. Meat products depict a lot of variation in the amount of the protein but most of the products are rich in protein content which is of very high quality due to the availability of essential amino acids. In fact, meat products could meet a major portion of recommended dietary allowance (RDA) of 56g protein per day as prescribed by the National Research Council. Since protein is needed to make up the day-to-day wear and tear of

Table 22.2: Mineral and vitamin content of some processed meat and poultry products (per 100 g edible portion)

Product details	*Ca (mg)*	*P (mg)*	*Fe (mg)*	*Na (mg)*	*K (mg)*	*Vit.B_1 (mg)*	*Vit.B_2 (mg)*	*Niacin (mg)*	*Vit.A (IU)*
Chicken, boneless, canned	21	247	1.5	250	138	0.04	0.12	4.4	230
Chicken with vegetables, canned	22	85	0.9	265	71	0.09	0.15	1.6	100
Egg yolks, with ham or bacon, canned	71	185	2.8	313	82	0.10	0.23	0.5	1900
Lamb, strained, canned	9	124	2.1	241	181	0.02	0.17	3.3	–
Liver, strained, canned	6	182	5.6	253	202	0.05	2.00	7.6	24,000
Pork strained, canned	8	130	1.5	223	178	0.19	2.00	2.7	–
Bacon, cured, cooked and drained	14	224	3.3	1021	238	0.51	0.34	5.2	1021
Ham, cured, medium fat, cooked	9	172	2.6	–	–	0.47	0.18	3.6	–
Pork with gravy (90% pork, 10% gravy) canned	13	183	2.4	–	–	0.49	0.17	3.5	–
Beef, corned, boneless, medium fat, canned	20	106	4.3	–	–	0.02	0.24	3.4	–

Source: USDA Handbook No. 8

body tissues in adults and large amount of protein can be stored in the body, consumption of meat products can ensure its availability to a large extent. Besides, protein supports the growth in children and pregnant ladies, consumption of enough protein products with high biological value becomes an absolute necessity. Since meat products contain ample amount of fatty acids that are essential in the diet of human, the recommended dietary allowance of fat is relatively less, it can be easily met. However, it should be emphasised that people with genetic disposition for obesity should restrict the consumption of animal fat.

Meat products contain enough of vital minerals such as iron, sodium, potassium and phosphorus. However, these are particularly deficient in calcium. Much of the requirement of iron which is an absolute necessity for health upkeep, can be made available by the meat products. Anaemic patients are usually recommended a liver diet because of its high iron content. A regular intake of iron is must for the proper synthesis of haemoglobin, myoglobin and certain enzymes due to very limited capacity to store iron in the body. All the water soluble vitamins are present in meat products but thiamine, riboflavin and niacin are present in significant quantities. Liver containing meat products are extremely rich in vitamin A content.

Though processed products vary in the relative proportion of nutrients, 100g serving of most meat products can supply 50% of recommended dietary allowance for protein, 25-50% of B-complex vitamins, 35% of recommended iron and 10% of the calorie requirement of an adult person. The nutritional attributes of meat products are highly acclaimed by the dieticians for a healthy living.

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